ZAPI® S.p.A.

ELECTRONIC • OLEODYNAMIC • INDUSTRIAL EQUIPMENTS CONSTRUCTION
Via Parma, 59 – 42028 – POVIGLIO (RE) – ITALY
Tel +39 0522 960050 – Fax +39 0522 960259
e-mail: zapi@zapispa.it – web: www.zapispa.it

EN

User Manual

HIGH VOLTAGE INVERTER



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APPROVAL SIGNS

COMPANY FUNCTION	INITIALS	SIGN
PROJECT MANAGER		
TECHNICAL ELECTRONIC MANAGER VISA		
SALES MANAGER VISA		

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1 INTRODUCTION

The Zapi High Voltage inverter is an electronic device for controlling traction motors in application where the dc-link voltage is beyond 120V.
This product is suitable for all kind of electric and hybrid vehicles where the battery voltage is in the range 160÷720V and the motor power up to 30 kW.
The logic section of the inverter is designed to operate at low voltage, 12V or 24V, and it is electrically isolated from the high voltage power section.

It has been designed to control:

- Asynchronous motors, with encoder for speed feedback,
- Permanent Magnet Synchronous motor (PMSM, IPM) with sin/cos, resolver, encoder+index position sensor.

The High Voltage Inverter is equipped with a Molex CMC 48 poles connector for low voltage supply, control signals, output and communication busses. For heat dissipation two different solutions are available:

- high efficiency finned heatsink.
- baseplate with conductors for passage of a liquid coolant. This solution is the preferred choice in a system where water cooling is already available.

2 CONTROLLER SPECIFICATION

2.1 General Features

Zapi HV Inverter, based on IGBT technology, is designed to control AC induction, BLDC and PMSM motors, mainly in the range from 25 kVA to 35 kVA continuous apparent power, for use in harsh environment like off-road application.

It's available with finned heatsink or liquid cooling base plate, an integrated unit with PM motor and inverter is also available

Features include:

- 16 bits microcontroller for motor control and main functions, 576+ Kbytes embedded Flash memory
- 16 bits microcontroller for safety functions, 320+ Kbytes embedded Flash memory
- Field-oriented motor control algorithm
- Smooth low speed control, including zero speed
- Driver for Line Contactor coil
- Driver for pre-charge circuit
- Pre-charge resistor not included
- Drivers for PWM voltage controlled electrovalves
- Short circuit and open load protection
- HVIL function
- Active discharge function
- Active Short Circuit function for PM motors
- Thermal cutback, warning, and automatic shutdown provide protection for motor and controller
- ESD protected CAN bus interface
- Software downloadable via serial link or CAN bus
- Diagnostic provided via CAN bus using Zapi CanConsole tool for PC
- Rugged sealed housing and connectors meet IP69K environmental sealing standards for use in harsh environments

2.2 Technical specifications

Inverter for AC asynchronous 3-phase motors or PM motors
Regenerative braking function
Dual CAN Bus interface
Digital control based on microcontroller
Redundant CPU for improved safety

Logic section Supply Voltage	12÷24 V
Power section Supply Voltage	160÷720 V
Maximum current (at 700 V)	70 Arms for 2'
Continuous current (at 700 V)	45 Arms
Maximum output power	45 kW
Continuous output power	30 kW
Operating frequency (automatic switching possible)	2,4,8 kHz
External temperature range	-40 °C ÷ 40 °C
Maximum inverter temperature (at full power)	70 °C
Connector	Molex CNC 48 pins
Environmental rating	IP69K



Internal algorithms automatically reduce maximum current limit when heatsink temperature is > 60°C and output frequency is less than 10Hz. Heatsink temperature is measured directly inside the IGBT module.



2-minute ratings are based on an initial controller heatsink temperature of 25°C and a maximum heatsink temperature of 60°C. No additional external heatsink is used for the 2-minute rating test. PWM frequency may be reduced to improve performances



Controller may be set to operate without alarm in the range 80% to 120% of the nominal low voltage supply and high voltage battery. Outside this range controller will raise an alarm condition at start-up.



Undervoltage threshold is defined in hardware, for the low voltage supply. Overvoltage is managed in hardware, for high voltage battery, Undervoltage may be set as a parameter. After start-up the controller is fully operative until the voltage supplies exceed that limits.



The Undervoltage is measured on +Key line CNA-1 (M1) or CNA-5 (L1). DC-Link Overvoltage is measured on the Power Capacitor (HVBATT+)

2.3 Special features

The HV inverter has some special features with the purpose of improving safety and optimal management of PM machines.

2.3.1 DC Link capacitor auto-discharge

Zapi HV Inverter includes a built-in circuit for discharging internal capacitors in short time, normally less than 10 seconds. Energy stored in the internal capacitors is dissipated through the base plate.

The circuit is automatically driven when DC Link voltage start to decrease after a short delay (500mS) to avoid false start. Auto-discharge is operative also when low voltage power supply is missing.

The internal power dissipation for auto-discharge function is designed for the total capacity (nominal 240uF) installed inside the HV Inverter. In case of other devices installed on the same DC Link, with capacitors inside, the discharge time will be longer.

If total capacity will result enough high, it's also possible that initial decrease of the DC Link voltage is so slow and won't trigger the auto-discharge function.

Auto-Discharge function is also triggered when the HVIL loop is interrupted.

2.3.2 Active Short Circuit

Active Short Circuit function is a fundamental feature when application needs to drive PM motor in deep field weakening area.

If PM motor is driven above the nominal speed and control is lost (for instance when low voltage supply fails), the voltage generated by the motor can reach dangerous level for inverter components.

In normal operation, when DC Link voltages reach the overvoltage threshold, microcontroller stops driving the IGBT bridge, but in case raising won't stop the Active Short Circuit comparator is activated.

The Active Short Circuit comparator switches on automatically the low side of IGBT bridge, putting motor phases in short circuit.

Motor inertia is then dissipated on the IGBT and possibly it will be stopped or decelerated bringing voltage again in the safe range.

Normally, the Active Short Circuit comparator has a pull in threshold of 840V and a pull out voltage of 750V.

The circuit is operative also in case low voltage power supply is lost because it's supplied directly from the DC Link. As soon as there's energy in the capacitor circuit is able to keep the IGBT in short.

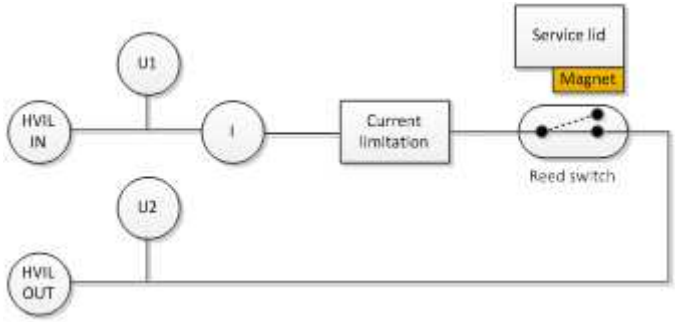
Short circuit current of PM motor must be checked not to exceed the maximum

current of IGBT devices because there isn't any way to control the motor current in the active short circuit phase.

Energy dissipated by the inverter must be enough to break the motor (it depends on motor short circuit current and IGBT characteristics), otherwise it must be ensured by design that maximum motor speed won't increase above a safe value.

2.3.3 HVIL

Zapi HV Inverter has a Hazardous Voltage Interlock (HVIL). The HVIL supervises all access points were high voltage live parts can be exposed. If any HV connector or service point is opened, the system must be de energized.



HVIL circuit concept

The HVIL shall be supplied with a 15 mA DC current source.

The current through the HVIL circuit is measured. While 8 mA < I < 30 mA the motor controller is considered electrically safe. The measurement circuit can handle common-mode voltage up to 0 – 10 VDC.

If the current is outside the valid range diagnostic is performed according to:

- U1 ≥5 V, U2 <5 V → HVIL internally open (a HV connector or service lid is not mounted).
- U1 ≥5 V, U2 ≥5 V → HVIL open after motor controller.
- U1 <5 V, U2 <5 V → HVIL open before motor controller.

3 EXTERNAL CONNECTIONS FOR ZAPI HV INVERTER

In some installations Zapi HV Inverter may require some external parts in order to operate properly. The following devices complete the kit for the installation in a typical application.

The use of special devices to be connected to the inverter, has to discussed with Zapi Technicians

3.1 Main contactor

The main contactor is a very important component for electrical safety so, from a general point of view, it must be installed.
The power contacts must be rated to sustain the high voltage of the battery, while the coil must operate at low voltage (24V or less).
For best electrical protection use two contactors, one to separate the HVBATT+ line and one to separate the HVBATT- line.

The standard connection of the MC coil is the following:

- Positive supply is the KEY voltage
- Negative supply is given by an output pin from connector CNA. CNA-11 (K3) , reference NLC, is dedicated for such a function (see 4.9)

Eventually, MC coil can receive the positive supply from the tiller switch.

CNA-11 (K3) is a low side driver, which drives to N_KEY, in two possible way:

- on/off control: coil is driven with the full low voltage supply
- pwm control: coil is switched on and off at fixed frequency (1 kHz) with a duty cycle which can be set by the user. This is useful to reduce the power dissipation of the contactor coil.

Please check the specific coil requirements to select the appropriate mode of operation. In the HV Invert there's already a built-in free-wheeling diode, but customer can connect an external diode as well, if preferred.

Wiring diagram in chapter 5.4.8 shows a possible implementation for contactors connection. For more details see chapters 4, 6 and 7.

3.2 Pre-charge circuit

Zapi HV Inverter doesn't have an embedded pre-charge circuit for internal capacitors connected between HVBATT+ and HVBATT-. The pre-charge process, if needed, must be carried out by external components.

Wiring diagram in chapter 5.4.8 shows a possible solution with a couple of relays and resistors: relay coils are controlled by a single output of the inverter.

Once the pre-charge is completed, the pre-charge relays are deactivated and main contactors will be closed.



Other solutions for pre-charge are possible. Please contact Zapi technicians for further details. Depending on DC Link source, pre-charge may not be needed.

4 I/O INTERFACE DESCRIPTION

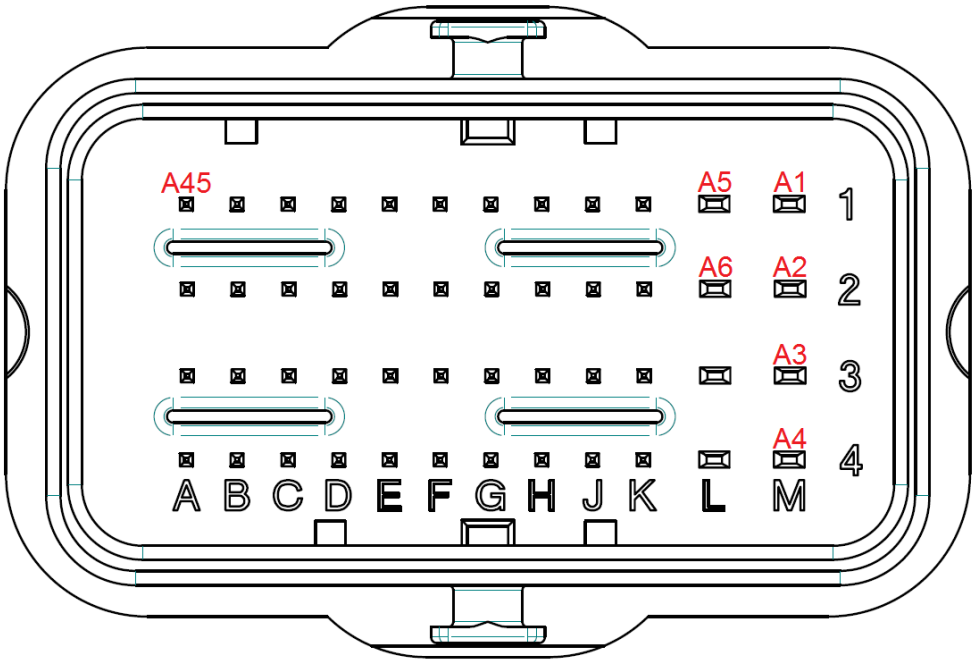
4.1 Motor and Battery Connections

Terminal Name	Description
HVBATT+	High Voltage DC Link positive
HVBATT-	High Voltage DC Link negative
U, V, W	Motor phases U, V, W connections

4.2 Connectors

4.2.1 Low voltage connector (Molex CNC 48 poles)

Low voltage signals are connected to the HV Inverter through a 48 pins Molex CNC connector



Inside this document, connector's pins are referenced in 2 ways:

- using the suffix "CNA-nn", with numbers from 1 to 48
- using a 2-coordinate alphanumeric system from M1..4, L1..4, … A1..4

For each I/O pin is indicated the default function using Zapi std software. The function of each pin can be changed in the customized software.

Some I/O pins can have special functionality depending by controller configuration .

Pin	Coor.	Type	Name	Description
A1	M1	Input	KEY	Input of the key switch signal (low voltage supply positive 12V or 24V)
A2	M2	Input	PEV	Connect the positive supply of electrovalves (EB, EVP and EVx) to this pin. Take the positive supply immediately after the main contactor.
A3	M3	Output	PANIN	Positive supply for external analog device (Sin/Cos sensor, potentiometers, ecc) factory selectable +12/+5V, maximum 100mA
A4	M4	Input	N_KEY	Low voltage power supply negative
A5	L1	Input	KEY	Input of the key switch signal (low voltage supply positive 12V or 24V)
A6	L2	Input	PEV	Connect the positive supply of electrovalves (EB, EVP and EVx) to this pin. Take the positive supply immediately after the main contactor.
A7	L3	Output	AGND	Ground for external analog devices (Sin/Cos sensor, potentiometers, ecc)
A8	L4	Input	N_KEY	Low voltage power supply negative
A9	K1	Output	EV1	Output of the ON/OFF or PWM voltage controlled electrovalve EV1. Maximum 1A continuous current, driving to N_KEY (pre-charge function).
A10	K2	Output	EV2	Output of the ON/OFF or PWM voltage controlled electrovalve EV2. Maximum 1A continuous current driving to N_KEY (HVBATT- main coil function).
A11	K3	Output	NLC	Main contactor coil driver output, ON/OFF or PWM voltage controlled. Maximum 1.5A continuous current driving to N_KEY.
A12	K4	Output	GND3	Ground for power I/O
A13	J1	Output	PENC	Positive supply for external device +12 /+5 V 100mA maximum (Encoder, potentiometers, ecc)
A14	J2	Input	ENC A	Traction motor encoder phase A.
A15	J3	Input	ENC B	Traction motor encoder phase B.
A16	J4	Output	NENC	Negative supply for external device (Encoder, Potentiometers, ecc)

Pin	Coor.	Type	Name	Description
A17	H1	Input	ANIN1	Analog input #1. The default function is the accelerator potentiometer wiper.
A18	H2	Input	DI0	Input of the switch DI0. The input is activated when connected to GND. The default function is the controller “TILLER” request input, it must be active before starting the motor
A19	H3	Input	DI1	Input of the switch DI1. The input is activated when connected to GND. The default function is the controller “FORWARD” request input, closing this input truck moves in forward direction.
A20	H4	Input	DI2	Input of the switch DI2. The input is activated when connected to GND. The default function is the controller “REV” request input, closing this input truck moves in backward direction.
A21	G1	Output	RES_ECC_N	Negative output for Resolver Excitation
A22	G2	Output	RES_ECC_P	Positive output for Resolver Excitation
A23	G3	Input	ENC_INDEX	Traction motor encoder INDEX signal
A24	G4	Output	GND	Negative ground for digital I/O
A25	F1	Input	RES_SINE_N	Negative input for resolver Sine secondary winding
A26	F2	Input	RES_SINE_P	Positive input for resolver Sine secondary winding
A27	F3	Input	RES_COS_N	Negative input for resolver Cosine secondary winding
A28	F4	Input	RES_COS_P	Positive input for resolver Cosine secondary winding
A29	E1	Output	NCLTXD	Serial Console negative serial transmission.
A30	E2	Input	NCLRXD	Serial Console negative serial reception.
A31	E3	Output	+12	Serial Console positive power supply.
A32	E4	Output	PCLTXD	Serial Console positive serial transmission.
A33	D1	Input	BOOT	Serial programming activation signal

Pin	Coor.	Type	Name	Description
A34	D2	Input	HVIL_IN	Input pin for HVIL current loop (referred to N_KEY)
A35	D3	Output	HVIL_OUT	Output pin for HVIL current loop (referred to N_KEY)
A36	D4	Output	PCAN	Isolated CAN power supply (12V not regulated)
A37	C1	Input	CANT_M	If connected with CANH_M introduces activate a 120 Ω termination resistance between CANL_M and CANH_M.
A38	C2	Output	CANH_M	“positive” isolated CAN bus bidirectional line for master microcontroller
A39	C3	Output	CANL_M	“negative” isolated CAN bus bidirectional line for master microcontroller
A40	C4	Output	NCAN	CAN bus isolated negative reference. Must be used because of CAN isolation
A41	B1	Input	CANT_S	If connected with CANH_S introduces activate a 120 Ω termination resistance between CANL_S and CANH_S.
A42	B2	Output	CANH_S	“positive” isolated CAN bus bidirectional line for slave microcontroller
A43	B3	Output	CANL_S	“negative” isolated CAN bus bidirectional line for slave microcontroller
A44	B4	Output	NCAN	CAN bus isolated negative reference. Must be used because of CAN isolation
A45	A1	Input	PTH2	Motor thermal sensor input #2. The internal pullup is a fixed 2mA (max 3V) source current, isolated from low voltage power supply
A46	A2	Output	GND_THM2	Isolated ground for thermal sensor #2
A47	A3	Output	GND_THM1	Isolated ground for thermal sensor #1
A48	A4	Input	PTH1	Motor thermal sensor input #1. The internal pullup is a fixed 2mA (max 3V) source current, isolated from low voltage power supply

4.2.2 Internal connector (CNG)

Pin	Type	Name	Description
1	Output	PCLRXD	Positive serial reception (Not used: leave disconnected).
2	Input	NCLRXD	Negative serial reception.

3	Output	PCLTXD	Positive serial transmission.
4	Output	NCLTXD	Negative serial transmission.
5	Output	GND	Negative console power supply.
6	Output	+12	Positive console power supply.
7	Input	BOOT	Serial programming activation signal
8	Output	GND	Negative console power supply.

4.3 Key Input

4.3.1 Function

KEY input brings battery voltage to the controller for internal processors and other functions. Normally it's connected in series to the vehicle start switch. After power-up, KEY voltage is monitored for overvoltage and undervoltage condition

4.3.2 Protection

A diode protects the KEY against reverse polarity. There's a significative capacitance internally connected at KEY input so current spike can arise at KEY input depending on the external circuit.
The Fuse Fu1 shown in the functional drawings, should be sized according to the number of controllers connected to the fuse and to protect the cable area in the circuit (recommended fuse size max 10 A).
Input power is normally under 10W.



The Key Switch connected to the KEY must handle the short inrush current spike. Peak value is depending on the external circuit and wire.



Cable length from the battery to the KEY input shall be as short as possible

4.4 Digital Inputs

4.4.1 Function

Digital inputs work in the low voltage range [N_KEY; KEY]. Related command devices (microswitches) must be typically connected to KEY or N_KEY depending by input configuration (refer to pin description in the related paragraph) because pull-down or pull-up resistance are built-in.

DI0..2 are "normally open", with internal pull-down, so related function becomes active when the microswitch closes.

The threshold level is 3.1V
Input impedence is 4.7kΩ

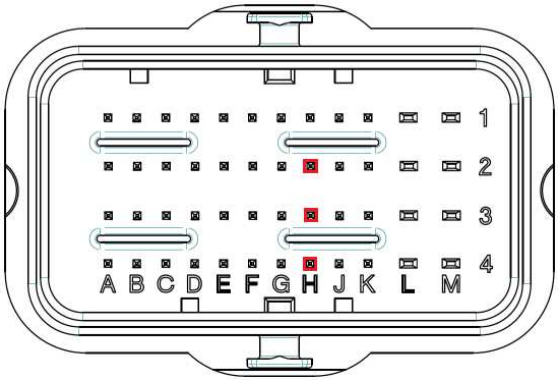
Voltage range 0..24V

For critical functions when good diagnostic coverage is necessary it is recommended to use two digital inputs for plausibility check, for example, use of both normally open and normally closed contacts.

4.4.2 Protection

The Digital Inputs have a 10 nF capacitor to N_KEY for ESD protection.

4.4.3 Connector position



CNA-18 (H2)	DI0
CNA-19 (H3)	DI1
CNA-20 (H4)	DI2

4.4.4 Microswitches

- It is suggested that microswitches have a contact resistance lower than 0.1Ω and a leakage current lower than 100µA.
- For correct operation, the voltage between the key switch contacts must be lower than 0.1V.
- To connect microswitches with different electrical characteristics, it is suggested to contact a Zapi technicians.

4.5 Analog Inputs

4.5.1 Function

The analog inputs are normally used for speed or brake potentiometers. They are internally connected to a 10 bit Analog to Digital Converter (ADC value range 0..1023).

Input range +0V to +10.7 V, give ADC value from 0 to 1023.
Analog inputs impedance = 99 kΩ, maximum input frequency 13 Hz.

The standard connection for the potentiometer is the one in the Left side of next figure (potentiometer on one end at rest) in combination with a couple of Travel demand switches. On request it is also possible the handling in the Right side of next figure (potentiometer in the middle at rest) still in combination with a couple

of Travel demand switches.

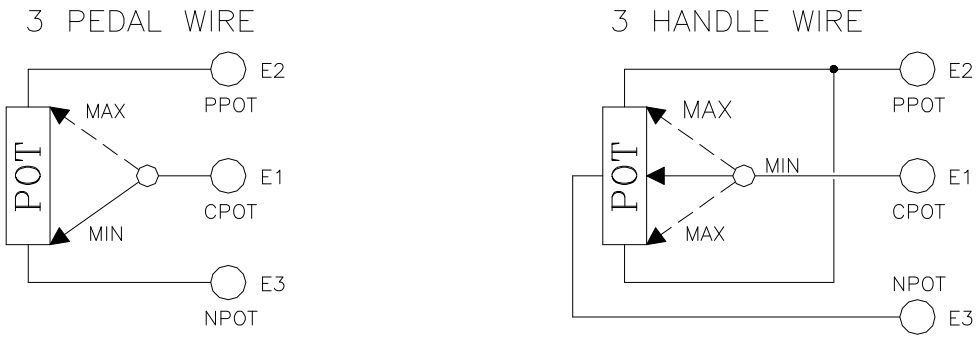


Figure 1: Potentiometer configuration

The negative supply of the potentiometer has to be taken from CNA-7 (L3) AGND. Potentiometer value should be in the 0.5 – 10 kΩ range; generally, the load should be in the 1.5 mA to 30 mA range.

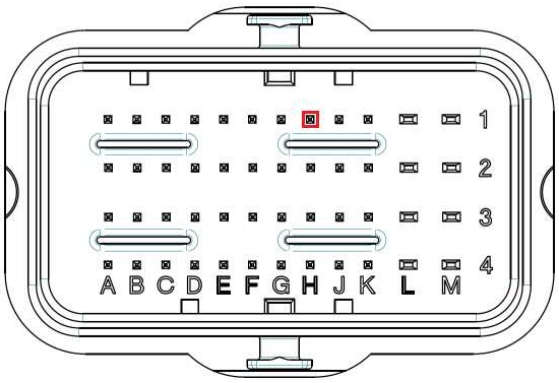
The Procedure for automatic potentiometer signal acquisition is carried out using the Console (see paragraphs 14.2.6).

The Analog inputs may also be used as extra Digital inputs. The ADC-value shall be used as the indicator of the input status. As an example: a proximity switch supplied from N_KEY is connected to an analog input.

4.5.2 Protection

Analog inputs are KEY and N_KEY protected and has a 100 nF capacitor to N_KEY for ESD protection.

4.5.3 Connector position



CNA-17 (H1) ANIN1

If an analog input is used as a speed reference to the motor controller, a system safety strategy must be defined.

The application software must take care of analog input errors such as: Vacc out of range, Vacc not ok.

4.6 Encoder A/B Inputs and Index signal

4.6.1 Function

Digital motor feedback sensor input has an internal 1kΩ pull-up to CNA-13 (J1) P_ENC. Inverter inputs are designed for open collector sensor output. Optional "index" signal has a dedicated input with the same characteristics.
Threshold levels are:

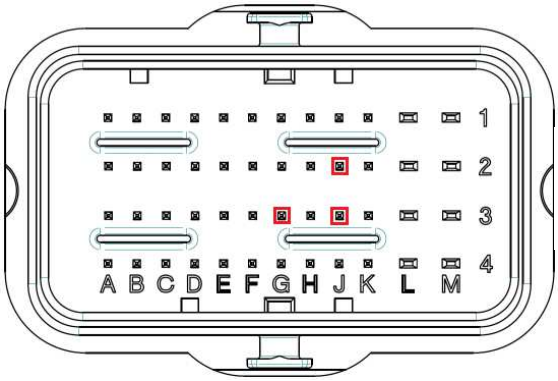
Encoder Supply Voltage	**5V**	**12V**
Logic low	1.2V	3.6V
Logic high	3.1V	5.8V

The speed sensor signals are connected to a CPU dedicated peripheral for quadrature signals.

4.6.2 Protection

Encoder inputs are KEY and N_KEY protected and have ESD suppressor to N_KEY.

4.6.3 Connector position



CNA-14 (J2) ENCA
CNA-15 (J3) ENCB
CNA-23 (G3) ENC_INDEX

It is important to verify the wiring by ensuring that the Encoder signals are not disturbed from the motor current or the electric motor brake.

For more details about encoder installation see also paragraph 5.4.5.

Note: The encoder resolution and the motor poles pair (the controller can handle), is specified in the home page display of the handset showing following headline:
HBMT3 R ZP0.01

That means:
HBMT = HV Inverter traction controller (M stands for "µC Master", S for "µC Supervisor")
(HBMP = ACE4 pump controller)
3 = number of pole pairs

R = configured for resolver position sensor

The encoder resolution is given by dedicated parameters. See paragraph 8.1.7.

4.7 Sine/Cosine input signal

4.7.1 Function

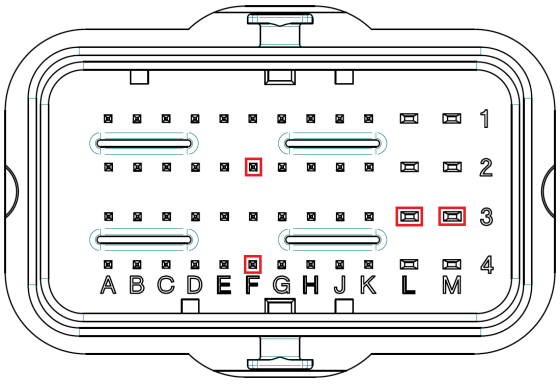
Position information, coming from an analog Sin/Cos sensor, can be connected to pins CNA-26 (F2) and CNA-28 (F4).
Sensor power supply must be referenced to the N_KEY potential, pin CNA-7 (L3) AGND can be used as positive power supply.
Power supply voltage for sin/cos sensor must be carefully select and tested to be compatible with sensor characteristics. Connecting a PANIN output from HV Inverter, configured for 12V, to a 5V sensor bearing may destroy the sensor.

Sin/Cos interface pins are shared with Resolver interface partially shared with Resolver interface

4.7.2 Protection

Sin/Cos inputs are N_KEY using ESD suppressor.

4.7.3 Connector position



CNA-26 (F2)	RES_SINE_P
CNA-28 (F4)	RES_COS_P
CNA-3 (M3)	PANIN
CNA-7 (L3)	AGND

It is important to verify the wiring by ensuring that the Encoder signals are not disturbed from the motor current or the electric motor brake.

For more details about encoder installation see also paragraph 5.4.6

Sin/Cos sensor bearing have normally a nominal power supply of 5V and don't tollerate overvoltage (maximum 7V). When povered directly from HV Inverter using PANIN, check

4.8 Resolver interface

4.8.1 Function

Position information, coming from a Resolver, can be connected to the HV Inverter Resolver Interface:

- CNA-21 (G1), CNA-22 (G2) resolver primary winding (excitation)
- CNA-25 (F1), CNA-26 (F2) resolver secondary “sine” winding
- CNA-27 (F3), CNA-28 (F4) resolver secondary “cosine” winding.

Resolver excitation is fully isolated from high/low voltage power supply. Excitation signal is a 100uS pulse up to 60V peak-to-peak with 2Khz of repetition rate.

Pins for secondary windings are referred to the low voltage power supply, they are managed as differential signals. Input impedance is greater than 50kΩ.

HV Inverter Resolver interface can accommodate a wide range of Resolver type considering impedance and transformer ratio. In any case Resolver electrical characteristics should be evaluated, in advance, with Zapi technician before.

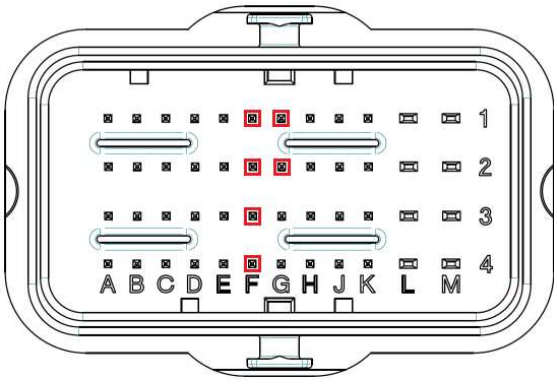
Motor and Resolver must have the same number of pole pairs, a single pole pair Resolver can be used as universal solution.

4.8.2 Protection

Resolver input pins have protection against voltage pulse with ESD suppressor connected to N_KEY. A temporary wrong connection to KEY or N_KEY will not harm the circuit.

Resolver excitation generates high voltage pulse, up to 60V peak-to-peak, with 100uS length and fully isolated from high/low voltage power supply. A temporary wrong connection to KEY or N_KEY will not harm the circuit.

4.8.3 Connector position



CNA-21 (G1)	RES_ECC_N
CNA-22 (G2)	RES_ECC_P
CNA-25 (F1)	RES_SINE_N
CNA-26 (F2)	RES_SINE_P
CNA-27 (F3)	RES_COS_N
CNA-28 (F4)	RES_COS_P

It is important to verify wirings by ensuring that Resolver signals are not

disturbed from the motor current or the electric motor brake.

Normally twisted pair cables are not needed for resolver signals as a consequence of high voltage excitation pulse.

For more details about resolver installation see also paragraph 5.4.5.

4.9 MC Output

4.9.1 Function

Open Drain PWM voltage controlled output used normally for operating Main Contactor.
Built-in free-wheeling diode can be used only if the load must is supplied from KEY pin (CNA-1 L1, CNA-5 M1).
In case the vehicle design does not allow usage of the built-in free-wheeling diodes, i.e. if the return path integrity cannot be guaranteed in all situations, external free-wheeling diodes must be applied directly over the inductive load.

Output features

- 1.5A continuous (hold current) and max 2.0 A peak (pull current) for a maximum of 200 ms.
- Individual hardware for driver shorted, driver open and coil open detection.
- 1kHz default PWM frequencies.
- Voltage controlled with separate pull and hold voltage software parameters.

PWM shall only be used for inductive loads such as relays, contactors, motor brakes or hydraulic valve

PWM frequency can be changed by software. If a different PWM frequency has to be used, it is suggested to discuss it with Zapi technicians.

4.9.2 Protection

Protected against inductive discharge with internal freewheeling diode connected to pin CNA-1 (L1), CNA-5 (M1) and ESD protection device.

Protected against reverse polarity of the battery.

Built-in diagnostics:

- Overcurrent
- Driver shorted
- Driver open
- Coil open

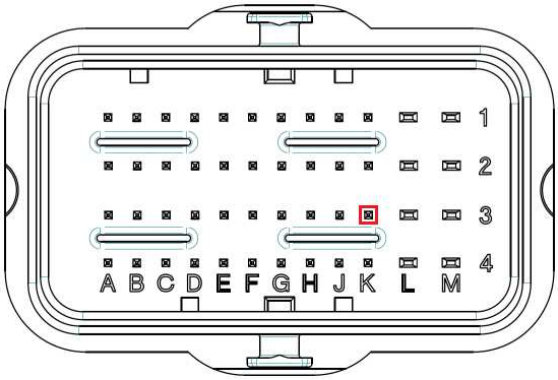
Refer to chapter 11 for more detailed description.

Overcurrent protection is applied by hardware

*MC output can be only a “PWM voltage controlled” output. It **can’t** be used as “current controlled” output.*

When driving inductive loads on PWM Open drain outputs there must always be a path for the current to the freewheeling diodes. Do not connect any switch or fuse in series with the diode.

4.9.3 Connector position



CNA-11 (K3) NLC

To protect the motor controller from overvoltage at inductive load, internal free-wheeling diode is mounted to the pins CNA-1 (L1) and CNA-5 (M1).

Please ensure that inductive loads are connected such that the path for the free-wheeling diode is always intact (see 4.9.1), or use an external free-wheeling diode if this is not possible.

Use of brushless fan or other loads with built-in capacitor can give high inrush current when turn ON which will give an Open Drain over current trip. The inrush current must be below the Open Drain peak current.

4.10 Auxiliary Outputs

4.10.1 Function

Open Drain outputs can be used for operating services such as relay, hydraulic valves, horn, etc. They can work in On/Off control or PWM Voltage Controlled

In order to utilize the built-in free-wheeling diodes, positive of the load must be connected to pins CNA-2 (M2) or CNA-6 (M6).

In case vehicle design does not allow usage of the built-in free-wheeling diodes, i.e. if the return path integrity cannot be guaranteed in all situations, external free-wheeling diodes must be applied directly over the inductive loads supplied by the open drain outputs.

ON/OFF Outputs features

- 1A continuous (hold current) and max 2.0 A peak (pull current) for a maximum of 200 ms.
- individual hardware for driver shorted and driver open detection.
- shared hardware coil shorted detection.

PWM Voltage controlled Outputs features

- 1A continuous (hold current) and max 2.0 A peak (pull current) for a maximum of 200 ms.
- Individual hardware for driver shorted and driver open detection.
- 1kHz PWM frequencies. It is applied to all PWM outputs.
- Each PWM voltage controlled outputs can be voltage controlled with separate voltage software parameters.

4.10.2 Protection

The auxiliary outputs are protected against inductive discharge with internal freewheeling diodes to pin CNA-2 (M2) or CNA-6 (M6).

Auxiliary outputs are NOT protected against reverse polarity of the battery.

Built-in diagnostics:

- Over Current
- Driver shorted
- Driver open
- Coil open (only for PWM Current Controlled Outputs)

Refer to section 11 for more detailed description

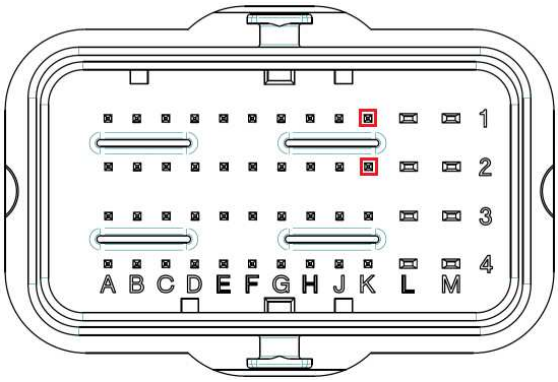


PWM shall only be used for inductive loads such as relays, contactors, motor brakes or hydraulic valves.



When driving inductive loads on PWM Open drain outputs there must always be a path for the current to the freewheeling diodes. Do not connect any switch or fuse in series with the diode.

4.10.3 Connector position



CNA-9 (K1)	NEV1
CNA-10 (K2)	NEV2

To protect the motor controller from overvoltage at inductive load, internal free-wheeling diodes are mounted to the A3 pin.

Please ensure that inductive loads are connected such that the path for the free-wheeling diode is always intact (see Function), or use an external free-wheeling diode if this is not possible.

Use of brushless fan or other loads with built-in capacitor can give high inrush current when turn ON which will give an Open Drain over current trip. The inrush current must be below the Open Drain peak current.

4.11 Motor Temperature

4.11.1 Function

HV Inverter is equipped with 2 motor temperature sensor input to avoid over heating of motor winding.

Support is intended for temperature sensors like

- KTY84 with 1000Ω @ 100°C
- KTY83 with 1670Ω @ 100°C
- PT1000 with 1385Ω @ 100°C
- ON/OFF digital sensor.

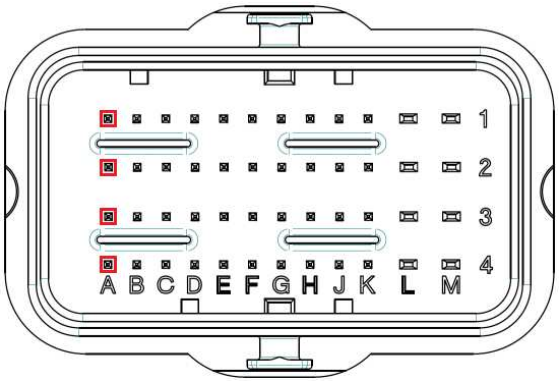
Circuits are fully isolated from high and low voltage power supply so thermal sensors (PTHx) must be connected using a dedicated signal ground (GND_THM1 or GND_THM2).

4.11.2 Protection

A 22 nF input capacitor provides ESD protection and filters disturbance noise from the motor.

Because of isolation, circuit is intrinsically protected again short circuit to KEY or N_KEY.

4.11.3 Connector position



CNA-45 (A1)	PTH2
CNA-46 (A2)	GND_THM2
CNA-47 (A3)	GND_THM1
CNA-48 (A4)	PTH1

4.12 Sensor Supply Output

4.12.1 Function

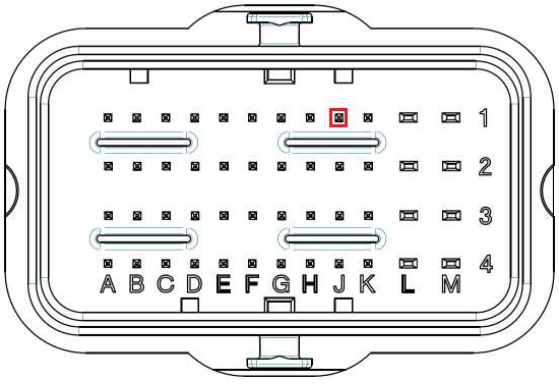
Supply for external motor speed sensors.
Output voltage is factory selectable in hardware using an internal jumper between “+12V” or “+5V” and total maximum output current is 100 mA.

Actual value for “+12V” and “+5V” are respectively 12V ± 0,2V and 5V ± 0,2V with a diode in series

4.12.2 Protection

Sensor supply output is protected for over current and against accidental connection to KEY with a diode.

4.12.3 Connector position



CNA-13 (J1)	PENC

4.13 Analog Supply Output

4.13.1 Function

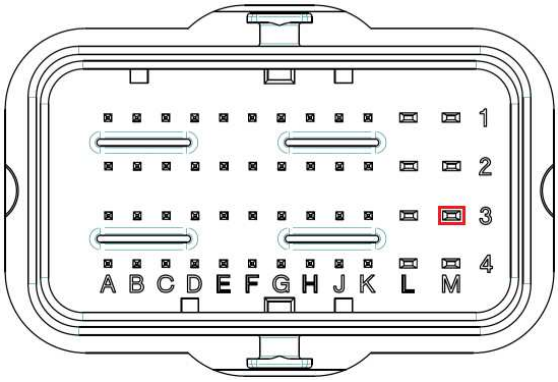
Second supply for external analog sensors, analog position sensor or potentiometers is available on pin CNA-3 (M3) PANIN.
Output voltage is factory selectable by internal jumper to “+12V” or “+5V” and maximum output current is 100 mA.

Actual value for “+12V” and “+5V” are respectively 12V ± 0,2V and 5V ± 0,2V with a diode in series

4.13.2 Protection

Sensor supply output is protected for over current and against accidental connection to KEY with a diode.

4.13.3 Connector position



CNA-3 (M3) PENC

4.14 CAN Bus

4.14.1 Function

Double physical CAN interface according to ISO 11898-2 with SW selectable data rate from 125 to 500 kBit/s. One interface is managed by Master microcontroller, the other by Supervisor microcontroller.

The CAN driver is +5V supplied and maximize signal amplitude on the CAN_H to CAN_L. A common-mode filter (resistors and capacitor) is installed.

CAN interfaces are fully isolated from HV Inverter low and high voltage power supply. For such a reason CAN negative reference must be used on pin CNA-40 (C4) or CNA-44 (B4).

An internal 120 Ω termination resistor can be inserted by closing pin CNA-37 (C1) to pin CNA-38 (C2) for Master microcontroller or CNA-41 (B1) to pin CNA-42 (B2) for Supervisor microcontroller .

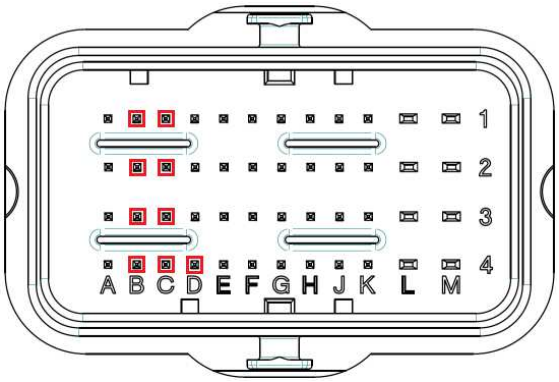
CAN isolated power supply is available on pin CNA-36 (D4) for a maximum current of 100mA. If needed, CAN isolated power supply can be factory set to

receive power from an external, using the same pin.

4.14.2 Protection

CAN interfaces are ESD protected using a transient suppressor

4.14.3 Connector position



CNA-36 (D4)	PCAN
CNA-37 (C1)	CANT_M
CNA-38 (C2)	CANH_M
CNA-39 (C3)	CANL_M
CNA-40 (C4)	NCAN
CNA-41 (B1)	CANT_S
CNA-42 (B2)	CANH_S
CNA-43 (B3)	CANL_S
CNA-44 (B4)	NCAN

It's possible to connect CAN Bus from master and supervisor microcontroller internally upon request.

CNA-wiring shall use a twisted pair for CAN_H and CAN_L.

The CAN Bus must have a characteristic impedance of 120 Ω and both physical ends of the CAN Bus must be terminated with 120 Ω between CAN_H and CAN_L for best possible noise immunity.

The HV Inverter CAN Bus is isolated so the use of NCAN reference is mandatory.

4.15 HVIL (High Voltage Interlock Loop)

4.15.1 Function

HV Inverter can be part of a HVIL (High Voltage Interlock Loop) for safety maintenance operation.

Current loop must enter in pin CNA-34 (D2) and exit from pin CNA-35 (D3). HVIL isn't isolated from low voltage power supply so, to be fully operational, negative of loop source must be referenced, in some way, to the N_KEY supply.

When the HV Inverter wiring compartment is open, the HVIL loop is interrupted to active the safety function of discharging the DC Link capacitors.

Please ask Zapi technicians for further information about the HVIL loop electrical

specification

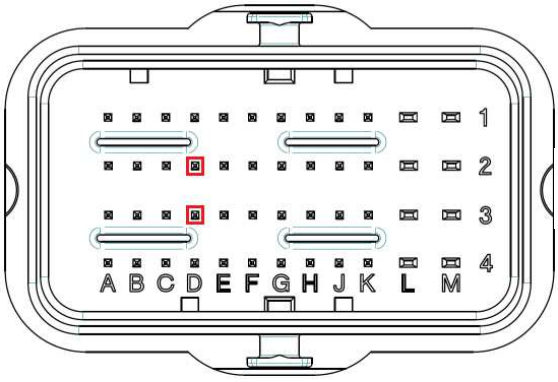
4.15.2 Protection

HVIL input and output are ESD protected using a transient suppressor.

Connecting HVIL pins to KEY or N_KEY won't damage the circuit

HVIL circuit is internally protected against overcurrent with fuse and solid state switches.

4.15.3 Connector position



CNA-34 (D2) HVIL_IN
CNA-35 (D3) HVIL_OUT

4.16 Smart Console

4.16.1 Function

Zapi Smart Console is a device developed to adjust parameters and read operative conditions from Zapi inverters.

It's widely used inside the commissioning phase, because it allows a deep control over parameters and behaviour of Zapi controllers.

Connection pins are:

- CNA-29 (E1 – NCLTXD)
- CNA-30 (E2 – NCLRXD)
- CNA-31 (E3 – 12V)
- CNA-32 (E4 – PCLTXD)
- CNA-33 (D1 – BOOT)
- use one of CNA-4 (M4), CNA-8 (L4), CNA-24 (G4) as a negative supply

Use of Zapi Smart Console should only be allowed to engineers and well trained

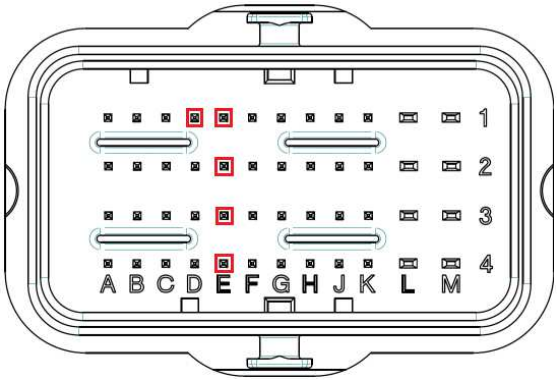
technicians!

4.16.2 Protection

NCLTXD, NCLRXD, PCLTXD and BOOT pins are ESD protected using a transient suppressor.

Connecting pins to KEY or N_KEY won't damage the circuit

4.16.3 Connector position



CNA-29 (E1)	NCLTXD
CNA-30 (E2)	NCLRXD
CNA-31 (E3)	12V
CNA-32 (E4)	PCLTXD
CNA-33 (D1)	BOOT

5 INSTALLATION HINTS

In the description of these installation suggestions you will find some boxes of different colours, they mean:

*These are **information** useful for anyone is working on the installation, or a deeper examination of the content*

These are Warning boxes, they describe:
- operations that can lead to a failure of the electronic device or can be dangerous or harmful for the operator;
- items which are important to guarantee system performance and safety

5.1 High and Low Voltage

The controller is divided into three different sections:

- Logic section which integrates microcontrollers, I/O interface and all control circuits. This board operates at low voltage, 12V or 24V
- Power section composed by 3-phase IGBT bridge and its driving circuit. It's supplied by the High Voltage DC Link, up to 720V. Greater voltage can be evaluated case by case, with hardware modifications
- Capacitors board, eventually with DC current sensor

Low voltage section and high voltage section are internally fully isolated (reinforced isolation) to guarantee the maximum level of electrical protection. External wiring and connected devices must respect good practice and safety rules:

- In the machine, never connect directly low voltage section and high voltage section.
- Follow good wiring practice.
- Respect creepage and clearance rules for the battery voltage in use.
- Use connectors and cables designed for the voltage that they must sustain.
- Never supply a low voltage device directly from the high voltage DC Link.

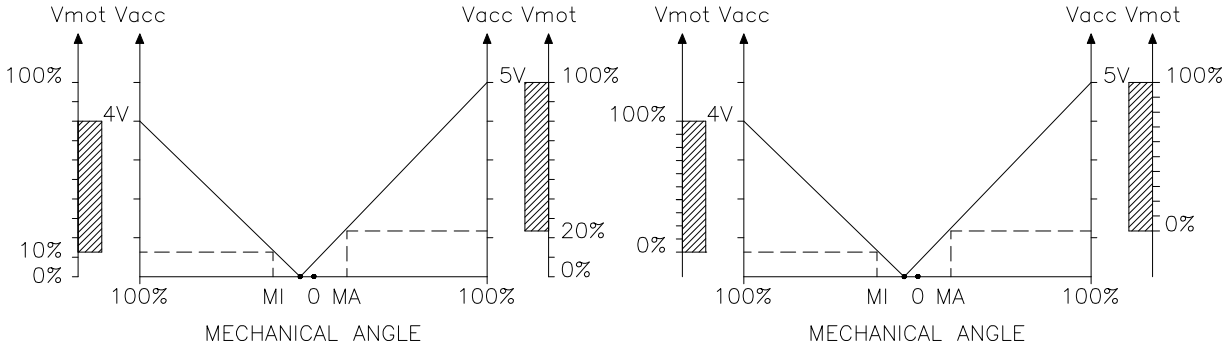
Be careful: high voltage levels pose a significant hazard to any operator which can access the wires

5.2 Setting up of the accelerator

The accelerator input must be programmed in order to have the correct operation. Set-up is divided into two parts.

5.2.1 Automatic accelerator acquisition

The accelerator output range is a voltage signal with a minimum and a maximum value that has to be acquired by the HV Inverter.
A procedure for automatic potentiometer signal acquisition is available both in Smart Console and CAN Console under the function PROGRAM VACC. This function allows adjustment of the minimum and maximum signal range level in either direction. The meaning is to compensate asymmetry with the mechanical elements associated with the potentiometer, especially relating to the minimum level. The sequence of procedure is described in the programming console manual.



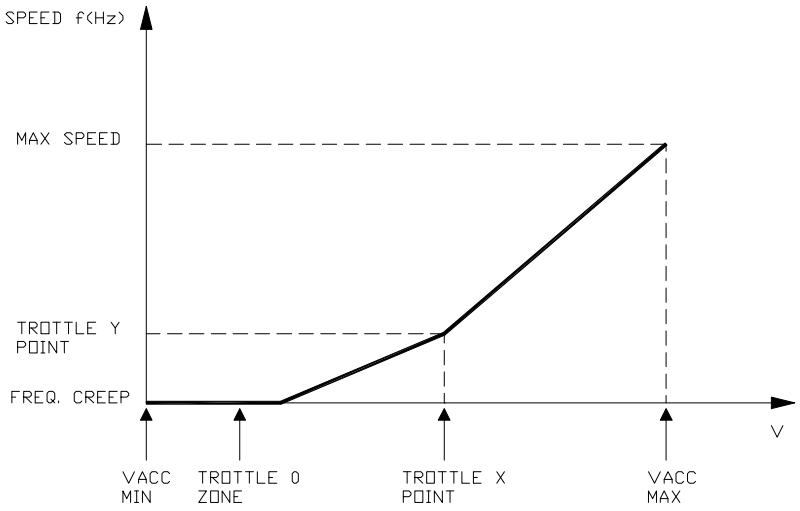
The two graphs show the output voltage from a non-calibrated potentiometer respect to the mechanical “zero” of the control lever. MI and MA indicate the point where the direction switches close. 0 represents the mechanical zero of the rotation.
The Left Hand graph shows the relationship of the motor voltage without signal acquisition, Right Hand graph shows the same relationship after potentiometer signal acquisition.

5.2.2 Management of the pedal-acceleration relation

The output voltage of the accelerator device sets the speed set-point of the truck. The relation between the output signal and the speed set-point is proportional but can be modified in the following way:

- Adding a dead-band around the “zero” output
- Adding an intermediate point in the proportional relation, in order to set two different slopes.

The following figure shows a possible configuration:



"Vacc min" and "Vacc max" are set by the Automatic Acquisition procedure.
The parameters THROTTLE 0 ZONE, THROTTLE X POINT and THROTTLE Y POINT are described in 8.2.3.

5.3 Wires and fuses

Before starting it is necessary to have the right material for a correct installation, otherwise a wrong choice of cables or other parts could lead to failures, misbehaviour or bad performances.

5.3.1 Connection cables

For the auxiliary circuits, use cables of 0.5 mm² section.
For power connections to motor and battery, use cables having section of 10 mm² (at least). The exact size depends on the average current of a typical application operative cycle.
For the optimum inverter performances, cables to the battery should be as short as possible, and run side by side.

5.3.2 Fuse for I/O circuits

Use a fuse for protection of the auxiliary circuits: normally this fuse is in series with the key switch (see connection diagrams). Typical fuse rating is 6.3÷10A and the voltage must be the same (or larger) than low voltage power supply (12V or 24V).

5.3.3 Fuse for power section

We suggest placing a power fuse to protect the battery cables and, eventually, the motor phases. Normally the fuse is installed on the cable carrying the positive of the battery (see connection diagrams).
The fuse rating depends on the average current of a typical operating cycle of the application and is not directly connected to the controller current.
A 80A fuse should cover the majority of applications, the voltage rating must be the same (or larger) as the battery voltage.

Notice that the purpose of the power fuse is not to protect the power section of the inverter. In case of an internal short circuit damage in the inverter will occur too quickly and the fuse will not blow. The purpose of the fuse is to protect the

power cables from burning in case the operating current rises too much.

For Safety reasons, we recommend the use of protected fuses in order to prevent the spread of fused particles in the case the fuse blows.

5.4 Installation of the hardware

Before doing any operation, ensure that the battery is disconnected and when all the installation is completed start the machine with the drive wheels raised from the floor to ensure that any installation error do not compromise safety.

After operation, even with the Key Switch open, the internal capacitors may remain charged for few seconds time. The HV Inverter has an internal discharge circuit but Zapi recommend that battery is disconnected, and operator wait more than 60 seconds before handling the device.

5.4.1 Positioning and cooling of the controller

The HV Inverter is available either with a finned heatsink or liquid cooling heatsink.
The choice of the type of heat dissipation and the characteristics of the chosen dissipation system (power of the fan, type of the coolant,…) depend on the performance requirement from the controller.
Remind that the thermal energy dissipated by the power block module is variable and depends on the current drawn and the duty cycle.
Nominal values are:

- for liquid cooling: 6 l/m of 50/50 water/glycol at 65° of inlet temperature
- for air cooling: a minimum flux of 150 m3/h

Abnormal operating conditions, like very hot ambient temperatures, should be considered. A bad heat dissipation is very detrimental for the performance of the inverter!

5.4.2 Connection of power cables

For safety reason the connection of the power cables is protected by a separate cover. Unless this cover is removed the live parts of the cable cannot be reached.

General recommendations are:

- Keep power cables length as short as possible to minimize power losses.
- Tight power cables end to the controller power connections using a torque of

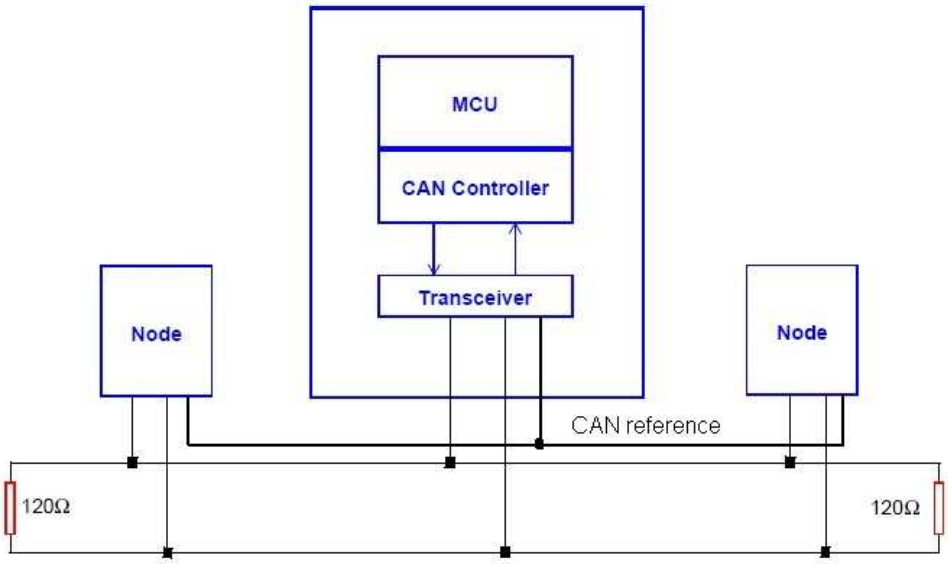
2.5÷3 Nm.

- Is recommended to place a Main Contactor along the positive of the battery cable. Its coil must operate at low voltage and must be connected to the logic board (see next chapters for more details about the connection).
- An internal protection inside the HV Inverter is designed to protect the device when battery voltage is higher than the nominal voltage. Obviously it's effective only if the main contactor is installed and driven by the inverter!

Do not connect the controller to a power source with a voltage larger than 725 V. A larger voltage may cause power section failure.

5.4.3 CAN connections and possible interferences

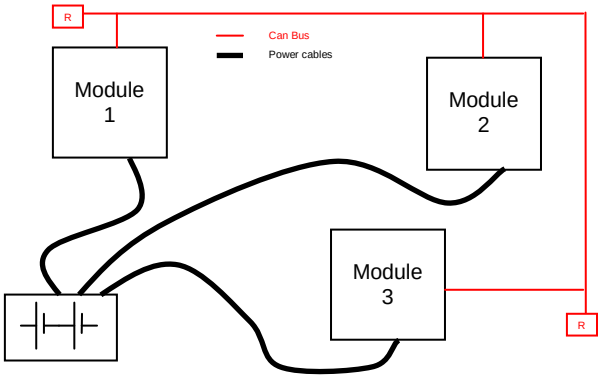
CAN stands for Controller Area Network. It is a communication protocol for real time control applications. CAN operates at data rate of up to 1 Megabits per second.
It was invented by the German company Bosch used in car industry to enable communication among the various electronic modules of a vehicle, connected as illustrated in this image:



General features:

- Best cable for a CAN bus is twisted pair; if it is necessary to increase the immunity of the system to disturbances, a good choice would be to use a cable with a shield connected to the frame of the truck. Sometimes it is sufficient a simple double wire cable or a duplex cable not shielded.
- The HV Inverter CAN bus is fully isolated from the low voltage power supply so it's mandatory to connect also the CAN reference signal.
- In a system like an industrial truck, where power cables carry hundreds of Ampere, there are voltage drops due to the impedance of the cables, and that could cause errors on the data transmitted through the can wires. In the following figures there is an overview of wrong and right layouts of the cables routing.

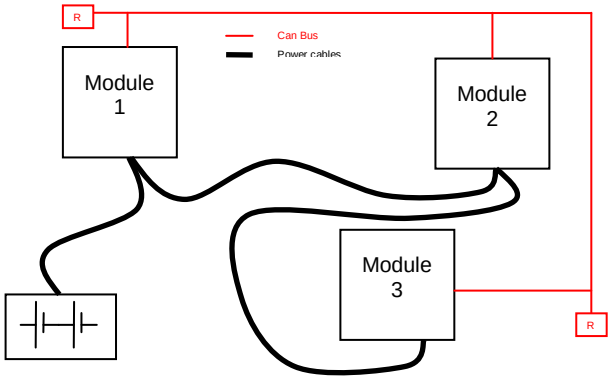
Wrong Layout:



The red lines are CAN bus wires.
The black boxes are different modules, for example traction controller, pump controller and display connected by CAN bus.
The black lines are the power cables.

This seems a good layout, but can bring to errors in the CAN bus.
The best solution depends on the type of nodes (modules) connected in the network.
If the modules are very different in terms of power, then the preferable connection is the daisy chain.

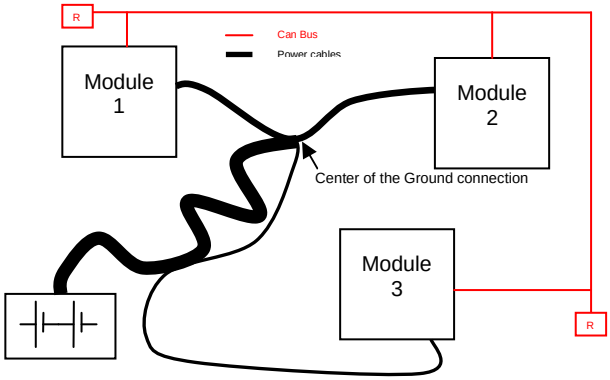
Correct Layout:



Note: Module 1 power > Module 2 power > Module 3 power

The chain starts from the HBATT- post of the controller that works with the highest current, and the others are connected in a decreasing order of power.
Otherwise, if two controllers are similar in power (for example a traction and a pump motor controller) and a third module works with less current, the best way to deal this configuration is to create a common ground point (star configuration).

Correct Layout:



Note: Module 1 power ≈ Module 2 power > Module 3 power

In this case the power cables starting from the two similar controllers must be as short as possible. Of course also the diameter of the cable concurs in the voltage drops described before (higher diameter means lower impedance), so in this last example the cable between the minus of the Battery and the common ground point (pointed by the arrow in the image) must be dimensioned taking into account thermal and voltage drop problems.

Can advantages:
The complexity of today systems needs more and more data, signal and information must flow from a node to another. CAN is the solution to different problems that arise from this complexity
- simplified design (readily available, multi sourced components and tools)
- lower costs (less and smaller cables)
- improved reliability (fewer connections)
- analysis of problems improved (easy connection with a pc to read the data flowing through the cable).

5.4.4 I/O connections

- After crimping the cable, verify that all strands are entrapped in the wire barrel.
- Verify that all the crimped contacts are completely inserted on the connector cavities.

A cable connected to the wrong pin can lead to short circuits and failure; so, before turning on the truck for the first time, verify with a multimeter the continuity between the starting point and the end of a signal wire.

For information about the mating connector pin assignment see the chapter 6 “.

5.4.5 Connection of standard encoder

Zapi HV Inverter can manage different types of encoder. To control AC motor with Zapi inverter, it is necessary to install an incremental encoder with 2 phases shifted of 90°. The encoder power supply can be +5 or +12 V (factory selected). Encoder output stages can be of different types.
For special application it is possible to install incremental encoder with Zero position signal.

A13	(J1) +5V/+12V	encoder positive power supply.
A16	(J4) GND	encoder negative power supply.
A14	(J2) ENC A	phase A of encoder.
A15	(J3) ENC B	phase B of encoder.
A23	(G3) Zero POS	encoder Zero Position.

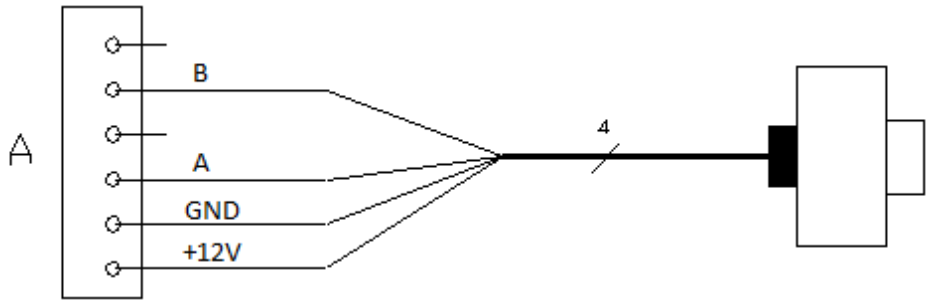


Figure 2: Standard Encoder

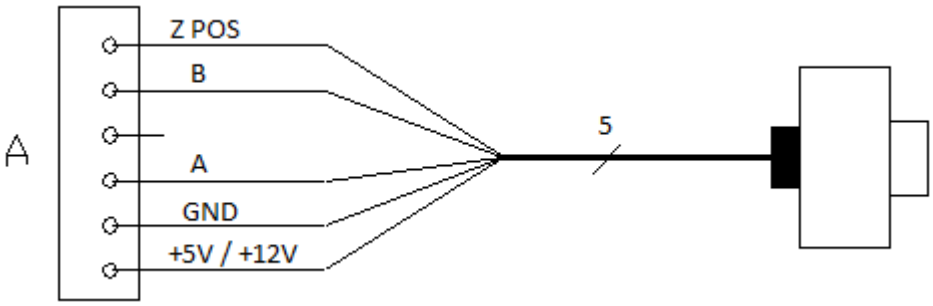


Figure 3: Encoder with Zero position signal

VERY IMPORTANT
It is necessary to specify in the order the type of encoder used, in terms of power supply, electronic output in order the logic can be properly set by Zapi.

VERY IMPORTANT
The number of pulse/rev can be properly set using the dedicated parameters (see paragraph 9.2.6).

The maximum speed detectable by standard Hardware configuration can be limited depending by number of pulse/rev. Contact Zapi technician for checking

VERY IMPORTANT
It is strongly suggested, for safety reason, to lift up the wheel from the floor

and set the correct value according to the type of sensor used before to perform any operation with the truck.

5.4.6 Connection of sin/cos sensor

To control PMSM motor with Zapi inverter, an absolute position sensor, like a Sin/Cos sensor, is needed. Normally power supply for Sin/Cos sensor is +5V, but eventually it's possible to use also +12V.
At the first key-on an auto-teaching procedure it is necessary to acquire the sensor signals range and origin from the controller.

A3	(M3)	+5V/+12V	analog positive power supply
A7	(L3)	AGND	analog negative power supply
A26	(F2)	SIN	sine signal
A28	(F4)	COS	cosine signal

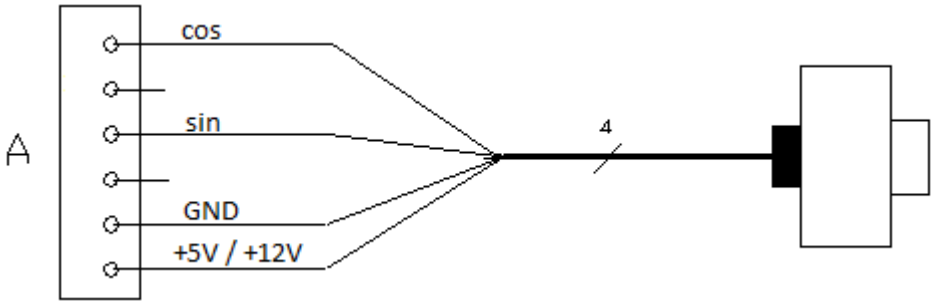


Figure 4: Sin/cos sensor connection

VERY IMPORTANT
It is necessary to specify in the order the type of sin/cos sensor used, in terms of power supply, electronic output in order the logic can be properly set by Zapi.

VERY IMPORTANT
An auto-teaching must be run at first to acquire the signal range and rotor origin position.

The maximum speed detectable by standard Hardware configuration can be limited depending by number of pulse/rev. Contact Zapi technician for checking

VERY IMPORTANT
It is strongly suggested, for safety reason, to lift up the wheel from the floor and set the correct value according to the type of sensor used before to perform any operation with the truck.

5.4.7 Connection of Resolver sensors

To control PMSM motor with Zapi inverter, an absolute position sensor, like a

Resolver sensor, is needed. Resolver is a sort of rotating transformer, equipped with one primary and two secondary windings.
HV Inverter can be adapted to every Resolver type but it's better to check sensor characteristics with Zapi technician to understand if hardware adjustments are needed.
At the first key-on an auto-teaching procedure it is necessary to acquire sensor signals range and origin from the controller. Number of poles must be set manually as the internal parameters, see paragraph 9.2.6

A22	(G2)	RES_ECC_P	resolver excitation (positive)
A21	(G1)	RES_ECC_N	resolver excitation (negative)
A26	(F2)	RES_SINE_P	resolver sine output (positive)
A25	(F1)	RES_SINE_N	resolver sine output (negative)
A28	(F4)	RES_COS_P	resolver cosine output (positive)
A27	(F3)	RES_COS_N	resolver cosine output (negative)

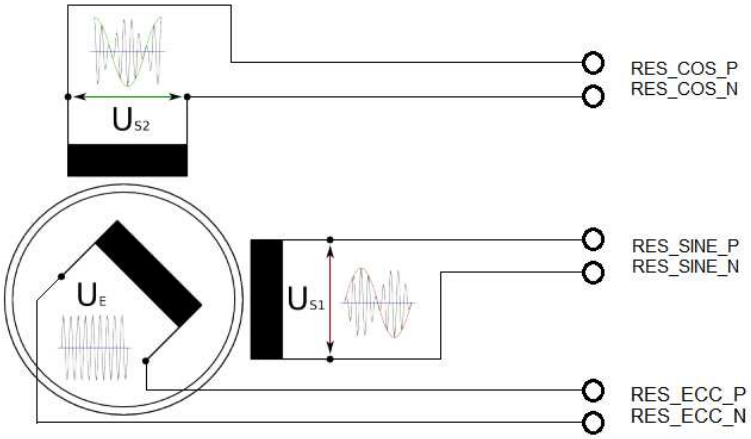


Figure 5: Resolver connection

VERY IMPORTANT
The number of Resolver poles must be manually set in the HV Inverter parameters. Number of pole pairs must be the same of the motor or a Resolver with only one pole pair must be employed as universal solution.

VERY IMPORTANT
HV Inverter is able to adjust the internal behaviour to automatically meet a wide range of Resolver's parameter. In any case customer had better to contact Zapi technician for a preliminary check of the chosen device.

5.4.8 Main Contactor and KEY connection

The connection of the main contactor can be carried out following the drawing in the figure

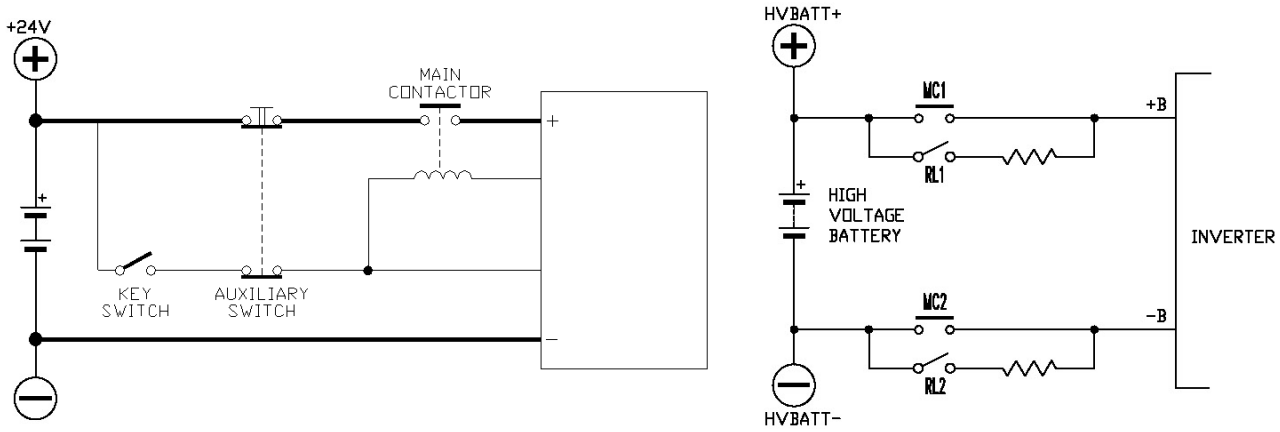


Figure 6: Main Contactor and Key connection

- The connection of the high voltage battery line switches must be carried out following ZAPI instructions.
- An intrinsic protection is present inside the logic when the voltage on the low battery power connection overtakes 40% more than the nominal voltage or if the key is switched off before the battery power line is disconnected.
- Also the High Voltage side is protected against over voltage condition

5.4.9 Insulation of truck frame

Respect all isolation practices when installing electric/electronic devices in a vehicle with a high voltage battery. A bad electric isolation is a serious hazard for any operator!

Refer to international standard rules where available for the application. Example: EN-1175-1 for industrial trucks states (chapter 5.7) "there shall be no electrical connection to the truck frame". So the truck frame has to be isolated from any electrical potential of the truck power line".

5.5 EMC

High voltage power sources, like this product, are strong emitters of radiated and conducted disturbances. The designer of the vehicle must be aware of EMC problems and of possible solutions.

EMC problems caused by a bad installation/design of the vehicle are beyond ZAPI's control. Zapi can offer assistance and suggestions, based on its experience of years on EMC related items. However, ZAPI declines any responsibility for non-compliance, malfunctions and failures, if correct testing is not made. The machine manufacturer holds the responsibility to

carry out machine validation, based on existing standards.

EMC stands for Electromagnetic Compatibility, and it represents the studies and the tests on the electromagnetic energy generated or received by an electrical device.
So the analysis works in two directions:

- The study of the **emission** problems, the disturbances generated by the device and the possible countermeasure to prevent the propagation of that energy; we talk about “conduction” issues when guiding structures such as wires and cables are involved, “radiated emissions” issues when it is studied the propagation of electromagnetic energy through the open space. In our case the origin of the disturbances can be found inside the controller with the switching of the IGBTs which are working at high frequency and generate RF energy, **but wires and cables have the key role to propagate the disturbs because they works as antennas**, so a good layout of the cables and their shielding can solve the majority of the emission problems.
- The study of the **immunity** can be divided in two main branches: protection from electromagnetic fields and from electrostatic discharge.
  The **electromagnetic immunity** concerns the susceptibility of the controller with regard to electromagnetic fields and their influence on the correct work made by the electronic device.
  There are well defined tests which the machine has to be exposed to.
  These tests are carried out at determined levels of electromagnetic fields, to simulate external undesired disturbances and verify the electronic devices response.

The second type of immunity, **ESD**, concerns the prevention of the effects of electric current due to excessive electric charge stored in an object. In fact, when a charge is created on a material and it remains there, it becomes an “electrostatic charge”; ESD happens when there is a rapid transfer from a charged object to another. This rapid transfer has, in turn, two important effects:

- this rapid charge transfer can determine, by induction, disturbs on the signal wiring and thus create malfunctions; **this effect is particularly critical in modern machines, with serial communications (CAN Bus) which are spread everywhere on the truck and which carry critical information.**
- in the worst case and when the amount of charge is very high, the discharge process can determine failures in the electronic devices; the type of failure can vary from an intermittently malfunction to a completely failure of the electronic device.

IMPORTANT NOTE: it is always much easier and cheaper to avoid ESD from being generated, than to increase the level of immunity of the electronic devices.



Normally EMC and ESD performances of an electronic system are strongly influenced by the installation. Special attention must be given to the lengths and the paths of the electric connections and the shields.

There are different solutions for EMC issues, depending on level of emissions/ immunity required, the type of controller, materials and position of the wires and electronic components.

EMISSIONS. Three ways can be followed to reduce the emissions:

- SOURCE OF EMISSIONS: finding the main source of disturb and work on it.
- SHIELDING: enclosing contactor and controller in a shielded box; using shielded cables;
- LAYOUT: a good layout of the cables can minimize the antenna effect; cables running nearby the truck frame or in iron channels connected to truck frames is generally a suggested not expensive solution to reduce the emission level.

ELECTROMAGNETIC IMMUNITY. The considerations made for emissions are valid also for immunity. Additionally, further protection can be achieved with ferrite beads and bypass capacitors.

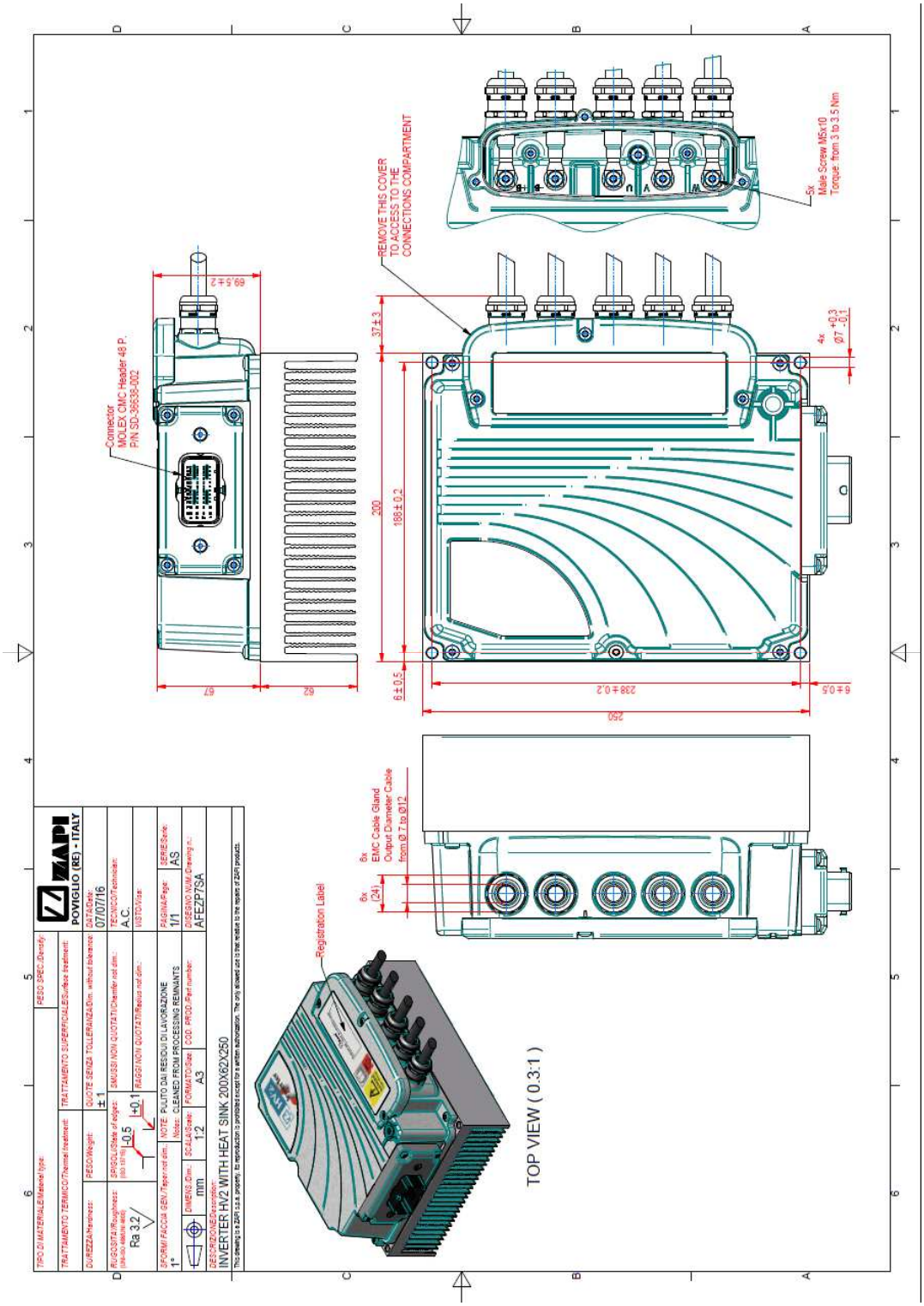
ELECTROSTATIC IMMUNITY. Three ways can be followed to prevent damages from ESD:

- PREVENTION: when handling ESD-sensitive electronic parts, ensure the operator is grounded; test grounding devices on a daily basis for correct functioning; this precaution is particularly important during controller handling in the storing and installation phase.
- ISOLATION: use anti-static containers when transferring ESD-sensitive material.
- GROUNDING: when a complete isolation cannot be achieved, a good grounding can divert the discharge current trough a “safe” path; the frame of a truck can works like a “local earth ground”, absorbing excess charge. **So it is strongly suggested to connect to truck frame all the parts of the truck which can be touched by the operator, who is most of the time the source of ESD.**

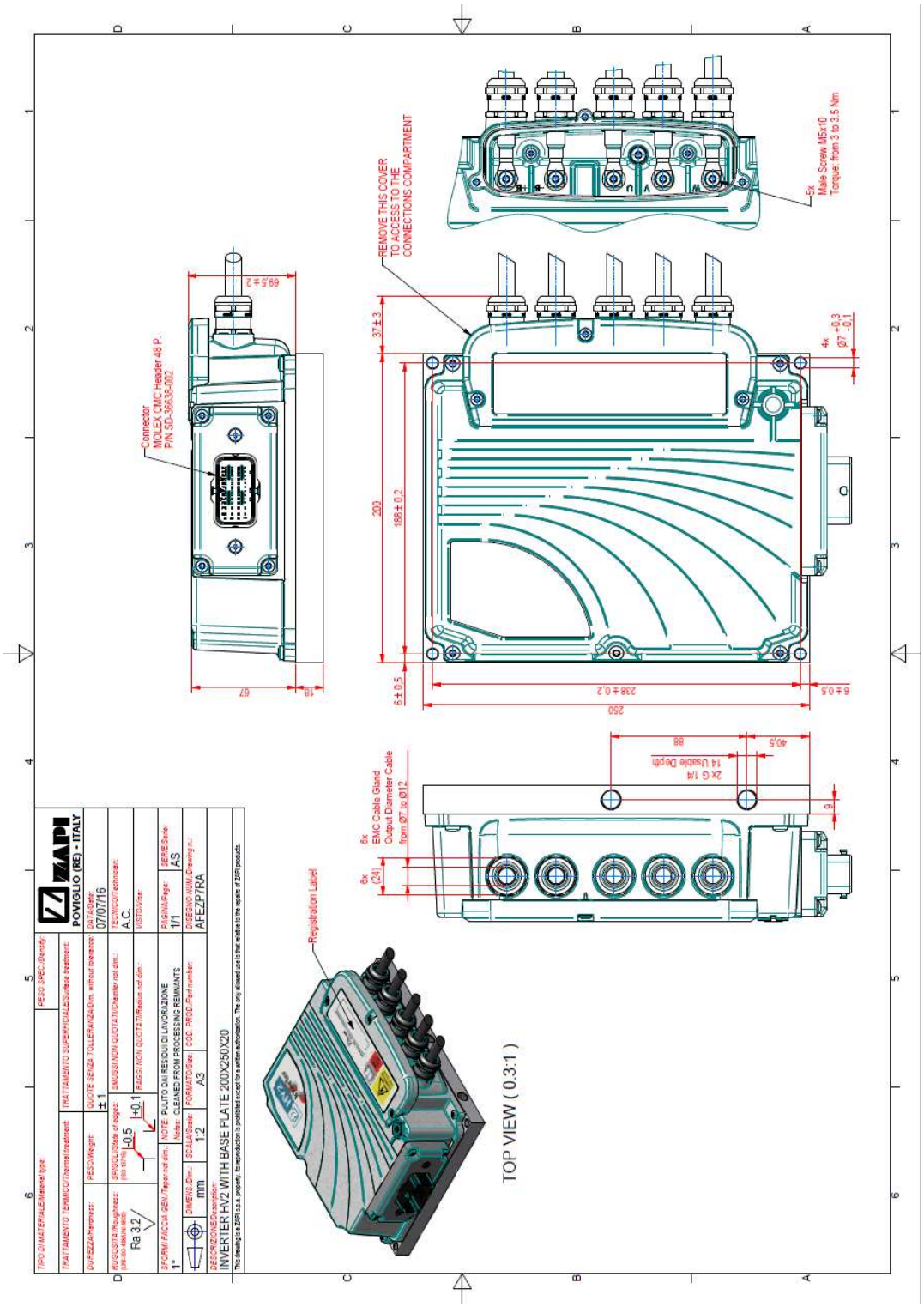
6 DRAWINGS

6.1 Mechanical drawings

6.1.1 Finned Heatsink version

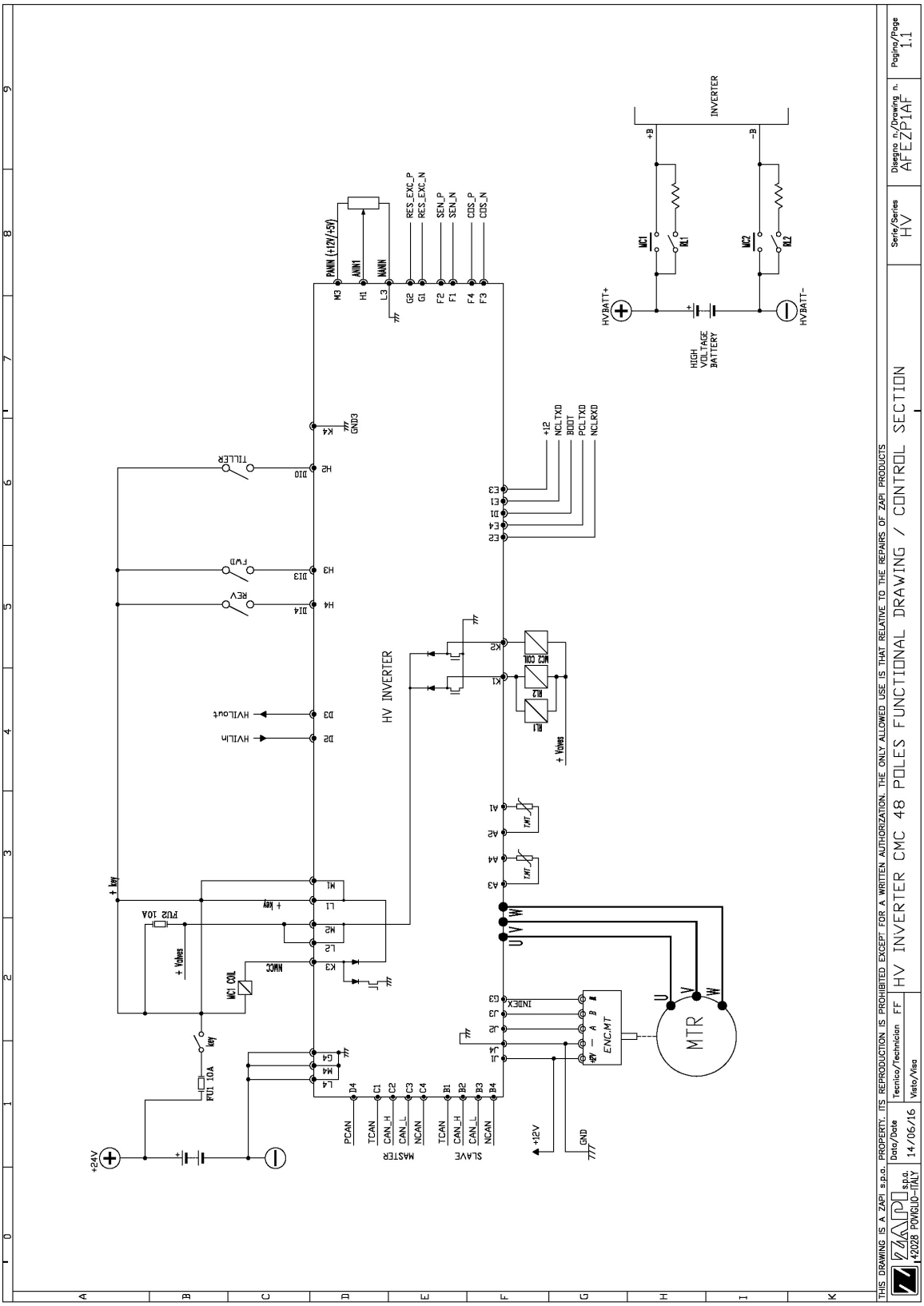


6.1.2 Liquid Cooled Base Plate version



6.2 Connection drawing

The following figure shows an example of basic I/O connection



7 FEATURES

7.1 Operational Features

- Speed control (three versions available: sensored, sense coil and sensorless as explained in the introduction section).
- Speed control with speed and position feedback for IM and PM motors
- Optimum behavior on slope due to the speed feedback:
  - the motor speed follows the accelerator, starting a regenerative braking if speed overtakes the speed set-point.
  - the system can perform an electrical stop on a ramp (the machine is electrically hold on a slope) for a programmable time (chapter 8.3).
- Stable speed in every position of the accelerator.
- Regenerative release braking based upon deceleration ramps.
- Regenerative braking when the accelerator pedal is partially released (deceleration).
- Direction inversion with regenerative braking based upon deceleration ramp.
- Regenerative braking and direction inversion without contactors: only the main contactor is present.
- The release braking ramp can be modulated by an analog input, so that a proportional brake feature is obtained.
- Optimum sensitivity at low speeds.
- The inverter can drive an electromechanical brake.
- Hydraulic steering function:
  - traction inverter: the traction inverter sends a "hydraulic steering function" request to the pump inverter on the CAN Bus line (see also OPTIONS chapter 8.2). Moreover, if the pump inverter is not present (for ex: tractor application), the traction inverter can manage an "hydraulic steering function" by driving a hydro contactor which drive an hydraulic steering motor, see also OPTIONS chapter.
  - pump inverter: the pump inverter manage an "hydraulic steering function". That is, it drives the pump motor at the programmed speed for the programmed time.
- High efficiency of motor and battery due to high frequency commutations.
- Double microcontroller for safety functions
- Self-diagnosis, the faults can be displayed through the console or Zapi's MDI/Display.
- Read and Modification of parameters through the PC CAN Console
- Internal hour-meter with values that can be displayed on the console.
- Memory of the last five alarms with relative hour-meter and temperature displayed on the console.

7.2 Dual traction motor

In the case of dual traction motors, there is additional processing of the associated steering signal (from a potentiometer or switches) in order to generate separate torque demands for the left and right motors of the vehicle. This allows the two motors to be operated at different speeds, which greatly assists in turning the vehicle and prevents wheel scrub. After the torque demands have been generated, the operation of each motor control system is as described in the

case of a single traction motor.

7.3 Pump motor

Pump motor control is similar to traction motor control, although motion is requested using a different combination of switches.

7.4 Torque mode

In this mode the controller maintains the motor torque output at a constant value for a given throttle position.
This is similar to DC motors (in particular, series wound DC motors) and provides a driving experience like a car. To prevent excessive speed when the load torque is low, for example when driving down hill, a maximum vehicle speed can be set.

7.5 Speed mode

In this mode the controller maintains the motor at a constant speed for a given throttle position as long as sufficient torque is available. Speed mode differs from torque mode in that the torque value applied to the motor is calculated by the controller based on the operator's requested speed (determined by throttle position) and the vehicle's actual speed.

7.6 Protection and safety features

The HV Inverter is protected against some controller installation erros and malfunctions:

- **Battery polarity inversion:** It is necessary to fit a MAIN CONTACTOR to protect the inverter against reverse battery polarity and for safety reasons.
- **Connection Errors**: all inputs are protected against wrong connection toward low voltage power supply
- **Voltage monitoring**: protected against battery undervoltage and overvoltage
- **Thermal protection**: the HV Inverter has a special protection against high temperature. For IGBT a different approach must be considered when power section become hot, because low output frequency can easily push junction temperature to critical level. HV Inverter is reducing maximum current at high temperature, but reduction will be greater if output frequency is low

I_max	***output frequency (Hz)***			
Temp. (°C)	**1**	**2**	**5**	**10**
60	68	70	70	70
70	56	57	61	70
80	44	45	48	56
90	32	33	36	42

100	19	20	24	28
110	7	8	11	14
120	0	0	0	0

Figure 7: Thermal cutback

- **External agents:** HV Inverter is protected against dust and the spray of liquid to a degree of protection IP69K. Nevertheless, it is suggested to carefully study controller installation and position.
- **Protection against uncontrolled swicth-on:** main contactor will not close if:
  - The power section is not working properly
  - The logic section is not working perfectly
  - The output voltage of the accelerator does not fall below the minimum voltage value stored, with 1 V added
  - Running microswitch in closed position (“on” position)
- **Low battery charge:** when the battery voltage is low, maximum current is reduced to the half of the rated current (MAX CURRENT param.)
- **Protection against accidental Start Up:** a precise sequence of operations are necessary before the machine starts. Operation cannot begin if these operations are not carried out correctly. Requests for drive, must be made after closing the key switch

7.6.1 Safety Features

ZAPI controllers are designed according to the prEN954-1 specifications for safety related parts of control system and to UNI EN1175-1 norm. The safety of the machine is strongly related to installation; length, layout and screening of electrical connections have to be carefully designed.
ZAPI is always available to cooperate with the customer in order to evaluate installation and connection solutions. Furthermore, ZAPI is available to develop new SW or HW solutions to improve the safety of the machine, according to customer requirements.
Machine manufacturer holds the responsibility for the truck safety features and related approval.

7.7 Diagnosis

Both master and slave microcontrollers continually monitor the inverter and check the main functions. We can consider 4 phases:

- Diagnosis before closing Main Contactor at power on involving: master/slave internal communication and congruence, current sensor, capacitor charging, phase's voltages, contactor drives, CAN Bus interface, power module short or open, status of microswitch and accelerator.
- Diagnosis in standby: master/slave internal communication and congruence, phase's voltages, contactor driver, current sensor, CAN Bus interface.

- Diagnosis during operation: master/slave internal communication and congruence, contactor driver, current sensors, CAN Bus interface.
- Continuous diagnosis: inverter temperature, motor temperature.

For troubleshooting the local console can be used or detailed diagnostic abiut errors and inverter status are available on CAN Bus.

7.8 Field Oriented Control

The HV Inverter features the latest Field Oriented Control algorithm for controlling of the following types of electric motors:

- Asynchronous induction motors
- Permanent Magnet Synchronous motors (PMSM)

The Field Oriented Controls gives best motor performance in terms of efficiency and dynamic control with and without load.

The algorithm uses the motor speed feedback to control the motor.

- For Asynchronous induction motors, a standard encoder is sufficient as a feedback devices
- For PMSM an absolute sensor, like a sin/cos sensor or a resolver is the best choice. It can be used alone or together with a standard encoder for optimum control



For best operation the Field Oriented Control requires a good setting of the inverter. Please ask Zapi engineers for further details

8 START-UP HINTS

8.1 Check prior to initial power up

For traction applications, raise up or otherwise disable drives wheels to prevent the possibility of unexpected vehicle motion or motion in the wrong direction during initial commissioning. For hydraulic applications, open the valve to prevent the possibility of excess pressure build-up (in the event of a pressure relief valve malfunction).

Take necessary precautions to do not compromise safety in order to prevent injury to personnel or damage to equipment

After operation, even with the Key Switch open, the internal capacitors may remain charged for some time. For safe operation, we recommend that the battery is disconnected, and a short circuit is made between Battery Positive and Battery Negative power terminals of the inverter using a Resistor greater than 100 ohm with appropriate power.

Perform the following checks before applying power to a motor controller for the first time:

- Verify that the proper motor controller item number for the application has been installed.
- Verify that the vehicle battery voltage matches the motor controllers Nominal DC Supply Voltage rating listed on the product identification label
- Verify that the correct software for the application has been loaded into the motor controller
- Verify that all power and signal wiring to the motor controller is correctly connected.
- Verify that connections to battery and motor terminals are tightened with appropriate torque.
- Verify that the control I/O plug is fully mated and latched into position with the mating connector on the motor controller.
- Verify that the motor controller is correctly fused for the application. Refer to the vehicle manufacturer's maintenance documentation for the correct fuse size.

8.2 Configuring motor controller for the application

Normally, motor controllers shipped for OEM series production are factory

programmed with correct parameters and do not require any further configuration.

Please refer to the OEM documentation for any further setup required during vehicle commissioning.

Setting up a prototype controller for a new vehicle, within a vehicle development program, may require extensive parameterization and possibly reprogramming of the motor controller via the CAN bus.

8.3 Sequence for IM Motor settings

This section describes the basic HV Inverter set-up procedure using local console or the CAN Console software. If the same configuration has to be repeated on different controllers the Save and Restore feature in the PC software can be used.

When the "Key Switch" is closed, if no alarms or errors are present, the display show the standard Zapi opening information's ("Home" display).

Minimum settings for every application are listed below, please remember to re-cycle the Key Switch if changes are made to the inverter's configuration.

- Fill your setting with the Options you need.
- Select the nominal low voltage Battery value.
- Select the nominal high voltage Battery value.
- Check that all input switches are functional using the TESTER function
- Perform the analog signals acquisition procedure using the PROGRAM VACC function (see paragraph)
- Set the "MAXIMUM CURRENT" current parameter.
- Insert motor and speed sensor parameters (it is suggested to double-check them with Zapi technicians).
- Set the "performance parameters" tailored on application (Acceleration delay, inversion braking, ecc).
- Test the application in all the operative condition (with/without load, on flat and on the maximum ramp)

8.4 Sequence for PM Motor settings

Preliminary settings are the same of IM motors, with obviously specific parameters for PM motor, but it's mandatory to acquire the position sensor features and origin in order to operate the motor properly.

Sin/Cos sensors have a sinusoidal DC output voltage, with variable amplitude and offset, and normally sin/cos wave origin has an arbitrary angle respect the magnetic field "0" position.

Offset, amplitude and angle must be known by the inverter software before starting a PM for the first time.

An automatic procedure, embedded in the inverter software, must be activate only one time at commissioning, to let the HV Inverter acquire the values.

Before starting the procedure please be sure motor is free to spin, with a minimum load on the shaft:

- from the “Options” menù select “ABS SENS. ACQUIRE”
- answer “NO” to the request of saving data (otherwise the main contactor will be opened)
- a console message “ACQUIRING ABS” indicate that acquisition procedure is ready to start
- it’s advisable to enter in the “Tester” menu and monitor motor speed for the further steps
- activate the “Tiller” and “Forward” (or “Backward”) microswitch: motor must start spinning in open loop mode
- because of the open loop mode, it’s normal if reported speed is not perfectly stable, but value on display must be, in any case, quite fixed.
- If motor doesn’t spin or is vibrating or speed in the display is oscillating too much, please stop the acquisition procedure removing the “Forward” command (see troubleshooting at the paragraph end)
- the first phase, where motor is spinning at low speed (something like 5Hz), allow the HV Inverter to acquire signal offset and amplitude for both sin/cos channels
- completed the first step, rotor is aligned to the magnetic field origin, and angle between sin/cos “0” value is acquired and stored
- final step is a sort of verification where motor is accelerated up to 50Hz in closed loop mode
- because of the closed loop, speed reported on display must be stable
- if something has gone wrong (for instance rotor didn’t align correctly because of friction on the shaft…), it’s possible that rotor start spinning at uncontrolled speed with high current: the only way to stop, is switching off the inverter using the “key” switch
- when procedure complete correctly the main contactor is open and display will show “ACQUIRE END” on the main screen
- turn off and then on again the “key” switch and verify that motor can move according to the accelerator in both direction

Please note Inverter is advancing step by step automatically, every phase is marked by a different message in the main display.

In case of problems, mainly in the first phase:

- check that PM motor pole pairs is set correctly
- in the “Hardware Settings” menu increase the “ABS.SENS. ACQ.ID”, it’s the

current used for the “open loop” phase that give more torque and perhaps solve a small friction problem (the “ID RMS MAX” must be set congruently)

- if increasing “ABS.SENS. ACQ.ID” is not enough try increasing, always in “Hardware Settings” menu, the “ABS.SENS.A.KTETA” parameter. It manages the speed in the open loop phase and in some situation a faster speed can help to achieve more uniformity in motor rotating.



Offset angle can also be refined manually using the “MAN.OFFSET ANGLE” parameter but, in any case, the sensor voltage range must be first acquired using the automatic procedure

8.5 PM Motor with Resolver

First of all, the resolver number of pole pairs must be set as a parameter in the “Hardware Settings” menu.

The automatic procedure to acquire resolver voltage span, voltage offset and angle offset is exactly the same used for sin/cos position sensor and described in the previous paragraph.

9 PROGRAMMING & ADJUSTEMENT VIA CONSOLE

To access and adjust all parameters it is necessary to use the Zapi console connected in various possible ways:

- connect the Smart Zapi console (or old hand console) using dedicated pins on CNC connector.
- use the Zapi Smart Console connected to the CAN Bus line (ask directly to Zapi for the dedicated User Manual)
- use the PC CAN Console software. This tool is more powerful than the standard serial console. The following paragraphs describe the controller configuration in the case the operator is using Zapi PC CAN console.
- connect the Smart Zapi console (or old hand console) through a remote module, like a Zapi tiller card of a Zapi display. This module has to be connected to the same CAN bus line of the inverter.

The Zapi Smart Console and PC CAN Console software are tool developed to improve setup and programming of all Zapi products installed in any application. It features a clean and easy-to-use interface in order to simplify access to parameters and troubleshooting.

See Appendix A and Appendix B to have a general general overview and basic knowledge about the use these tools



The Zapi Tools give a deep control over the parameters and behaviour of Zapi controllers. Its use is restricted to engineers and well trained technicians!

9.1 Menu Overview

PARAMETER CHANGE	SET OPTION	ADJUSTMENT	SPEC ADJUSTMENT	HARDWARE SETTING
ACC. TORQUE DEL.	HM DISPLAY OPT.	SET KEY VOLT.	ADJUSTMENT #01	TOP MAX SPEED
DEC. TORQUE DEL.	HM CUSTOM 1 OPT.	SET VDC LINK	ADJUSTMENT #02	FEEDBACK SENSOR
ACCELER. DELAY	HM CUSTOM 2 OPT.	ADJUST KEY VOLT	ADJUSTMENT #03	DIAG. ENC. LOCK
RELEASE BRAKING	TILL/SEAT SWITCH	ADJUST BATTERY	SET TEMPERATURE	DEAD TIME COMP.
TILLER BRAKING	EB ON TILLER BRK	SET POSITIVE PEB	DITHER AMPLITUDE	DECOUPLING Q-D
INVERS. BRAKING	BATTERY CHECK	THROTTLE 0 ZONE	DITHER FREQUENCY	DC-LINK COMP.
DECEL. BRAKING	STOP ON RAMP	THROTTLE X1 MAP	HIGH ADDRESS	ENABLE WEAKENING
PEDAL BRAKING	QUICK INVERSION	THROTTLE Y1 MAP	CAN BUS SPEED	EN. VANGLE WEAK
SPEED LIMIT BRK.	HARD & SOFT	THROTTLE X2 MAP	DEBUG CANMESSAGE	ENABLE MOTION
STEER BRAKING	MAIN POT. TYPE	THROTTLE Y2 MAP	CONTROLLER TYPE	OPEN LOOP
MAX SPEED FORW	AUX POT. TYPE	THROTTLE X3 MAP	SAFETY LEVEL	TORQUE CONTROL
MAX SPEED BACK	SET MOT. TEMPER.1	THROTTLE Y3 MAP	RS232 CONSOLLE	BRAKING TORQUE
MAX SPEED LIFT	SET MOT. TEMPER.2	BAT. MIN ADJ.	ID CANOPEN OFST	MOT RES AT 25DEG
1ST PUMP SPEED	EPS TYPE	BAT. MAX ADJ.	2ND SDO ID OFST	LD INDUCTANCE UH
2ND PUMP SPEED	M.C. FUNCTION	BDI ADJ STARTUP		LQ INDUCTANCE UH
3RD PUMP SPEED	EBRAKE ON APPL.	BDI RESET		ID RMS MAX
4TH PUMP SPEED	ACCEL MODULATION	BATT.LOW THRESHLD		MAX FLUX WB/1000
5TH PUMP SPEED	EVP TYPE	STEER RIGHT VOLT		ABS.SENS. ACQ.ID
HYD PUMP SPEED	EV1	STEER LEFT VOLT		ABS.SENS.A.KTETA
CUTBACK SPEED 1	EV2	STEER ZERO VOLT		KP GAIN IQSP
CUTBACK SPEED 2	HIGH DYNAMIC	MAX ANGLE RIGHT		KI GAIN IQSP SM
H&S CUTBACK	INVERSION MODE	MAX ANGLE LEFT		KI GAIN IQSP LA
CURVE SPEED 1	DISPLAY TYPE	STEER DEAD ANGLE		KI GAIN IQSP RIL
CURVE CUTBACK	HVIL	STEER ANGLE 1		KI GAIN IQSP STO
FREQUENCY CREEP	ABS.SENS.ACQUIRE	STEER ANGLE 2		KI GAIN IQSP ROL
TORQUE CREEP		SPEED FACTOR		KP GAIN VQSP
ACC SMOOTH		SPEED ON MDI		KI GAIN VQSP
INV SMOOTH		LOAD HM FROM MDI		KP GAIN VDSP
STOP SMOOTH		CHECK UP DONE		KI GAIN VDSP
BRK SMOOTH		CHECK UP TYPE		KP GAIN VANGLE
STOP BRK SMOOTH		MC VOLTAGE		KI GAIN VANGLE
BACKING SPEED		MC VOLTAGE RED.		FREQMAX FOR IMAX
BACKING TIME		EB VOLTAGE		IQMAX1
AUXILIARY TIME		EB VOLTAGE RED.		FREQ1 FOR IQMAX1
ROLLING DW SPEED		PWM EV2		IQMAX2
MIN EVP		PWM EV3		FREQ1 FOR IQMAX2
MAX EVP		MAX MOTOR TEMP.		IQMAX3
EVP OPEN DELAY		TEMP. MOT. STOP		FREQ1 FOR IQMAX3
EVP CLOSE DELAY		A.SENS.OFFSET SE		ROTATION CW ENC
		A.SENS.OFFSET CE		ROTATION CW MOT
		MAN.OFFSET ANGLE		ROTATION CW POS
		MAN.OFFS.ANG.DEC		ENCODER PULSES 1
		VACC SETTINGS		ENCODER PULSES 2
				MOTOR P. PAIRS 1
				MOTOR P. PAIRS 2
				PWM AT LOW FREQ
				PWM AT HIGH FREQ
				FREQ TO SWITCH

9.2 Function configuration

In the following chapters, the parameters are presented as following indicated:

Parameter	Allowable range	Description
Name of the parameter as indicated in the CAN Console tool (Availability)	Allowable range of values for the parameter	Description of the parameter and, when applicable, suggestion of how to set it.

In the “Parameter” column, the availability field (between parentheses) is the list of controller types where the parameter is available. The followings code can be indicated:

A = All controller types
T = Traction controllers (in single motor applications)
TM = Traction master controllers (in multiple motor applications)
TS = Traction supervisor controllers (in multiple motor applications)
P = AC pump controllers
CO= CANopen controllers
N = none



The parameters and the functionalities described in the following paragraphs are referred to ZAPI Standard software. They could be different in any other customized software releases depending by customer requests.

9.2.1 Menu “SET OPTIONS” functions list

SET OPTIONS menu		
Parameter	**Allowable range**	**Description**
HM DISPLAY OPT. (T, TM, P, CO)	0 ÷ 6	This parameter decides the configuration for the hour meter shown on a display (i.e. MDI). The possible settings are the same described for HM CUSTOM 1 OPT. parameter.
HM CUSTOM 1 OPT. (T, TM, P, CO)	0 ÷ 6	This parameter decides the configuration for the customer hour meter no.1. The possible settings are: **0:** The hour meter counts since the controller is on. **1:** The hour meter counts when the 3-phase power bridge is active **2:** The hour meter counts when the DC motor power bridge is active **3:** The hour meter counts when one of the valve outputs is active **4:** The hour meter counts when the 3-phase power bridge is active or the DC motor power bridge is active **5:** The hour meter counts when the DC motor power bridge is active or one of the valve outputs is active **6:** The hour meter counts when the 3-phase power bridge is active or the DC motor power bridge is active or one of the valve outputs is active **Note: options 2, 4, 5 and 6 are not effective on HV Inv.**
HM CUSTOM 2 OPT. (T, TM, P, CO)	0 ÷ 6	This parameter decides the configuration for the customer hour meter no. 2. The possible settings are the same described for HM CUSTOM 1 OPT. parameter.
TILL/SEAT SWITCH (T, TM, P)	HANDLE ÷ SEAT	This option handles the input CNA-18 (H2). This input opens when the operator leaves the truck. It is connected to a KEY voltage when the operator is present. **HANDLE =** CNA-18 (H2) is managed as tiller input (no delay when released). **DEADMAN =** CNA-18 (H2) is managed as deadman input (no delay when released) **SEAT =** CNA-18 (H2) is managed as seat input (with a delay when released, debouncing function)
EB ON TILLER BRK (T)	OFF ÷ ON	This option defines how it is managed the Electromechanical brake dependently by the status of Tiller/seat input: **ON =** the Electromechanical brake is engaged as soon as the tiller input goes in OFF state. The deceleration ramp defined by “tiller braking” parameter has no effect. **OFF =** when the tiller input goes in OFF state the “tiller braking” ramp is applied before to engage the Electromechanical brake.

SET OPTIONS menu		
Parameter	**Allowable range**	**Description**
BATTERY CHECK (T, TM, P, CO)	0 ÷ 3	This option specifies the handling of the high voltage battery charge status. There are four levels: **0 =** Nothing happens, the battery charge level is calculated but is ignored, it means no action is taken when the battery is discharged. **1 =** BATTERY LOW alarm is raised when the battery level is calculated being less than or equal to 10% of the full charge. The BATTERY LOW alarm reduces the maximum speed down to 24% of the full speed and reduces the maximum current down to 50% of the full current. **2 =** BATTERY LOW alarm is raised when the battery level is calculated being less or equal to 10% of the full charge. **3 =** BATTERY LOW alarm is raised when the battery level is calculated being less or equal to 10% of the full charge. The BATTERY LOW alarm reduces the maximum speed down to 24% of the full speed.
STOP ON RAMP (T, TM, P, CO)	OFF ÷ ON	This parameter allows to enable or disable the functionality that electrically holds the track on a slope for a defined time. **ON =** the stop on ramp feature (truck electrically hold on a ramp) is managed for a time established by "AUXILIARY TIME" parameter. After this time, the behaviour depends on the "aux out function" option programming (see also 9.2.3). **OFF =** the stop on ramp feature is not performed. A controlled slowdown is performed for minimum time established by "AUXILIARY TIME" parameter. After this time, the behaviour depends on the "aux out function" option programming (see also 9.2.3).
QUICK INVERSION (T, TM, P)	NONE ÷ BELLY	This parameters allow to enable or disable the quick inversion functionality managed by BELLY input. **NONE =** The quick inversion function is not managed (no effect when BELLY switches over). **BRAKE =** the motor is braked **TIMED =** The quick inversion function is timed. When the functionality is activated by switching over of input BELLY the controller drives the motor in the opposite direction for a fixed time (1,5sec as default) **BELLY =** The quick inversion function is managed but not timed. When the functionality is activated by switching over of input BELLY the controller drives the motor in the opposite until the input BELLY switches over again (not timed). **Note: function not present as default on HV Inverter controller**

SET OPTIONS menu		
Parameter	**Allowable range**	**Description**
HARD & SOFT (T, TM)	OFF ÷ ON	Enable or disable the Hard & Soft functionality. With H&S, it is possible to turn the truck moving (at reduced speed) only by activating the H&S switch, and the accelerator, without the tiller. **ON =** H&S function is enabled **OFF =** H&S function is disabled **Note: function not present as default on HV Inverter controller**
MAIN POT. TYPE (T, TM)	0 ÷ 11	This parameter decides the type of the main pot CNA-17 (H1). The possible settings are: **0:** V-type pot , low to high value , with direction switches, without enable switch, without enable dead band **1:** V-type pot , low to high value , with direction switches, without enable switch, with enable dead band **2:** V-type pot , high to low value , with direction switches, without enable switch, without enable dead band **3:** V-type pot , high to low value , with direction switches, without enable switch, with enable dead band **4:** Z-type pot , low to high value , with direction switches, without enable switch, without enable dead band **5:** Z-type pot , low to high value , with direction switches, without enable switch, with enable dead band **6:** Z-type pot , low to high value , without direction switches, with enable switch, with enable dead band **7:** Z-type pot , low to high value , without direction switches, without enable switch, with enable dead band **8:** Z-type pot , high to low value , with direction switches, without enable switch, without enable dead band **9:** Z-type pot , high to low value , with direction switches, without enable switch, with enable dead band **10:** Z-type pot , high to low value , without direction switches, with enable switch, with enable dead band **11:** Z-type pot , high to low value , without direction switches, without enable switch, with enable dead band

SET OPTIONS menu		
Parameter	**Allowable range**	**Description**
AUX POT. TYPE (T, TM, TS, P)	0 ÷ 12	This parameter decides the type of the auxiliary. The possible settings are: **0:** V-type pot , low to high value , with direction switches, without enable switch, without enable dead band **1:** V-type pot , low to high value , with direction switches, without enable switch, with enable dead band **2:** V-type pot , high to low value , with direction switches, without enable switch, without enable dead band **3:** V-type pot , high to low value , with direction switches, without enable switch, with enable dead band **4:** Z-type pot , low to high value , with direction switches, without enable switch, without enable dead band **5:** Z-type pot , low to high value , with direction switches, without enable switch, with enable dead band **6:** Z-type pot , lo to hi value , without direction switches, with enable switch, with enable dead band **7:** Z-type pot , low to high value , without direction switches, without enable switch, with enable dead band **8:** Z-type pot , high to low value , with direction switches, without enable switch, without enable dead band **9:** Z-type pot , high to low value , with direction switches, without enable switch, with enable dead band **10:** Z-type pot , high to low value , without direction switches, with enable switch, with enable dead band **11:** Z-type pot , high to low value , without direction switches, without enable switch, with enable dead band **12:** No pot, with direction switches, without enable switch **Note: function not present as default on HV Inverter controller**
SET MOT. TEMP.1 (T, TM, P, CO)	NONE ÷ OPTION#2	Sets the motor temperature sensor type. **NONE =** no motor thermal sensor switch is connected. **DIGITAL =** a digital (ON/OFF) motor thermal sensor is connected to CNA-48 (A4) / CNA-47 (A3). **OPTION#1 =** an analogue motor thermal sensor is connected to CNA-48 (A4) / CNA-47 (A3). The temperature sensor is a KTY 84-130 PTC (positive thermal coefficient resistance). **OPTION#2 =** an analogue motor thermal sensor is connected to CNA-48 (A4) / CNA-47 (A3). The temperature sensor is a KTY 83-130 PTC (positive thermal coefficient resistance) **OPTION#3 =** an analogue motor thermal sensor is connected to CNA-48 (A4) / CNA-47 (A3). The temperature sensor is a PT1000 PTC (positive thermal coefficient resistance). **Note: motor temperature sensor circuit is isolated so it must be connected to the proper groung CNA-47 (A3) and not to the low voltage power supply negative**

SET OPTIONS menu		
Parameter	**Allowable range**	**Description**
SET MOT. TEMP.2 (T, TM, P, CO)	NONE ÷ OPTION#2	Sets the motor temperature sensor type. **NONE =** no motor thermal sensor switch is connected. **DIGITAL =** a digital (ON/OFF) motor thermal sensor is connected to CNA-45 (A1) / CNA-46 (A2). **OPTION#1 =** an analogue motor thermal sensor is connected to CNA-45 (A1) / CNA-46 (A2). The temperature sensor is a KTY 84-130 PTC (positive thermal coefficient resistance). **OPTION#2 =** an analogue motor thermal sensor is connected to CNA-45 (A1) / CNA-46 (A2). The temperature sensor is a KTY 83-130 PTC (positive thermal coefficient resistance) **OPTION#3 =** an analogue motor thermal sensor is connected to CNA-45 (A1) / CNA-46 (A2). The temperature sensor is a PT1000 PTC (positive thermal coefficient resistance). **Note: motor temperature sensor circuit is isolated so it must be connected to the proper groung CNA-46 (A2) and not to the low voltage power supply negative**
EPS TYPE (T, TM)	NONE ÷ OPTION#2	It allows to select which type of EPS (Electric power steering) module is connected to the controller. **NONE =** NO EPS is present on the truck, HV Inverter doesn't wait for CAN message by the EPS and it does not apply EPS and braking steer cutback. **OPTION#1 =** EPS is present and it is an EPS-AC0, ENCODER + TOGGLE SWITCHES. **OPTION#2 =** EPS is present and it is an EPS-AC0, POT + ENCODER type.
M.C. FUNCTION (T, TM, P, CO)	OFF ÷ OPTION#2	It set the configuration for output CNA-11 (K3) NLC (Negative Line Contactor): **OFF =** Main Contactor is not present. If is set OFF the diagnosis are masked and M.C. is not closed after controller Key-on **ON =** Main Contactor in standalone configuration **OPTION#1 =** Traction + Pump only one Main Contactor for both **OPTION#2 =** Traction + Pump with double Main Contactor. Each one manages its own main contactor.
EBRAKE ON APPL. (T, TM, P, CO)	ABSENT ÷ PRESENT	The parameter enables or disables the management of the electro mechanic brake: **ABSENT =** the diagnosis are masked and E.B. is not closed after a traction request **PRESENT =** the Electro mechanic Brake is driven after a traction request if all the related diagnosis pass

SET OPTIONS menu		
Parameter	**Allowable range**	**Description**
AUX OUT FUNCTION (A)	NONE ÷ BRAKE	It enables the output for negative of Electro mechanic Brake (NEB): **NONE =** the diagnosis are masked and E.B. is not closed after a traction request. **BRAKE =** the Electro mechanic Brake is driven after a traction request if all the related diagnosis pass. The behaviour on a slope depends on the "STOP ON RAMP" setting as detailed in **Errore. L'origine riferimento non è stata trovata.**. **Note: function not present as default on HV Inverter controller**
SYNCRO (CO)	OFF ÷ ON	It enables or disables the syncro message **OFF =** the syncro message is not used **ON =** the Syncro message is enabled
AUTO PARK BRAKE (CO)	OFF ÷ ON	It enables or disables the autonomous management of the Brake output: **OFF =** the output is activated or deactivated according by command received by CAN Bus **ON =** the output is manged autonomously by the controller itself ignoring any activation/deactivation command received by CAN Bus
AUTO LINE CONT. (CO)	OFF ÷ ON	It enables or disables the autonomous management of the Line Contatctor output: **OFF =** the output is activated or deactivated according to command received by CAN Bus **ON =** the output is manged autonomously by the controller itself ignoring any activation/deactivation command received by CAN Bus
ACCEL MODULATION (T, TM, P, CO)	OFF ÷ ON	Enable or disable the accel modulation function **OFF =** the acceleration ramp is only dependent by final speed setpoint and by Accel Delay **ON =** the acceleration ramp is re-scaled by software dependently by speed setpoint and by Accel Delay See Paragraph 10.2
EVP TYPE (A)	NONE ÷ DIGITAL	It defines how it has to be managed the output for a proportional electrovalve (EVP): **NONE =** output not enabled, no load connected on dedicated pin. **ANALOG =** the related output manages a proportional valve, PWM current controlled. **DIGITAL =** the related output manages an on/off valve. **Note: function not present as default on HV Inverter controller** See the related parameter description in the menu "Parameter change"

SET OPTIONS menu		
Parameter	**Allowable range**	**Description**
EV1 (A)	ABSENT ÷ OPTION#2	It enables the control of output NEV1 pin CNA-9 (K1): **ABSENT =** output not enabled, no load connected to CNA-9 (K1). **OPTION#1 =** the output manages an ON/OFF valve. As default it is activated by 1st speed command **OPTION#2 =** free for future use.
EV2 (A)	ABSENT ÷ DIGITAL	It enables the control of output NEV2 pin CNA-10 (K2) : **ABSENT =** output not enabled, no load connected to CNA-10 (K2). **DIGITAL =** the output manages a PWM voltage controlled valve. The PWM frequency is 1kHz and the duty cycle depends by “PWM EV2” parameter (menu adjustment).
HIGH DYNAMIC (T, TM, P, CO)	OFF ÷ ON	It enables the High Dynamic function: **ON =** all the ramp set by dedicated parameters are ignored and the controller work always with maximum performance. **OFF =** standard behaviour
INVERSION MODE (T, TM)	OFF ÷ ON	This parameter sets the logic of the Quick Inversion input: **ON =** the Quick Inversion switch is Normally Closed (function active when switch opens). **OFF =** the Quick Inversion switch is Normally Open (function active when switch closes). **Note: function not present as default on HV Inverter controller**
STEER TABLE (TM)	OPTION#1 ÷ OPTION#3	It enables a pre-defined steer table: **OPTION#1 =** 3-Wheel steer table 1 **OPTION#2 =** 3-Wheel steer table 2 **OPTION#3=** 4-Wheel steer table 2 **The steering table depends by Truck geometry. The three options available as default could not fit the requirements of your truck.** **It is strongly recommended to consult Zapi technicians in order to establish if a default table can be used or if a proper steering table has to be created.**
DISPLAY TYPE (T, TM, P)	0 ÷ 9	This parameter decides which display is connected to the inverter. **0 =** No Display **1 =** MDI PRC connected **2 =** ECO DISPLAY connected **3 =** SMART DISPLAY connected **4 =** MDI CAN connected **5..9 =** Reserved for future developments

SET OPTIONS menu		
Parameter	**Allowable range**	**Description**
ABS.SENS.ACQUIRE (Only for PM machine with sin/cos, resolver or PWM sens) (A)	OFF ÷ ON	This parameters activates the acquisition of motor speed sensor used for PMSM (Permanent Magnets Synchronous Motor). See paragraph 8.4, 8.5 **Ask to Zapi Technicians for a detailed description of acquisition procedure.**

9.2.2 Menu “ADJUSTMENTS” functions list

ADJUSTMENTS menu		
Parameter	**Allowable range**	**Description**
SET KEY VOLT. (A)	12V ÷ 24V	This parameter must be set to the nominal low voltage power supply. The available options are: **12V 24V**
SET VDC LINK (A)	1600V ÷ 630V	This parameter must be set to the nominal battery voltage. The available options are: **160V 320V 560V 630V**
ADJUST KEY VOLT (A)		Fine adjustment of the KEY voltage measured by the controller. Calibrated by Zapi production department during the end of line test.
ADJUST BATTERY (A)		Fine adjustment of the battery voltage measured by the controller. Calibrated by Zapi production department during the end of line test.
SET POSITIVE PEB (A)	12V ÷ 80V	Allow to set the voltage supply value connected to CNA-3. The available values are: **12V 24V 36V 40V 48V 72V 80V**
THROTTLE 0 ZONE (T, TM, P)	0% ÷ 100% (step of 1%)	It establishes a deadband in the accelerator input curve. **See paragraph 10.6**
THROTTLE X1 MAP (T, TM, P)	0% ÷ 100% (step of 1%)	This parameter changes the characteristic of the accelerator input curve **See paragraph 10.6**
THROTTLE Y1 MAP (T, TM, P)	0% ÷ 100% (step of 1%)	This parameter changes the characteristic of the accelerator input curve. **See paragraph 10.6**
THROTTLE X2 MAP (T, TM, P)	0% ÷ 100% (step of 1%)	This parameter changes the characteristic of the accelerator input curve **See paragraph 10.6**
THROTTLE Y2 MAP (T, TM, P)	0% ÷ 100% (step of 1%)	This parameter changes the characteristic of the accelerator input curve. **See paragraph 10.6**
THROTTLE X3 MAP (T, TM, P)	0% ÷ 100% (step of 1%)	This parameter changes the characteristic of the accelerator input curve. **See paragraph 10.6**
THROTTLE Y3 MAP (T, TM, P)	0% ÷ 100% (step of 1%)	This parameter changes the characteristic of the accelerator input curve. **See paragraph 10.6**

ADJUSTMENTS menu		
Parameter	**Allowable range**	**Description**
BAT. MIN ADJ. (T, TM, P, CO)	-12.8% ÷ 12.7% (step of 0.1%)	It adjusts the lower level of the battery discharge table. It is used to calibrate the discharge algorithm with the battery of the application. **See paragraph 10.8**
BAT. MAX ADJ. (T, TM, P, CO)	-12.8% ÷ 12.7% (step of 0.1%)	It adjusts the upper level of the battery discharge table. It is used to calibrate the discharge algorithm with the battery of the application. **See paragraph 10.8**
BDI ADJ STARTUP (T, TM, P, CO)	-12.8% ÷ 12.7% (step of 0.1%)	Adjusts the level of the battery charge table at startup, in order to calculate the battery charge at keyon. **See paragraph 10.8**
BDI RESET (T, TM, P, CO)	0% ÷ 100% (step of 1%)	It adjusts the minimum variation of the battery discharge table to update the battery % at the start up. It is used to calibrate the discharge algorithm with the battery of the application. **See paragraph 10.8**
BATT.LOW THRESHLD (T, TM, P, CO)	1% ÷ 50% (step of 1%)	It defines the minimum percentage of charge under that the Battery low alarms rises.
STEER RIGHT VOLT (T,TM)	0V ÷ 25.5V (step of 0.1V)	It records the maximum value of steering command in right turning **See paragraph** Errore. L'origine riferimento non è stata trovata.
STEER LEFT VOLT (T,TM)	0V ÷ 25.5V (step of 0.1V)	It records the maximum value of steering command in left turning **See paragraph** Errore. L'origine riferimento non è stata trovata.
STEER ZERO VOLT (T,TM)	0V ÷ 25.5V (step of 0.1V)	It records the maximum value of steering command in straight head position **See paragraph** Errore. L'origine riferimento non è stata trovata.
MAX ANGLE RIGHT (T,TM)	0° ÷ 90° (step of 1°)	It defines the maximum angle of steered wheel in right turning .
MAX ANGLE LEFT (T,TM)	0° ÷ 90° (step of 1°)	It defines the maximum angle of steered wheel in right turning
STEER DEAD ANGLE (T, TM)	1° ÷ 50° (step of 1°)	It defines the maximum steered wheel angle up to which the traction speed permitted is 100%. **See paragraph 10.5**

ADJUSTMENTS menu		
Parameter	**Allowable range**	**Description**
STEER ANGLE 1 (T, TM)	1° ÷ 90° (step of 1°)	It defines the steered wheel angle at that the traction speed is reduced to CURVE SPEED 1. In between STEER DEAD ANGLE and STEER ANGLE 1 the traction speed is reduce linearly from 100% to CURVE SPEED 1. **See paragraph 10.5**
STEER ANGLE 2 (T, TM)	1° ÷ 90° (step of 1°)	It defines the steered wheel angle after that the traction speed is reduced to CURVE CUTBACK. In between STEER ANGLE1 and STEER ANGLE 2 the traction speed is reduce linearly from CURVE SPEED 1 to CURVE CUTBACK **See paragraph 10.5**
SPEED FACTOR (T, TM, CO)	0 ÷ 255 (step of 1)	Speed factor coefficient. It has to be set in order to have the correct speed visualization. The formula used internally by the software to transduce the speed frequency (expressed in Hz) in km/h is the following: $Km/h = \frac{Hz \cdot 10}{Speed\ Factor}$
SPEED ON MDI (T, TM, CO)	OFF ÷ ON	It enables the speed visualization on MDI display: **ON =** MDI shows traction speed when the truck is moving. In steady-state condition the speed indication is replaced by hour meter indication. **OFF =** Standard MDI functionality.
LOAD HM FROM MDI (T, TM, P, CO)	OFF ÷ ON	Allows to enable or disable the hour meter transfer to the MDI. **OFF =** The HourMeter of the Controller is not transferred and recorded on the HourMeter of the Standard MDI **ON =** The HourMeter of the Controller is transferred and recorded on the HourMeter of the Standard MDI (connected on the Serial Link).
CHECK UP DONE (T, TM, P, CO)	OFF ÷ ON	Turn it On when the required Maintenance service has been executed to cancel the CHECK UP NEEDED warning.
CHECK UP TYPE (T, TM, P, CO)	NONE ÷ OPTION#3	It specifies the handling of the CHECK UP NEEDED warning: **NONE =** No CHECK UP NEENED warning **OPTION#1 =** CHECK UP NEENED warning shown on the hand set and MDI after 300 hours **OPTION#2 =** Equal to OPTION#1 but Speed reduction after 340 hours **OPTION#3 =** Equal to OPTION#2 but the truck definitively stops after 380 hours
MC VOLTAGE (A)	0% ÷ 100% (step of 1%)	It specifies the duty cycle (t_{on}) of the PWM applied to the output CNA-11 (K3) during the first 1 second after the activation command in order to close the main contactor.

ADJUSTMENTS menu		
Parameter	**Allowable range**	**Description**
MC VOLTAGE RED. (A)	0% ÷ 100% (step of 1%)	It specifies the percentage of MC VOLTAGE parameter and it determines the duty cycle applied after the first second to keep the contactor closed. **Example 1** MC VOLTAGE = 100% MC VOLTAGE RED = 70% The contactor will be closed applying a 100% of duty cycle to the coil and then the duty cycle will be reduced to 70%. **Example 2** MC VOLTAGE = 70% MC VOLTAGE RED. = 100% The contactor will be closed applying a 70% of duty cycle to the coil and then the voltage will be kept at the same value. **Example 3** MC VOLTAGE = 70% MC VOLTAGE RED = 70% The contactor will be closed applying a 70% of duty cycle to the coil and then the duty cycle will be reduced to 49%.
EB VOLTAGE (A)	0% ÷ 100% (step of 1%)	It specifies the duty cycle (t_{on}) of the PWM applied to the EB output during the first 1 second after the activation command in order to release the electromechanical brake. **Note: function not present as default on HV Inverter controller**
EB VOLTAGE RED. (A)	0% ÷ 100% (step of 1%)	It specifies the percentage of EB VOLTAGE parameter, supplied to EB coil to keep the electro mechanic brake applied. **Example 1** EB VOLTAGE = 100% EB VOLTAGE RED = 70% The electromechanical brake will be released applying a 100% of duty cycle to the coil and then the duty cycle will be reduced to 70%. **Example 2** EB VOLTAGE = 70% EB VOLTAGE RED. = 100% The electromechanical brake will be released applying a 70% of duty cycle to the coil and then the voltage will be kept at the same value. **Example 3** MC VOLTAGE = 70% MC VOLTAGE RED = 70% The electromechanical brake will be released applying a 70% of duty cycle to the coil and then the duty cycle will be reduced to 49%. **Note: function not present as default on HV Inverter controller**
PWM EV2 (A)	0% ÷ 100% (255 steps)	It defines the duty cycle of the PWM signal applied on the EV2 pin CNA-9 (K1).
PWM EV3 (A)	0% ÷ 100% (255 steps)	It defines the duty cycle of the PWM signal applied on the EV3 pin CNA-10 (K2).

ADJUSTMENTS menu		
Parameter	**Allowable range**	**Description**
MAX MOTOR TEMP. (T, TM, P, CO)	60°C ÷ 175°C (step of 1°C)	It defines the motor temperature above that it is applied a cutback of 50% to the maximum current. This cutback is valid only in motoring features; during braking the 100% of the maximum current is always available independently by the temperature.
TEMP. MOT. STOP (T, TM, P, CO)	60°C ÷ 190°C (step of 1°C)	It defines the maximum motor temperature permitted. Over this limit the controller will stop to drive the motor.
A.SENS.OFFSET SE (PM motor with sin/cos, resolver sens) (A)	0..1023	It is the values of the acquired voltage offset at the sine analog inputs during the autoteaching procedure. It can be compared with the values of the A.SENS.OFFSET SR entry
A.SENS.OFFSET CE (PM motor with sin/cos, resolver sens) (A)	0..1023	It is the values of the acquired voltage offset at the cosine analog inputs during the autoteaching procedure. It can be compared with the values of the A.SENS.OFFSET CR entry
MAN.OFFSET ANGLE (Only for PM motor) (A)	0-180 Degree (step of 1°)	This parameter, together with MAN.OFFS.ANG.DEC, gives the possibility to manually adjust the angle offset between the absolute position sensor and the PMSM rotor orientation. The unit is degrees and the max value is 180°.
MAN.OFFS.ANG.DEC (Only for PM motor) (A)	0-9 tenths of Degree (step of 0.1°)	This parameter, together with MAN.OFFS.ANG.DEC, gives the possibility to manually adjust the angle offset between the absolute position sensor and the PMSM rotor orientation. The unit is tenths of degrees and the max value is 9 (0.9°).

9.2.3 Menu “PARAMETER CHANGE” functions list

PARAMETER CHANGE menu		
Parameter	**Allowable range**	**Description**
ACC. TORQUE DEL. (T, TM, P, CO)	0.1sec ÷ 10sec (step of 0.1sec)	It determines the acceleration ramp when “torque control” is set to ON. The parameter sets the time needed to increase the torque from minimum value up to the maximum.
DEC. TORQUE DEL. (T, TM, P, CO)	0.1sec ÷ 10sec (step of 0.1sec)	It determines the deceleration ramp when “torque control” is set to ON. The parameter sets the time needed to move from maximum torque down to minimum torque.
ACCELER. DELAY (T, TM, P, CO)	0.1sec ÷ 25.5sec (step of 0.1sec)	It determines the acceleration ramp. The parameter sets the time needed to speed up the traction motor from 0Hz to 100Hz. A special software features manages the acceleration ramp depending by speed set point (see explanation Graph at the end of paragraph)
RELEASE BRAKING (T, TM, P, CO)	0.1sec ÷ 25.5sec (step of 0.1sec)	It controls the deceleration ramp when the travel request is released. The parameter sets the time needed to decelerate the traction motor from 100Hz to 0Hz. A special software features manages the deceleration ramp depending by starting speed (see explanation Graph at the end of paragraph).
TILLER BRAKING (T, TM)	0.1sec ÷ 25.5sec (step of 0.1sec)	It controls the deceleration ramp when the tiller/seat switch is released. The parameter sets the time needed to decelerate the traction motor from 100Hz to 0Hz. A special software features manages the deceleration ramp depending by starting speed (see explanation Graph at the end of paragraph).
INVERS. BRAKING (T, TM, CO)	0.1sec ÷ 25.5sec (step of 0.1sec)	It controls the deceleration ramp when the direction switch is inverted during travel. The parameter sets the time needed to decelerate the traction motor from 100Hz to 0Hz. A special software features manages the deceleration ramp depending by starting speed (see explanation Graph at the end of paragraph).
DECEL. BRAKING (T, TM, CO)	0.1sec ÷ 25.5sec (step of 0.1sec)	It controls the deceleration ramp when the accelerator has turned down but not completely released. The parameter sets the time needed to decelerate the traction motor from 100Hz to 0Hz. A special software features manages the deceleration ramp depending by starting speed and final speed set point (see explanation Graph at the end of paragraph).
PEDAL BRAKING (T, TM, CO)	0.1sec ÷ 25.5sec (step of 0.1sec)	It controls the deceleration ramp when the braking pedal position is pressed. The parameter sets the time needed to decelerate the traction motor from 100Hz to 0Hz. A special software features manages the deceleration ramp depending by starting speed (see explanation Graph at the end of paragraph).

PARAMETER CHANGE menu		
Parameter	**Allowable range**	**Description**
SPEED LIMIT BRK. (T, TM)	0.1sec ÷ 25.5sec (step of 0.1sec)	It controls the deceleration ramp when a speed reduction has been activated. The parameter sets the time needed to decelerate the traction motor from 100Hz to 0Hz. A special software features manages the deceleration ramp depending by starting speed and final speed set point (see explanation Graph at the end of paragraph).
STEER BRAKING (T, TM)	0.1sec ÷ 25.5sec (step of 0.1sec)	It controls the deceleration ramp when a speed reduction has been activated. The parameter sets the time needed to decelerate the traction motor from 100Hz to 0Hz. A special software features manages the deceleration ramp depending by starting speed and final speed set point (see explanation Graph at the end of paragraph).
MAX SPEED FORW (T, TM)	0% ÷ 100% (step of 1%)	It determines the maximum speed in forward direction. Percentage of the TOP MAX SPEED parameter.
MAX SPEED BACK (T, TM)	0% ÷ 100% (step of 1%)	It determines the maximum speed in backward direction. Percentage of the TOP MAX SPEED parameter.
MAX SPEED LIFT (P)	0% ÷ 100% (step of 1%)	It determines the maximum speed of the pump during a lift request. Percentage of the maximum voltage applied to the pump motor.
1ST PUMP SPEED (P)	0% ÷ 100% (step of 1%)	Defines the pump speed when the 1st speed request input is active. Percentage of the maximum pump speed.
2ND PUMP SPEED (P)	0% ÷ 100% (step of 1%)	Defines the pump speed when the 2nd speed request input is active. Percentage of the maximum pump speed.
3RD PUMP SPEED (P)	0% ÷ 100% (step of 1%)	Defines the pump speed when the 3rd speed request input is active. Percentage of the maximum pump speed.
4TH PUMP SPEED (P)	0% ÷ 100% (step of 1%)	Defines the pump speed when the 4th speed request input is active. Percentage of the maximum pump speed.
5TH PUMP SPEED (P)	0% ÷ 100% (step of 1%)	Defines the pump speed when the 5th speed request input is active. Percentage of the maximum pump speed.
HYD PUMP SPEED (P)	0% ÷ 100% (step of 1%)	Fine adjustment of the pump motor steering function speed when the parameter “HYDRO FUNCTION” is set to ON. Percentage of the maximum pump speed.
CUTBACK SPEED 1 (T, TM, P)	10% ÷ 100% (step of 1%)	Speed reduction when the cutback input 1 is active. Percentage of the TOP MAX SPEED parameter. When set to 100% the speed reduction is ineffective.
CUTBACK SPEED 2 (T, TM, P)	10% ÷ 100% (step of 1%)	Speed reduction when the cutback input 2 is active. Percentage of the TOP MAX SPEED parameter. When set to 100% the speed reduction is ineffective.

PARAMETER CHANGE menu		
Parameter	**Allowable range**	**Description**
H&S CUTBACK (T, TM)	10% ÷ 100% (step of 1%)	It determines the percentage of the TOP MAX SPEED parameter applied when the Hard & Soft function, is active. When set to 100% the speed reduction is ineffective. **Note: H&S function not present as default in the HV Inverter**
CURVE SPEED 1 (T, TM)	0% ÷ 100% (step of 1%)	It set the maximum speed when the steering angle corresponds to the angle set by the parameter STEER ANGLE 1. Percentage of the TOP MAX SPEED parameter.
CURVE CUTBACK (T, TM)	1% ÷ 100% (step of 1%)	It set the maximum speed when the steering angle corresponds to the angle set by the parameter STEER ANGLE 2. Percentage of the TOP MAX SPEED parameter.
FREQUENCY CREEP (T, TM, P)	0.6Hz ÷ 25Hz (step of 0.1Hz)	Minimum speed when the forward or reverse switch is closed, but the accelerator is at the a minimum value.
TORQUE CREEP (T, TM, P, CO)	0% ÷ 100% (step of 1/255)	Minimum torque when torque control is enabled and the forward or reverse switch is closed, but the accelerator is on a minimum position.
MAXIMUM CURRENT (T, TM, P, CO)	0% ÷ 100% (step of 1%)	This parameter changes the maximum current for the traction motor. Percentage of the factory calibrated maximum current.
ACC SMOOTH (T, TM, P, CO)	1 ÷ 5 (step of 0.1)	It gives a parabolic form to the acceleration ramp near to 0rpm. 1 means linear ramp, higher the value smoother the acceleration ramp.
INV SMOOTH (T, TM, CO)	1 ÷ 5 (step of 0.1)	It gives a parabolic form to the acceleration ramp near to 0rpm. 1 means linear ramp, higher the value smoother the ramp across 0 rpm.
STOP SMOOTH (T, TM, P, CO)	3Hz ÷ 100Hz (step of 1Hz)	It sets the level of frequency where the smooth effect of the acceleration parabolic form ends.
BRK SMOOTH (T, TM, CO)	1 ÷ 5 (step of 0.1)	It gives a parabolic form to the deceleration ramp. 1 means linear ramp, higher the value smoother the ramp close to 0 rpm.
STOP BRK SMOOTH (T, TM, CO)	3Hz ÷ 100Hz (step of 1Hz)	It sets the level of frequency where the smooth effect of the deceleration parabolic form ends.
BACKING SPEED (T, TM)	0% ÷ 100% (step of 1%)	This is the percentage of the TOP MAX SPEED parameter that the traction can reach when the inching function is active.
BACKING TIME (T, TM)	0sec ÷ 10sec (step of 0.1sec)	Set the duration of the inching function
AUXILIARY TIME (T, TM, P, CO)	0sec ÷ 10sec (step of 0.1sec)	For the encoder version, it determines the time duration the truck is hold on the ramp if the STOP ON RAMP option is ON.

PARAMETER CHANGE menu		
Parameter	**Allowable range**	**Description**
ROLLING DW SPEED (T, TM, P, CO)	1Hz ÷ 50Hz (step of 1Hz)	It determines the maximum speed for the rolling down function.
HYDRO TIME (P)	0sec ÷ 20sec (step of 0.1sec)	It specifies how much time the hydro guide must remain active after the traction command has been released.
MIN EVP (A)	0% ÷ 100% (step of 1/255)	This parameter determines the minimum current applied on the EVP when the position of the potentiometer is at the minimum. This parameter is not effective if the EVP is programmed like an On/Off valve. **Note: The HV Inverter doesn't have a EVP output**
MAX EVP (A)	0% ÷ 100% (step of 1/255)	This parameter determines the maximum current applied to the EVP when the position of the potentiometer is at the maximum. This parameter also determines the current value when the EVP is programmed like an ON/OFF valve. **Note: The HV Inverter doesn't have a EVP output**
EVP OPEN DELAY (A)	0sec ÷ 12.75sec (step of 0.05sec)	It determines the acceleration ramp on EVP. The parameter sets the time needed to increase the current to the maximum possible value. **Note: The HV Inverter doesn't have a EVP output**
EVP CLOSE DELAY (A)	0sec ÷ 12.75sec (step of 0.05sec)	It determines the deceleration ramp on EVP. The parameter sets the time needed to decrease the current from the maximum possible value to zero. **Note: The HV Inverter doesn't have a EVP output**

After changing, press STORE to save data.

9.2.4 Menu “SPECIAL ADJUSTMENTS” functions list

Note: the below set-up description is for skilled people only so, if you aren’t, please keep your hands off. This hidden menu needs a special sequence of operations to be displayed. Ask for this procedure, directly to a Zapi technician. In the SPECIAL ADJUSTMENTS functions list, there are factory adjustable parameters.

SPECIAL ADJUSTMENTS menu		
Parameter	**Allowable range**	**Description**
ADJUSTMENT #01 **(Read Only)** (A)	0% ÷ 255% (step of 1%)	(Factory adjusted). This is the Gain of the first Current Sensing Amplifier of traction motor. NOTE: only Zapi technicians can change this value through a special procedure.
ADJUSTMENT #02 **(Read Only)** (A)	0% ÷ 255% (step of 1%)	(Factory adjusted). This is the Gain of the second Current Sensing Amplifier of traction motor. NOTE: only Zapi technicians can change this value through a special procedure.
ADJUSTMENT #03 **(Read Only)** (A)	0% ÷ 255% (step of 1%)	(Factory adjusted). This is the Gain of the second Current Sensing Amplifier of traction motor. NOTE: only Zapi technicians can change this value through a special procedure.
SET TEMPERATURE (A)	0°C ÷ 255°C (step of 1°C)	(Factory adjusted). Set the temperature offset to have the correct value reading. This is a fine calibration of the controller temperature sensor.
DITHER AMPLITUDE (A)	0% ÷ 13%	It is the dither signal amplitude. The dither signal is a square wave which is overlapped to the proportional valves set point. In this way the proportional valves response to set point variations is optimized. This parameter is a percentage of the valves maximum current. Setting the parameter to 0% means the dither isn’t used. The available values are: **0.0% 1.0% 2.5% 4.0% 5.5% 7.0% 8.5% 10% 11.5% 13.0%**
DITHER FREQUENCY (A)	20.8Hz ÷ 83.3Hz	It is the dither signal frequency. The available values are: **20.8 22.7 25 27.7 31.2 35.7 41.6 50 62.5 83.3**
HIGH ADDRESS (A)	0 ÷ 4	Reserved. Used to have access to special memory address. NOTE: only Zapi technicians can change this value.
CAN BUS SPEED (A)	20Kbs ÷ 500Kbs	It set the CAN bus speed. The available options are the following: **20 50 125 250 500**
DEBUG CANMESSAGE (A)	OFF ÷ ON	Reserved. Used to activate special debug messages

SPECIAL ADJUSTMENTS menu		
Parameter	**Allowable range**	**Description**
CONTROLLER TYPE (A)	0 ÷ 9	Reserved. NOTE: only Zapi technicians can change this value.
SAFETY LEVEL (T, TM, P, CO)	0 ÷ 3	Defines the safety level of the controller (i.e. the functionality of the supervisor microcontroller) **0 =** the Supervisor µC does not check any signal **1 =** the Supervisor µC checks the inputs and the outputs **2 =** the Supervisor µC checks the inputs and the set-point of the motor **3 =** the Supervisor µC checks the inputs, the outputs and the set-point of the motor.
RS232 CONSOLLE (A)	OFF ÷ ON	Reserved. Used to enable parameter changing using serial interface NOTE: only Zapi technicians can change this value.
ID CANOPEN OFST (CO)	0 ÷ 56 (step of 8)	Defines the offset of the CANopen frame IDs.
2ND SDO ID OFST (A)	0 ÷ 126 (step of 2)	It decides if another SDO channel communication has to be added. Specify an ID offset different from 0 to enable channel.

9.2.5 Menu “TESTER” functions list

The TESTER functions are real time feedback measurements of the state of the controller/motor/command devices. It is possible to know the state (active / off) of the digital I/Os, the voltage value of the analog inputs and the state of the main variables used in the motor and hydraulics control. Enter in “Function” -> “Tester”. Some of the values may not be displayed depending on selected motor and input configuration.

9.2.5.1 Tester menu for master microcontroller

The following table lists the master microcontroller data that is possible to view form the tester menu.

TESTER menu (Master)		
Parameter	**Unit of measure (resolution)**	**Description**
KEY VOLTAGE (A)	Volt (0.1V)	KEY voltage value measured in real time from pins CNA-1 (M1) or CNA-5 (L1).
BATTERY VOLTAGE (A)	Volt (0.1V)	Battery voltage value measured in real time (DC Link).
MOTOR VOLTAGE (A)	% (1%)	It is the phase to phase voltage applied by the inverter to the motor expressed in percentage. 100% means maximum sine wave width; 0% means the sine wave width is null.
INDEX OVERMOD. (A)	% (1%)	It is an indication about the correction applied to the motor voltage theoretical set point in order to over-modulate or compensate the battery voltage drop. The actual motor voltage applied is the product between “Motor Voltage” and “Index Overmod”:
FREQUENCY (A)	Hertz (0.1Hz)	This is the frequency of the sine waves generated by the inverter.
MEASURED SPEED (A)	Hertz (0.1Hz)	This is the speed of the motor measured with the encoder and expressed in the same unit of the FREQUENCY reading.
SLIP VALUE (A)	Hertz (0.01Hz)	This is the slip between the frequency and the speed of the motor $Slip[Hz] = Frequency - Encoder$
CURRENT RMS (A)	Ampere (1A)	Root Mean Square value of the line current in the traction motor. $Current\ [Arms] = \sqrt{I_Q{}^2 + I_D{}^2}$
ID FILTERED RMS (A)	Ampere (1A)	Root Mean Square value of the current in d axle.

TESTER menu (Master)		
Parameter	**Unit of measure (resolution)**	**Description**
IQ FILTERED RMS (A)	Ampere (1A)	Root Mean Square value of the current in q axle.
IQ IMAX LIMIT (A)	Ampere (1A)	Root Mean Square maximum value of the current in q axle according to parameters set.
MOT. POWER WATT (A)	Watt (1W)	Estimated power supplied to the motor.
FLUX WB/1000 (A)	milliWeber (0.1mWb)	Estimated flux into the traction motor.
MOTION TORQUE (A)	NewtonMeter (0.1Nm)	Estimated motion torque value.
STEER ANGLE (T, TM)	Degrees (1°)	Current steered wheel angle. 0° when the wheels are straight ahead.
BATTERY CHARGE (A)	% (1%)	It shows the residual charge of the battery as a percentage of the full charge level.
TEMPERATURE (A)	°C (1°C)	This is the temperature of the inverter base plate. This temperature is used for the HIGH TEMPERATURE alarm detection.
MOTOR TEMP. 1 (A)	°C (1°C)	This is the temperature of the motor windings picked up with an analog sensor inside the motor. Normally the sensor is a PTC Philips KTY84-130. This temperature is used only to raise a warning when the motor temperature overtakes the MOTOR OVERTEMP setting.
MOTOR TEMP. 2 (A)	°C (1°C)	This is the temperature of the motor windings picked up with an analog sensor inside the motor. Normally the sensor is a PTC Philips KTY84-130. This temperature is used only to raise a warning when the motor temperature overtakes the MOTOR OVERTEMP setting.
DI0-A18 TILLER SW (T, TM)	OFF/ON	It is the status of the Digital Input on CNA-18 (H2)
DI1-A19 FWD SW (T, TM, TS)	OFF/ON	It is the status of the Digital Input on CNA-19 (H3)
DI2-A20 REV SW (T, TM)	OFF/ON	It is the status of the Digital Input on CNA-20 (H4)
A17 POT#1 (A)	Volt (0.01V)	It is the voltage of the analog signal on CNA-17 (H1)

TESTER menu (Master)		
Parameter	**Unit of measure (resolution)**	**Description**
SIN FB. INPUT **(PM motor with sin/cos sensor or resolver)** (A)	Volt (0.01V)	It is the voltage of Sine signal on CNA-26 (F2). In case of resolver it could the differential signal between CNA-26 (F2) and CNA-25 (F1)
COS FB. INPUT **(PM motor with sin/cos sensor or resolver)** (A)	Volt (0.01V)	It is the voltage of Cosine signal on CNA-28 (F4). In case of resolver it could the differential signal between CNA-28 (F4) and CNA-27 (F3)
A9 OUTPUT EV1 (A)	OFF/ON	It is the status of the NEV1 output on CNA-9 (K1).
A10 OUTPUT EV2 (A)	OFF/ON	It is the status of the NEV2 output on CNA-10 (K2).
MAIN CONT. (A)	% (1%)	This value shows the voltage applied on main contactor coil. The value the duty cycle of PWM applied and is expressed in percentage..
CTRAP HW (A)	\	This is a counter of the number of occurrence of hardware overcurrent detection.
CTRAP THRESOLD (A)	Volt (0.01V)	This is the voltage of the hardware overcurrent threshold **Note: feature not implemented in Zapi HV Inverter**
A.SENS.OFFSET SR **(PM motor with sin/cos sensor or resolver)** (A)	\	This parameter shows the voltage offset of the sin signals, in digital units, acquired during the absolute sensor acquisition automatic procedure.
A.SENS.OFFSET CR **(PM motor with sin/cos sensor or resolver)** (A)	\	This parameter shows the voltage offset of the cos signals, in digital units, acquired during the absolute sensor acquisition automatic procedure.
ANGLE OFFSET **(PM motor with sin/cos sensor or resolver)** (A)	Degree	This is the angle offset between the orientation of the rotor and the position sensor
ROTOR POSITION **(PM motor with sin/cos sensor or resolver)** (A)	Degree	This is current absolute orientation of the rotor, in degrees
CPU TIME F US (A)	\	Reserved for Zapi technicians use.

TESTER menu (Master)		
Parameter	**Unit of measure (resolution)**	**Description**
CPU TIME M US (A)	\	Reserved for Zapi technicians use.

9.2.5.2 Tester menu for supervisor microcontroller

The following table lists the master microcontroller data that is possible to view form the tester menu.

TESTER menu (Supervisor)		
Parameter	**Unit of measure (resolution)**	**Description**
MEASURED SPEED (A)	Hertz (0.1Hz)	This is the speed of the motor measured with the encoder and expressed in the same unit of the FREQUENCY reading.
DI0 (A)	OFF/ON	It is the status of the Digital Input on CNA-18 (H2)
DI1 (A)	OFF/ON	It is the status of the Digital Input on CNA-19 (H3)
DI2 (A)	OFF/ON	It is the status of the Digital Input on CNA-20 (H4)
POT#1 (A)	Volt (0.01V)	It is the voltage of the analog signal on CNA-17 (H1)
WARNING SYSTEM (CO)	\	In case of warning it shows the correspondent warning code.

9.2.6 Menu “HARDWARE SETTING” functions list

The Hardware setting menu includes parameters of motor control algorithm.



Only the parameters that the user can work on are described in this paragraph.



For description and teaching about missing parameters contact Zapi technician.

HARDWARE SETTING menu		
Parameter	**Allowable range**	**Description**
TOP MAX SPEED (T, TM, P, CO)	0Hz ÷ 600Hz (step of 10Hz)	Defines the maximum tractions motor speed.
FEEDBACK SENSOR **(Only for PM motor)** (T, TM, P, CO)	0 ÷ 4 (step of 1)	Defines the type of the feedback sensor used: **0:** Incremental encoder **1:** sin/cos sensor **2:** Incremental encoder + sin/cos sensor **3:** Incremental encoder + sin/cos sensor + index **4:** PWM absolute sensor + incremental encoder + index
ROTATION CW ENC (A)	OPTION#1 ÷ OPTION#2	It defines the sequence of the encoder channels expected by controller **OPTION#1:** channel A anticipates channel B **OPTION#2:** channel B anticipates channel A
ROTATION CW MOT (A)	OPTION#1 ÷ OPTION#2	It allows to change the sequence in that the motor phases are controlled. **OPTION#1:** U-V-W in forward direction. **OPTION#2:** V-U-W in forward direction.
ROTATION CW POS **(Only for PM motor)** (A)	OPTION#1 ÷ OPTION#2	It allows to change the direction of absolute position sensor. **OPTION#1** sin anticipates cos **OPTION#2** cos anticipates sin
ENCODER PULSES 1 (T, TM, P, CO)	32 ÷ 1024	First copy of the number of encoder pulses per revolution. It must be set equal to ENCODER PULSES 2 or the controller will rise an alarm. The available options are: **32 48 64 80 64 128 256 512 1024 124** **NOTE: with standard HW the capability to use high number of pulses encoder could be limited depending by the speed. Ask to Zapi technician before to operate on this parameter**

HARDWARE SETTING menu		
Parameter	**Allowable range**	**Description**
ENCODER PULSES 2 (T, TM, P, CO)	32 ÷ 1024	First copy of the number of encoder pulses per revolution. It must be set equal to ENCODER PULSES 2 or the controller will rise an alarm. The available options are: **32 48 64 80 64 128 256 512 1024 124** **NOTE: with standard HW the capability to use high number of pulses encoder could be limited depending by the speed. Ask to Zapi technician before to operate on this parameter**
MOTOR P. PAIRS 1 (T, TM, P, CO)	1 ÷ 30 (step of 1)	First copy of the number of traction motor pole pairs. It must be set equal to MOTOR P. PAIRS 2 or the controller will rise an alarm.
MOTOR P. PAIRS 2 (T, TM, P, CO)	1 ÷ 30 (step of 1)	Second copy of the number of traction motor pole pairs. It must be set equal to MOTOR P. PAIRS 1 or the controller will rise an alarm.

10 OTHER FEATURES & EXAMPLES

10.1 "PROGRAM VACC" function

This enables adjustment of the minimum and maximum useful signal level, in either direction. This function is unique when it is necessary to compensate for asymmetry with the mechanical elements associated with the potentiometer, especially relating to the minimum level.

The two graphs show the output voltage from a non-calibrated potentiometer with respect to the mechanical "zero" of the control lever. MI and MA indicate the point where the direction switches close. 0 represents the mechanical zero of the rotation.
The Left Hand graph shows the relationship of the motor voltage without signal acquisition being made. The Right Hand Graph shows the same relationship after signal acquisition of the potentiometer.

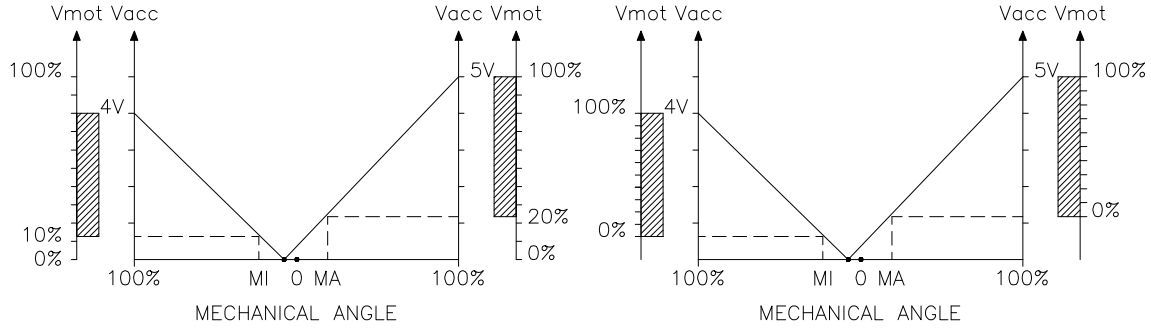


Figure 8: Program VACC

This function looks for and remembers the minimum and maximum potentiometer wiper voltage over the full mechanical range of the pedal. It enables compensation for dissymmetry of the mechanical system between directions.
The operation is performed by operating the pedal after entering the PROGRAM VACC function.

See paragraphs 14.1.4 and 14.2.6 for specific procedure description

10.2 ACCELERATION DELAY

The Accel Delay parameter permits to define the acceleration time depending by final speed set point and by ACCEL MODULATION parameter setting:

- If ACCEL MODULATION=OFF the actual acceleration time can be obtained applying this formula:

$$Accel\ time\ [sec] = \frac{Speed\ Set\ point\ [Hz]}{100Hz} \cdot Acceler\ delay$$

- If ACCEL MODULATION = ON the actual acceleration time is manged

differently by software depending by final speed set point:

Examples:

Case1:
- Final speed set point = 100Hz
- Accel Delay parameter = 2,5sec

the accel time is 2,5sec as expected (black trace in the graph)

Case2:
- Final speed set point = 60Hz
- Accel Delay parameter = 2,5sec

the accel time is re-scaled by software and the time to reach the final speed is still 2,5sec (red trace in the graph)

Case3:
- Final speed set point = 150Hz
- Accel Delay parameter = 2,5sec

the accel time is:

$$Accel\ time\ [sec] = \frac{150Hz}{100Hz} \cdot Acceler\ delay = 3{,}75sec$$

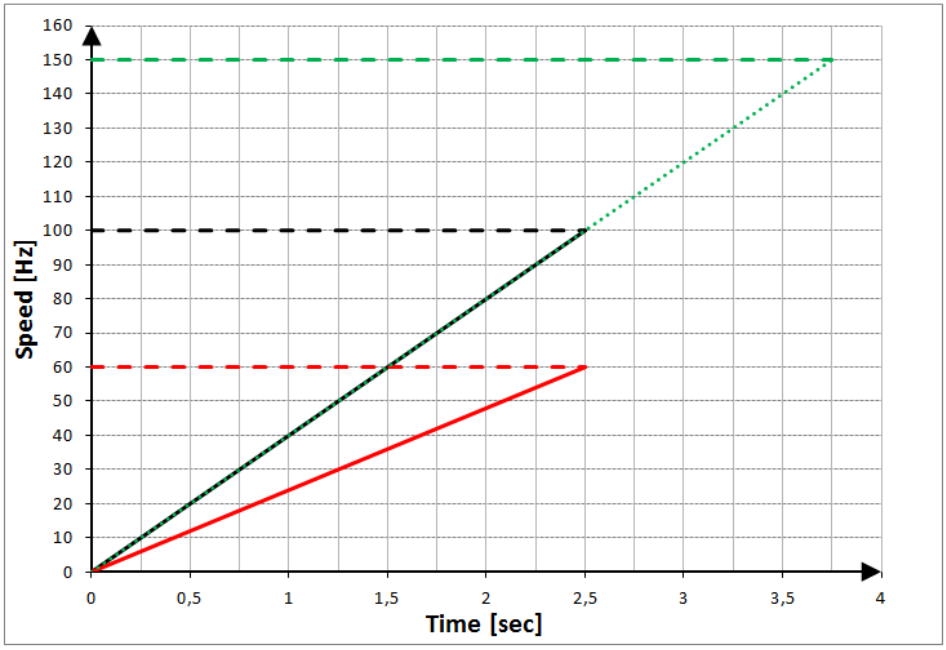


Figure 9: Accel delay

10.3 BRAKING DELAY

The “braking” parameters permit to define the deceleration time depending by final speed set point in different operating condition

Examples:

Case1:
- Initial speed = 110Hz

- Final speed set point = 10Hz
- Decel Braking parameter=2,5sec

the time to reach the final speed is 2,5sec (black trace in the graph)

Case2:

- Initial speed = 60Hz
- Final speed set point = 10Hz
- Decel Braking parameter=2,5sec

The ramp is re-scaled by software and the time to reach the final speed is still 2,5sec (red trace in the graph)

Case3:

- Initial speed = 150Hz
- Final speed set point = 10Hz
- Decel Braking parameter=2,5sec

the deceleration time is:

$$Decel\ time\ [sec] = \frac{150Hz}{100Hz} \cdot Decel\ Braking = 3{,}75sec$$

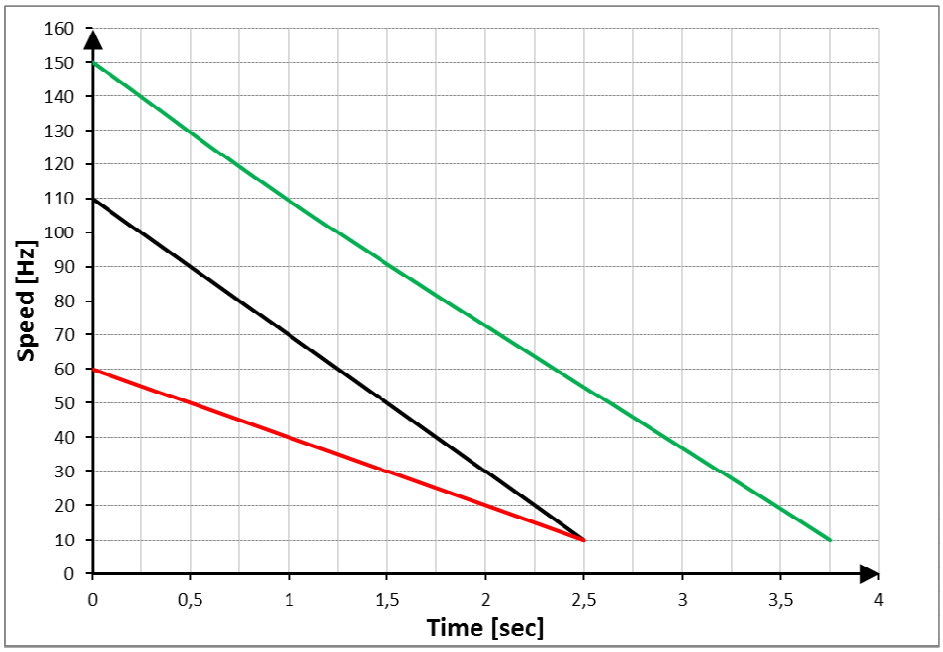


Figure 10: Braking delay

Note: This example is valid for all the "braking" parameters: Decel Braking, Inver Braking, Release Braking, Tiller Braking, Pedal Braking, Speed Limit BRK, Steer Braking

10.4 SMOOTHNESS

The "smooth" parameters give a parabolic form to the acceleration or deceleration ramp near to 0rpm. The value has not a phisycal meaning, simply 1 means linear ramp, higher the value smoother the acceleration ramp

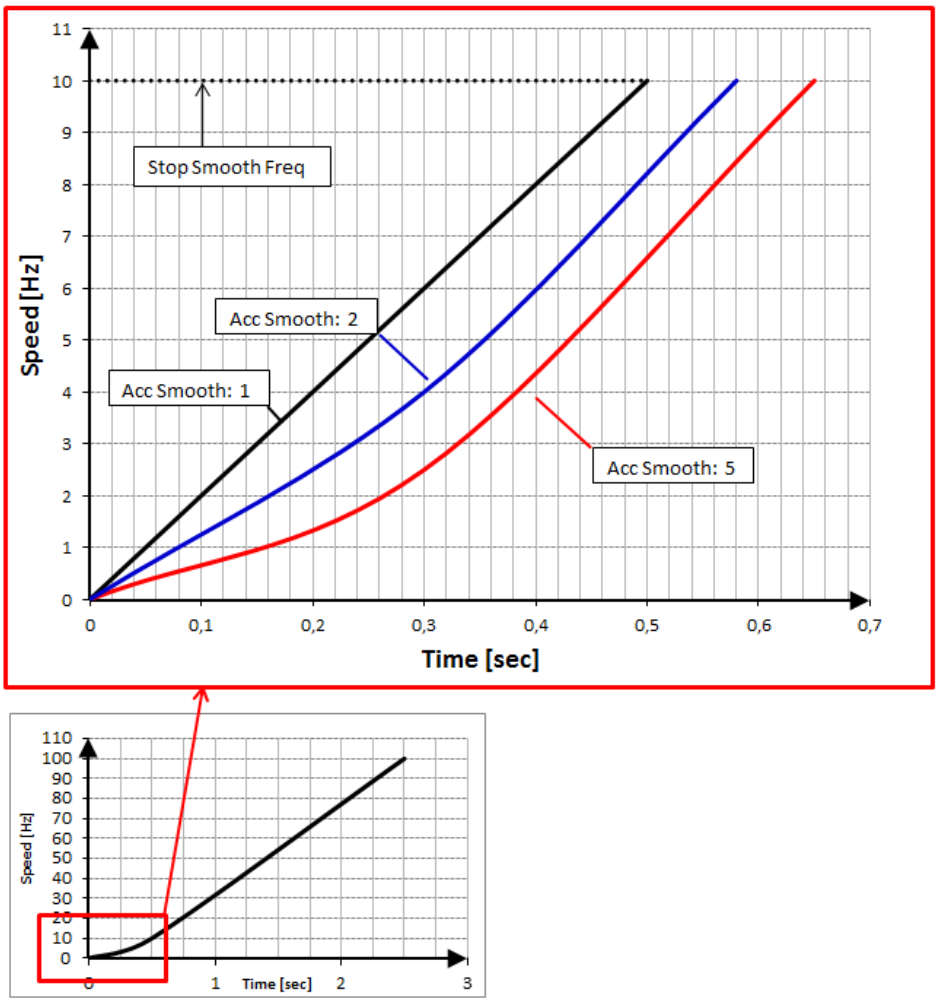


Figure 11: Smoothness

Note: This example is valid for ACC Smooth, BRK Smooth, INV Smooth

10.5 STEERING CURVE

Adjusting properly the dedicated parameters (Curve speed 1, Curve cutback, Steer dead Angle, Steer Angle 1 and Steer Angle 2), it is possible create a sort of speed profile dependent by steered wheel angle.
The profile created is valid both for positive angle values and negative angle values.

Example:

- 3 wheel CB truck
- Permitted steered wheel angle = -90°÷90°
- Curve Cutback = 30%
- Steer Dead Angle= 40°
- Steer Angle 1= 50°
- Steer Angle 2 = 80°

It will be obtained a speed profile as represented in the graph below

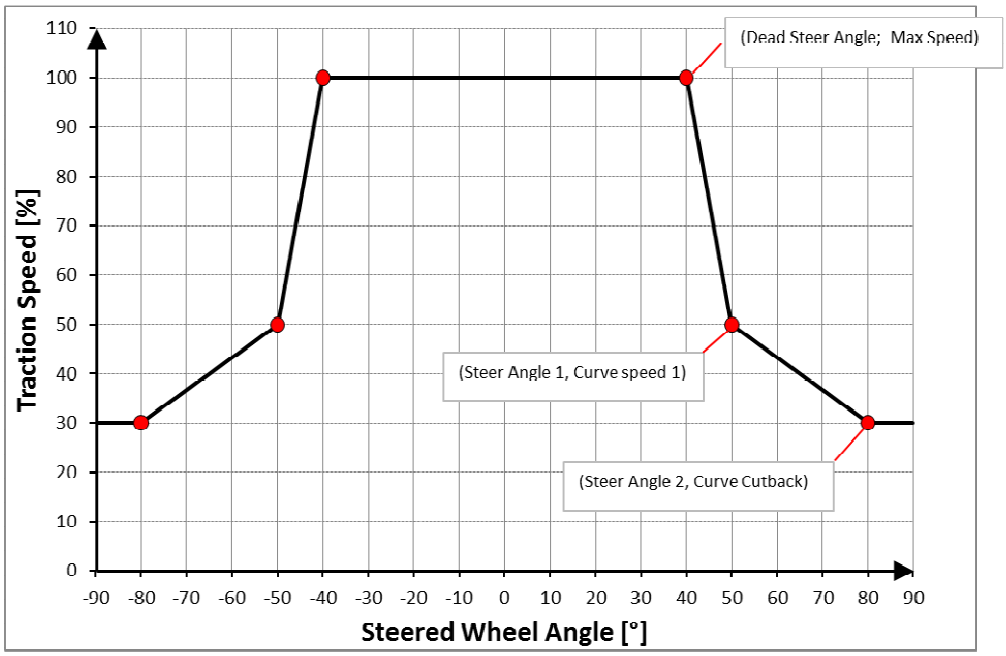


Figure 12: Steering curve

10.6 Description of the throttle regulation

This regulation applies a not linear relationship between the position of the accelerator and the speed of the truck. The main goal is to increase the resolution for the speed modulation when the truck is slowly moving.
Three adjustments are used for the throttle regulation:
1) THROTTLE 0 ZONE
2) THROTTLE X1 POINT
3) THROTTLE Y1 POINT
4) THROTTLE X2 POINT
5) THROTTLE Y2 POINT
6) THROTTLE X3 POINT
7) THROTTLE Y3 POINT

THROTTLE 0 ZONE: the speed of the truck remains at frequency creep meanwhile the voltage from the accelerator potentiometer is lower than this percentage of the MAX VACC setting. This adjustment defines the width of a dead zone close to the neutral position.
THROTTLE X1 POINT & THROTTLE Y1 POINT: the speed of the truck grows up with a fixed slope (linear relationship) from the THROTTLE 0 ZONE up to THROTTLE X1 POINT. This slope is defined by the matching between the X1 point percentages of the MAX VACC setting with the Y1 point percentage of the full truck speed.
Same as for the pairs (X2;Y2) and (X3;Y3)
From the X3 point up to the MAX VACC point, the slope of the relationship between the truck speed and the accelerator position is different to match to match the full speed in the truck with the MAX VACC voltage in the accelerator position.

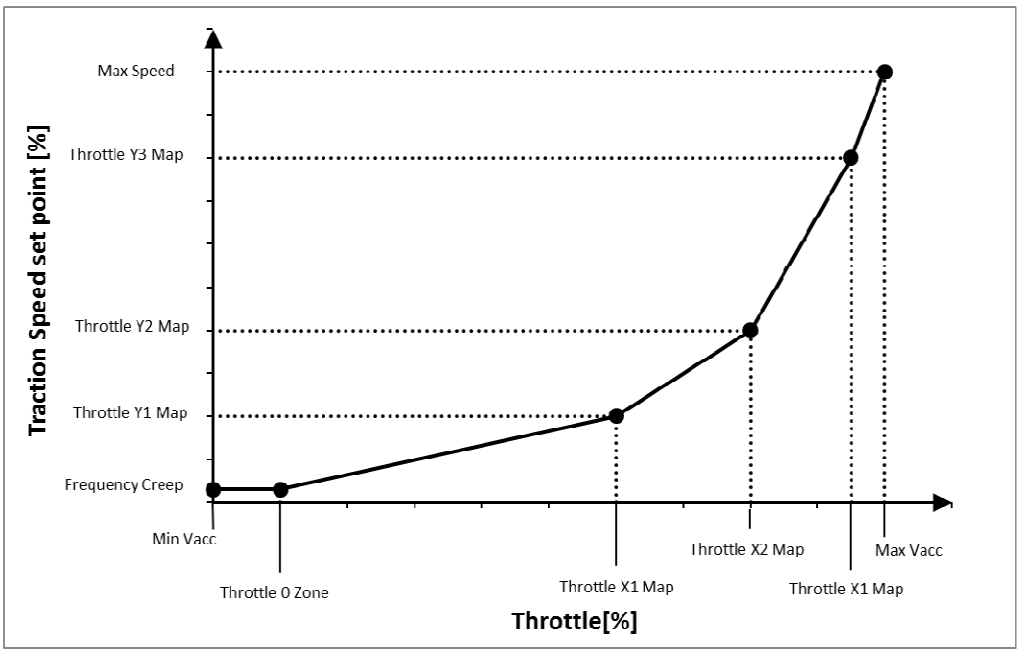


Figure 13: throttle profile

10.7 Description NLC output management

For the NLC output CNA-11 (K3) there is the possibility to set a Pull-in voltage and define a maintenance voltage applied continuously to the coil.
MC VOLTAGE parameter specifies the duty cycle applied in the first second after activation and MC VOLT RED set the duty cycle applied after the first second to keep the contactor closed according to this formula:

$$Final\ Duty\ Cycle\ [\%] = MC\ VOLTAGE\ \cdot MC\ VOLT\ RED$$

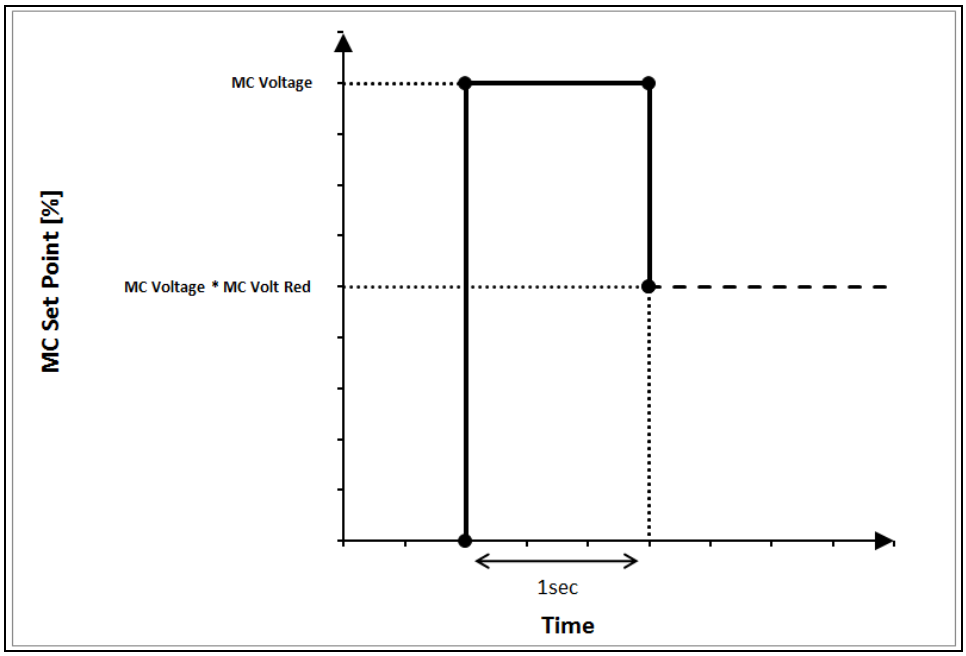


Figure 14: NLC Output management

Example 1
MC VOLTAGE = 100%
MC VOLTAGE RED = 70%
The contactor will be closed applying a 100% of duty cycle to the coil and then the duty cycle will be reduced to 70%.

Example 2
MC VOLTAGE = 70%
MC VOLTAGE RED. = 100%
The contactor will be closed applying a 70% of duty cycle to the coil and then the voltage will be kept at the same value.

Example 3
MC VOLTAGE = 70%
MC VOLTAGE RED = 70%
The contactor will be closed applying a 70% of duty cycle to the coil and then the duty cycle will be reduced to 49%.

10.8 Description of the battery charge detection setting

In operational condition the Battery Charge detection uses two setting that specify the Full Charge Voltage Level (100%) and the Discharge Voltage Level (10%). These two settings are the Bat.Max.Adj and the Bat.Min.Adj. It is possible to adapt the Battery Charge Detection to your specific battery, by changing the above two settings (e.g. if the Battery Discharged Detection occurs when the battery is not totally discharged, it is necessary to reduce the Bat.Min.Adj setting).

BDI adj Startup adjusts instead the level of the battery charge table at the start-up, in order to calculate the battery charge at keyon. The minimum variation of the battery charge that can be detected depends by the BDI reset parameter

The Battery Charge detection follows this algorithm:

Start up:
1) Battery voltages is read when the Battery current is zero, that is when the output power stage is not driven.
2) Vbatt is the average of the least samples measured by the microcontroller (the samples are took on key input).
3) Vbatt is compared with a threshold value (function of the actual charge percentage) in a table and with comparison is found a new charge percentage

- Threshold value can be changed with parameters BDI Adj Startup
- If the new charge percentage is included in the range " last percentage (last value stored in Eeprom) ± BDI reset" the percentage will not updated. Otherwise it will update to the new percentage

Operational condition:
The basic idea is following: the measure of the actual battery voltage, together with the discharge % at that time when the voltage is sampled, gives an unique information of the instantaneous battery current

- Battery voltages is read when the Battery current is not zero, that is when the

output power stage is driven.

- Vbatt is the average of the least samples measured by the microcontroller converter (the samples are took on key input).
- Vbatt is compared with a threshold value (function of the actual charge percentage) in a table and with comparison is found the current provided by the battery.
- The discharge current A, obtained by said relationship, integrated in the time, gives the Ah taken from the battery
- The charge is updated dynamically according to the Ah taken from battery.
- Thresholds value can be changed with parameters Bat. Max. Adj. and Bat. Min.Adj. With these two settings are used to adapt the Battery Charge detection to your specific battery

10.9 Torque Profile

Setting the proper parameter is possible define the a clamp for the max torque demand (set point) in the weakening area to match two goals:

1. Do not overtake the inner motor max torque profile.
2. To superimpose a limiting profile to the max torque to get different drive performances (Eco mode, Medium performance, High performance)

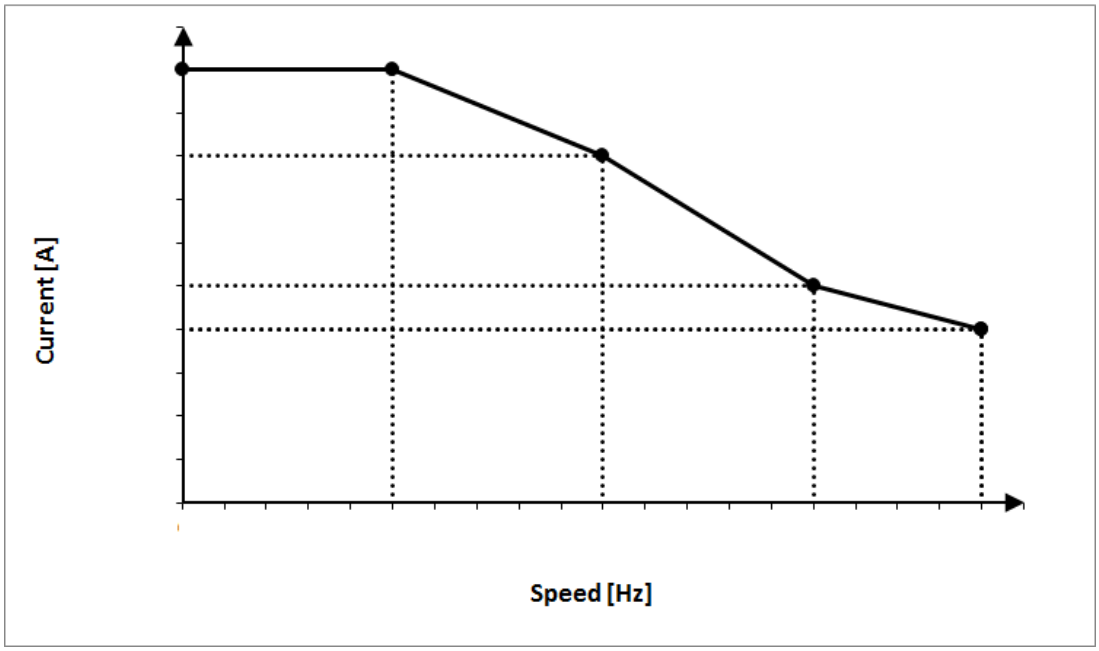


Figure 15: Torque profile

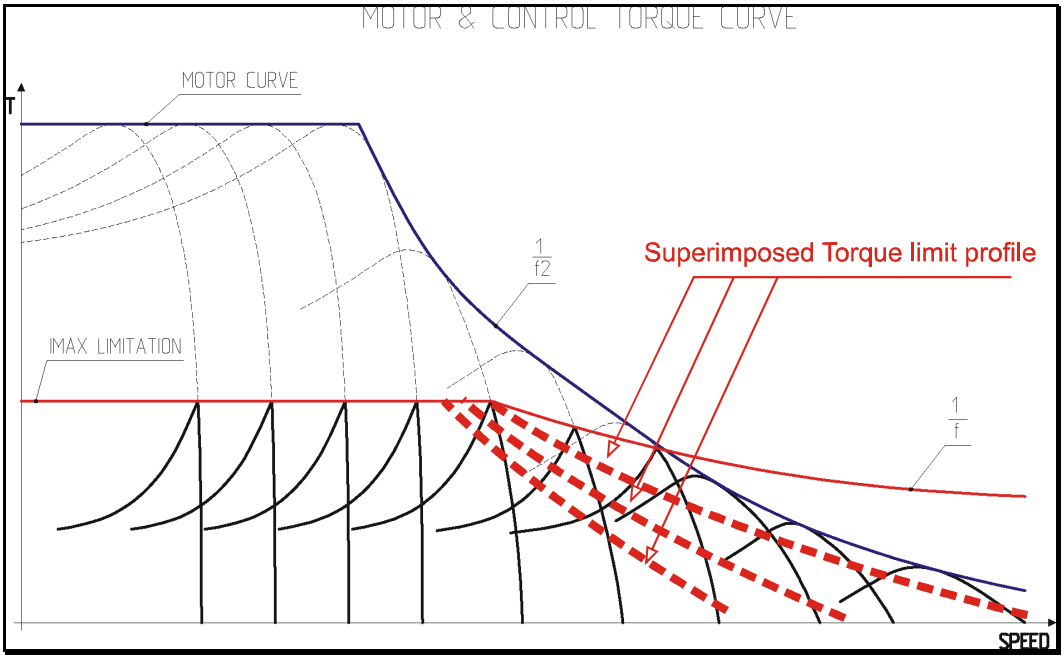


Figure 16: Torque curves

10.10 Description of “ALARMS” menu

The ALARMS logbook in the MAIN MENÙ records the alarms of the controller. It has a FIFO (First Input First Output) structure so the oldest alarm is lost when the database is full and a new alarm occurs. The logbook is composed of locations getting possible to stack different type of alarms with the following information:

- The alarm code
- Number of alarms occurs consecutively
- The Hour Meter value when the first event of every alarm occurred
- Inverter temperature when the first event of every alarm occurred.

This function permits a deeper diagnosis of problems as the recent history can be revised.



NOTE: if the same alarm is continuously happening, the controller does not use new memory of the logbook, but only updates the last memory cell increasing the related counter (point 2) of previous list). Nevertheless, the number of hours indicated in this memory refers to the first time the alarm occurred. In this way, comparing this number with the controller life, it is possible to determine:
- When this alarm occurred the first time.
- How many hours are elapsed from the first occurrence.
- How many times it has occurred in a said period.

11 FAULTS DIAGNOSTIC SYSTEM

The fault diagnostic system of HV Inverter is composed into 2 main groups of faults:

ALARMS: these are the faults which stops the power section, that means power bridge is switched off and, if installed, the main contactor is opened. These are faults related to:

- failures in the motor/controller that the power system is likely not anymore able to drive the truck
- safety related failures

WARNINGS: these are faults which do not stop the truck or stop it by a controlled regenerative braking. In other words, the controller is working well, but it has detected conditions to reduce the performances or to stop the truck without opening the power devices. These warnings are related to:

- wrong operator sequences
- conditions which require performance reduction (like high temperatures, ….)

11.1 Master Microcontroller alarms overview

Error Code	Effect	Machine status When the test is done	Restart procedure	Indication			
				CAN OPEN CODE	MDI CODE	ZAPI CODE	LED CODE
VDC LINK OVERV.	MC is not closed, EB is applied, Traction/Pump, valves stopped	Stand-by, running	Valves or Traction/Pump Request	0XFFCA	77	77	77
HOME SENSOR CORR	MC is opened , EB is applied, EVP stopped	Running	Key re-cycle	0xFFB0	3	170	3
IMS ERROR	MC is not closed, EB is applied, Traction/Pump, valves stopped	Start-up	Key re-cycle	0XFFA7	4	167	4
SHORT CIRCUIT	MC is not closed, EB is applied, Traction/Pump, valves stopped	Running	Valves or Traction/Pump Request	0xFFA6	5	166	5
SHORT CIRCUIT KO	MC is not closed, EB is applied, Traction/Pump, valves stopped	Start-up, stand-by	Key re-cycle	0xFFA5	5	165	5
PWM ACQ.ERROR	MC is not closed, EB is applied, Traction/Pump, valves stopped	Start-up	Key re-cycle	0xFFA4	6	164	6
WATCHDOG	MC is opened, EB is applied, Traction/Pump, valves stopped	Start-up, stand-by, running	Key re-cycle	0x6010	8	8	8

Error Code	Effect	Machine status When the test is done	Restart procedure	Indication			
				CAN OPEN CODE	MDI CODE	ZAPI CODE	LED CODE
EVP DRIV. SHORT.	MC is closed or opened , EB is applied, EVP stopped	EVP off	Traction/ Pump request	0x5003	9	215	9
CONTROLLER MISM.	MC is not closed, EB is applied, Traction/Pump, valves stopped	Start-up	Install the correct software and Key re-cycle	0xFFEF	12	239	12
SEAT MISMATCH	MC is not closed , EB is applied, Traction/Pump stopped	Start-up, stand-by, running	Valves or Traction/ Pump request	0xFFDE	15	222	15
LOGIC FAILURE #1	MC is not closed, EB is applied, Traction/Pump, valves stopped	Stand-by, running	Valves or Traction/Pump Request	0XFF11	17	17	17
LOGIC FAILURE #2	MC is not closed, EB is applied, Traction/Pump, valves stopped	Start-up, stand-by,	Valves or Traction/Pump Request	0XFF12	18	18	18
LOGIC FAILURE #3	MC is not closed, EB is applied, Traction/Pump, valves stopped	Start-up, stand-by	Valves or Traction/Pump Request	0x5114	19	19	19
MC COIL OPEN	MC is not closed, EB is applied, Traction/Pump, valves stopped	Start-up, stand-by, running	Valves or Traction/Pump Request	0xFFE6	22	230	22
IQ MISMATCHED	Traction is stopped	Running	Valves or Traction/Pump Request	0xFFF5	24	245	24
INIT VMN LOW 01 INIT VMN LOW 02 INIT VMN LOW 03	MC is not closed, EB is applied, Traction/Pump, valves stopped	Start-up	Key re-cycle	0x3121	30	207	30
INIT VMN HIGH 81 INIT VMN HIGH 82 INIT VMN HIGH 83	MC is not closed, EB is applied, Traction/Pump, valves stopped	Start-up	Key re-cycle	0x3111	31	206	30
VMN HIGH	MC is not closed, EB is applied, Traction/Pump, valves stopped	Start-up, stand-by	Valves or Traction/Pump Request	0x3110	31	31	30
VMN LOW	MC is not closed, EB is applied, Traction/Pump, valves stopped	Start-up	Valves or Traction/Pump Request	0x3120	30	30	31
HW FAULT 11 HW FAULT 12 HW FAULT 13	MC is not closed , EB is applied, Traction/Pump stopped	Start-up	Key re-cycle	0xFFE3	32	227	32
HW FAULT 01 HW FAULT 02 HW FAULT 03	MC is not closed , EB is applied, Traction/Pump stopped	Start-up	Key re-cycle	0xFFE3	32	227	32

Error Code	Effect	Machine status When the test is done	Restart procedure	Indication			
				CAN OPEN CODE	MDI CODE	ZAPI CODE	LED CODE
POSITIVE MC OPEN	MC is not closed, EB is applied, Traction/Pump, valves stopped	Start-up, stand-by, running	Key re-cycle	0xFFD5	35	213	
FIELD ORIENT KO	MC is opened, EB is applied, Traction/Pump, valves stopped	Running	Valves or Traction/Pump Request	0xFFFD	36	253	
CONTACTOR CLOSED	MC is not closed (command is not activated), EB is applied, Traction/Pump stopped	Start-up	Valves or Traction/Pump Request	0x5442	37	37	
CONTACTOR OPEN	MC is opened , EB is applied, Traction/Pump, valves stopped	Start-up, stand-by, running	Valves or Traction/Pump Request	0x5441	38	38	
POWER MISMATCH	Traction is stopped EB is applied, MC is opened	Running	Traction/ Pump request	0xFFD4	39	212	
WRONG SET BAT	MC is not closed, EB is applied, Traction/Pump, valves stopped	Start-up	The alarms disappears as soon as the voltage come back into the correct range	0x3100	41	251	
MOT.PHASE SH.36 MOT.PHASE SH.37 MOT.PHASE SH.38	MC is not closed, EB is applied, Traction/Pump, valves stopped	Start-up	Traction/ Pump request	0xFFC4	47	196	
STBY I HIGH	MC is not closed, EB is applied, Traction/Pump stopped	Start-up, stand-by	Valves or Traction/Pump Request	0x2311	53	53	
OVERLOAD	Traction is stopped	Running	Key re-cycle	0xFFB4	57	180	
CAPACITOR CHARGE	MC is not closed, EB is applied, Traction/Pump, valves stopped	Start-up	Valves or Traction/Pump Request	0x3130	60	60	
TILLER ERROR	Traction stopped, Eb applied	Stand-by, running	Valves or Traction/Pump Request	0xFFB9	64	185	
NO CAN MSG.	MC is opened, EB is applied, Traction/Pump, valves stopped	Start-up, stand-by, running	Valves or Traction/Pump Request	0X8130	67	248	
WRONG RAM MEM.	MC is opened, EB is applied, Traction/Pump, valves stopped	Stand-by	Key re-cycle	0xFFD2	71	210	
DRIVER SHORTED	MC is opened (the command is released), EB is applied, Traction/Pump, valves stopped	Start-up, stand-by, running	Valves or Traction/Pump Request	0x3211	74	74	

Error Code	Effect	Machine status When the test is done	Restart procedure	Indication			
				CAN OPEN CODE	MDI CODE	ZAPI CODE	LED CODE
CONTACTOR DRIVER	MC is opened (the command is released), EB is applied, Traction/Pump, valves stopped	Start-up, stand-by, running	Valves or Traction/Pump Request	0x3221	75	75	
MC-EF COIL SHOR.	MC is opened, EB is applied, Traction/Pump, valves stopped	Start-up (immediately after MC closing), stand-by, running	Valves or Traction/Pump Request	0x2250	76	223	
SPEED FB. ERROR	MC is opened , EB is applied, EVP stopped	Running	Key re-cycle	0xFFAF	81	175	
ENCODER ERROR	MC is opened, EB is applied, Traction/Pump, valves stopped	Running	Valves or Traction/Pump Request	0xFF52	82	82	
WRONG ENC SET	MC is not closed, EB is applied, Traction/Pump, valves stopped	Start-up	Key re-cycle	0xFF51	83	181	
POS. EB. SHORTED 03	MC is not closed, EB is applied, Traction/Pump, valves stopped	Start-up	Valves or Traction/Pump Request	0x3223	84	195	
ANALOG INPUT	MC is opened, EB is applied, traction/pump stopped	stand-by, running	Key re-cycle	0xFFFA	96	237	
CTRAP THRESHOLD	MC is opened (the command is released), EB is applied, Traction/Pump, valves stopped	Start-up, stand-by, running	Valves or Traction/Pump Request	0xFFEB	99	235	

11.2 Analysis and troubleshooting of alarms displayed on console (Master)

- **VDC LINK OVERV.**
  Cause:
  This fault is displayed when the controller detects an over voltage condition in hardware. The overvoltage threshold is normally set at 720V.
  As soon as the condition occurs the IGBT bridge is switched off and the main contactor is opened.

  Troubleshooting:
  If the alarm happens during release braking, check line contactor contact, check battery power cable connection looking for high series resistance.

  VDC LINK UNDERV.
  Cause:

This fault is displayed when the DC Link voltage has decreased under the undervoltage threshold. Possible causes are a strong battery current consumption, battery with low Ah capacity, high resistance in the battery connection.
As soon as the condition occurs the IGBT bridge is switched off and the main contactor is opened.

Troubleshooting:
If the alarm happens during release braking, check line contactor contact, check MC contacts and battery power cable connection looking for high series resistance.

- **LOGIC FAILURE #1**
  Cause:
  This fault is displayed when the controller detects an under voltage condition on KEY input pin CNA-1 (M1) or CNA-5 (L1).
  Under voltage threshold is 17V for 24V of nominal low voltage power supply and 9V for 12V of nominal low voltage power supply. It can be adjusted in factory by hardware modification for special requirements.

  Troubleshooting of fault displayed at startup or in standby:
  - Key input signal down-going pulses (below undervoltage threshold) due to external loads, like DC/DC converters starting-up, relays or contactor switching, solenoids energizing / de-energizing.
  - If no voltage transient is detected on the supply line and the alarm is present every time the key is switched ON, the failure is probably in the controller hardware, so it is necessary to replace the logic board.

- **LOGIC FAILURE #2**
  Cause:
  Fault in the hardware section of the logic board which manages the phase voltages feedback.

  Troubleshooting:
  This type of fault is not related to external components, so when it happens it is necessary to replace the Controller.

- **LOGIC FAILURE #3**
  Cause:
  Hardware problem in the logic card circuit for high current (overload) protection. An overcurrent condition is triggered even if the power bridge is not driven.

  Troubleshooting:
  This type of fault is not related to external components, so, when it is present it is necessary to replace the logic board.

- **IGBT SHORTED**
  Cause:
  Before switching the MC on, the software checks the correct operation of the power bridge. In case of failure inside the IGBT module the alarm is raised.

  Troubleshooting:
  This is an internal failure, controller has to be replaced.

- **FAULT DRV POWER**
  Cause:
  The driver's circuit has a problem.

  Troubleshooting:
  Check the motor connection and the motor insulation. If no problem is found it is an internal failure, replace the controller.

- **WATCH DOG**
  Cause:
  This is a safety related test. It is a self-diagnosis test within the logic between Master and Supervisor microcontrollers.

  Troubleshooting:
  This alarm could be caused by an internal CAN Bus malfunctioning, which blinds master-supervisor communication. Try to switch the key off and then on, if the alarm is still present replace the logic board

- **WRONG RAM MEM. 05**
  Cause:
  The algorithm implemented to check the main RAM registers finds a wrong contents: the register is "dirty". This alarm inhibit the machine operations.

  Troubleshooting:
  Try to switch the key off and then on, if the alarm is still present replace the logic board.

- **OVERLOAD**
  Cause:
  The motor current has overcomes the limit fixed by hardware.

  Troubleshooting:
  Reset the alarms by KEY re-cycling. If the alarms condition occurs again ask the assistance of a Zapi technician.
  The fault condition could be affected by wrong adjustment of motor parameters.

- **FIELD ORIENT. KO**
  Cause:
  The error between Id set point and Id estimated is out of range.

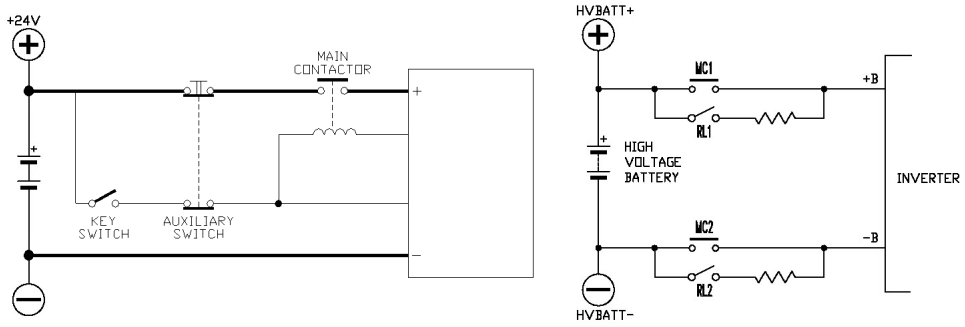
  Troubleshooting:
  Ask the assistance of a Zapi technician to do the correct adjustment motor of the parameters.

- **IQ MISMATCHED**
  Cause:
  The error between Id set point and Id estimated is out of range.

  Troubleshooting:
  Ask the assistance of a Zapi technician to do the correct adjustment motor of the parameters.

- **CAPACITOR CHARGE**

Follows the charging capacitor system:



Cause:
When the key is switched ON, the inverter tries to charge the DC Link capacitors activating an external pre-charge system (external power resistors must be installed), and check if the capacitor are enough charged within a timeout. If the capacitor voltage measured is less than 20% of the nominal battery voltage, an alarm is raised and main contactor is not closed.

Troubleshooting:
- There is an external load in parallel to capacitor bank, which sinks current from the controller capacitors pre-charging circuit, thus preventing the caps from charging. Check if an unexpected load is installed in parallel to capacitor bank.
- Failure in the pre-charge circuit, like charging resistors open or auxiliary contactors not correctly installed.
- The HVBATT polarity is not correct
- Failure in the controller power section

- **MOT.PHASE SH.**
  Cause:
  Short circuit between two of the motor phases

  Troubleshooting:
  - Verify the motor phases connection on motor side
  - Verify the motor phases connection on inverter side
  - Checks the power motor cables.
  - Replace the controller and if the alarm doesn't disappears the problem is inside the motor and it must be changed.

- **INIT VMN LOW**
  Cause:
  Before switching the MC on, the software checks the power bridge voltage without driving it.
  The software expects the voltage to be at a "steady state" value.
  If it is too low, an alarm is raised.

  Troubleshooting:
  Please check
  - Motor power cables connections
  - Impedance between motor power phase columns and HVBATT-
  - Motor leakage to truck frame
  - If the motor connections are OK and not low impedance paths are present externally, the problem is inside the controller. It has to be replace

- **INIT VMN HIGH**
  Cause:
  Before switching the MC on, the software checks the power bridge voltage without driving it.
  The software expects the voltage to be at a “steady state” value.
  If it is too high, this alarm is raised.

  Troubleshooting:
  Please check
  - Motor power cables connections
  - Impedance between motor power phase columns and HVBATT+
  - Motor leakage to truck frame
  - If the motor connections are OK and not low impedance paths are present externally, the problem is inside the controller. It has to be replaced.

- **HW FAULT 11/12/13**
  Cause:
  The controller checks at each start up that the hardware circuit for enabling and disabling of Main Contactor driver CNA-11 (K3) by Supervisor uC works properly.

  Troubleshooting:
  This type of fault is not related to external components, so, when appears it is necessary to replace the logic board.

- **HW FAULT 01/02/03**
  Cause:
  The controller checks at each start up that the hardware circuit for enabling and disabling of Power Bridge by Supervisor uC works properly.

  Troubleshooting:
  This type of fault is not related to external components, so, when appears it is necessary to replace the logic board.

- **POWER MISMATCH**
  Cause:
  The error between Power set point and Power estimated is out of range.

  Troubleshooting:
  Ask the assistance of a Zapi technician to adjust of the motor parameters.

- **SEAT MISMATCH**
  Cause:
  This alarm can appear only in a Traction + Pump configuration.
  Input mismatch between Traction controller and Pump controller: the two inputs are different.

  Troubleshooting:
  - Check if there is wrong connection in the external wiring.
  - Using the “Tester” menu of the controller verify that what the two controller see in input is in accordance with the actual state of the external switch input.
  - In case no failures/problems have been found, the problem is in the controller, which has to be replaced.

- **STBY I HIGH**
  Cause:
  The current sensor or the current feedback circuit is damaged in the controller. The sensor seems to detect current different than Zero

  Troubleshooting:
  This type of fault is not related to external components so, when it is present, it is necessary to replace the controller.

- **CONTROLLER MISM.**
  Cause:
  The software is not compatible with the Hardware. Each controller produced is “signed” during the end of line test with a specific mark code saved in eeprom according to the customized Part Number.
  According with this “sign”, only the customized firmware can be uploaded.

  Troubleshooting:
  - Upload the correct firmware
  - Ask the assistance of a Zapi technician in order to verify that the firmware is correct

- **ENCODER ERROR**
  Cause:
  This fault is signaled in following conditions: the frequency supplied to the motor is higher than 40 Hz and the signal feedback from the encoder has a jump higher than 40 Hz in few tens of msec. This condition is related to a malfunctioning of the encoder.

  Troubleshooting:
  - Check both the electric and the mechanical encoder functionality, the wires crimping.
  - Check the encoder mechanical installation, if the encoder slips inside its compartment this alarm condition could be generated.
  - Also the electromagnetic noise on the sensor bearing can a cause for the alarm. In these cases try to replace the encoder.
  - If the problem is still present after replacing the encoder, the failure is in the controller.

- **SPEED FB. ERROR**
  Cause:
  This alarm appears if the absolute position sensor is used also for the speed estimation. If signaled, it means that the controller measured too high speed.

  Troubleshooting:
  - Check that the sensor used is compatible with the software release
  - Check the sensor mechanical installation and if it works properly.
  - Also the electromagnetic noise on the sensor can be a cause for the alarm.
  - If no problem are found on the motors, the problem is inside the controller, it is necessary to replace the logic board.

- **WRONG ENC SET**
  Cause:
  Mismatch between ENCODER PULSES 1 parameter and ENCODER

PULSES 2 parameter (see paragraph 9.2.6).

Troubleshooting:
Set the correct parameters

- **CONTACTOR CLOSED**
  Cause:
  Before driving the MC coil, the controller checks if the contactor is stuck. The controller drives the bridge for some tens milliseconds, trying to discharge the capacitors bank. If the capacitor voltage does decrease by 20% of the KEY voltage the alarm is generated.
  Troubleshooting:
  It is suggested to verify the power contacts of MC; to replace the MC is necessary.

- **CONTACTOR OPEN**
  Cause:
  The main contactor coil has been driven by the controller, but the power contacts seem don't close.
  In order to detect this condition the controller injects a DC current into the motor and checks the voltage on power capacitor. If the power capacitors are discharged it means that the main contactor is open.

  Troubleshooting:
  - It could be a problem of the contacts in the MC that are not working (does not pull-in), try replacing the MC.
  - If the contactors of MC are working correctly contacts a Zapi technician.

- **CONTACTOR DRIVER**
  Cause:
  The MC coil driver is not able to drive the load. The device itself or its driving circuit is damaged.
  Troubleshooting:
  This type of fault is not related to external components; replace the logic board.

- **MC COIL OPEN**
  Cause:
  This fault appears when no load is connected between the output NLC CNA-11 (K3) and the positive (for example +KEY).

  Troubleshooting:
  - It is suggested to check the harness, in order to verify if MC coil is connected to the right connector pin and if it is not interrupted.
  - If, even connecting the coil to the right pin or replacing it, the alarm is still present than the problem is inside the controller logic board, replace it.

- **MC COIL SHOR.**
  Cause:
  This alarm occurs when there is an overload of the MC driver CNA-11 (K3). As soon as the overload condition has been removed, the alarm exits automatically by releasing and then enabling a travel demand.

  Troubleshooting:
  - The typical root cause is in the harness or in the load coil. So the very first

check to carry out concerns connections between controller outputs and loads.
- Collect information about characteristics of the coils connected to the two driver and ask the assistance of a Zapi technician in order to verify that the limit of the hardware is not exceeded.
- In case no failures/problems have been found, the problem is in the controller, which has to be replaced.

- **ANALOG INPUT**
  Cause:
  This alarm occurs when the A/D conversion of the analog inputs gives frozen value, on all of the converted signals, for more than 400msec. The goal of this diagnosis is to detect a failure of the A/D converter or a problem in the code flow that omits the refreshing of the analog signal conversion.

  Troubleshooting:
  If the problem occurs permanently it is necessary to substitute logic board.

- **DRIVER SHORTED**
  Cause:
  The driver of the main contactor coil is shorted.

  Troubleshooting:
  - Check if there is a short or a low impedance pull-down between NLC CNA-11 (K3) and NKEY.
  - The driver circuit is damaged in the logic board, which has to be replaced.

- **POWER ACQ. ERROR**
  Cause:
  This alarm occurs only when the controller is configured as PMSM and the feedback sensor selected is Encoder ABI + PWM
  The controller doesn't detect a correct information on PWM input at start up.

  Troubleshooting:
  - Re-cycle the key
  - Check the sensor to verify that it works properly.
  - Check the wiring

  If the problem occurs permanently it is necessary to substitute logic board.

- **HOME SENSOR CORR**
  Cause:
  The controller detected a difference between the estimated absolute orientation of the rotor, and the position of the index signal (ABI encoder). Erroneous acquisition of the angle offset between the orientation of the rotor and the index signal.

  Troubleshooting:
  Repeat the autoteaching procedure.

- **NO CAN MSG. 09**
  Cause:
  This fault is signaled when the communication with the supervisor uC is not present.

  Troubleshooting:

This type of fault is not related to external components, so, when it is permanently present it is necessary to replace the logic board.

- **WRONG SET BAT. 05**
  Cause:
  At start-up, the controller checks the battery voltage, measured on KEY input pin CNA-1 (M1) or CNA-5 (L1), and verify it is within a window of ±20% around the nominal value.

  Troubleshooting:
  - Check that the controller SET KEY VOLTAGE parameter value matches the battery nominal voltage.
  - Check that TESTER MENU / KEY VOLTAGE parameter shows a value close to the KEY voltage measured with a voltmeter on pin CNA-1 (M1) or CNA-5 (L1). If it does not match, use the “ADJUST BATTERY” function.
  - Replace the low voltage power supply source.

- **SHORT CIRCUIT KO**
  Cause:
  The HW to detect the fault on power bridge it doesn’t work properly

  Troubleshooting:
  - It is an internal fault, replace the controller.

11.3 Microcontroller Supervisor alarms overview

Error Code	Effect	Machine status When the test is done	Restart procedure	Indication			
				CAN OPEN CODE	MDI CODE	ZAPI CODE	LED BLINKS
WATCHDOG	MC is opened, EB is applied, traction/pump stopped	Stand-by, running	Key re-cycle	0X6010	8	8	8
CONTROLLER MISM.	MC is not closed, EB is applied, Traction/Pump, valves stopped	Start-up	Install the correct software and Key re-cycle	0XFFEF	12	239	12
SP MISMATCH xx	MC is opened, EB is applied, traction/pump stopped	Running	Key re-cycle	0XFFF2	15	242	15
OUT MISMATCH xx	MC is opened, EB is applied, traction/pump stopped	Running	Key re-cycle	0XFFE3	16	227	16
LOGIC FAILURE #1	MC is opened, EB is applied, traction/pump stopped	Stand-by, running	Valves or Traction/Pump Request	0X5514	17	17	17
LOGIC FAILURE #3	MC is opened, EB is applied, traction/pump stopped	Stand-by, running	Valves or Traction/Pump Request	0XFF11	19	19	19
INPUT MISMATCH	MC is opened, EB is applied, Traction/Pump stopped	Start-up, standby, running	Key re-cycle	0XFFD5	58	213	58
W.SET. TG-EB	Traction/ Pump motor is stopped	Start-up, stand-by, running	Key re-cycle	0XFFD4	59	212	59
NO CAN MSG.	MC is opened, EB is applied, Traction/Pump, valves stopped	Start-up, stand-by, running	Valves or Traction/Pump Request	0X8130	67	248	67
WRONG RAM MEM. 05	MC is opened, EB is applied, Traction/Pump, valves stopped	Stand-by	Key re-cycle	0XFFD2	71	210	71
VDC LINK OVERV.	MC is not closed, EB is applied, Traction/Pump, valves stopped	Stand-by, running	Valves or Traction/Pump Request	0XFFCA	77	202	77
WRONG ENC SET	MC is not closed, EB is applied, Traction/Pump, valves stopped	Start-up	Key re-cycle	0XFF51	85	201	85
ANALOG INPUT	MC is opened, EB is applied, traction/pump stopped	Stand-by, running	Key re-cycle	0XFFFA	96	237	96

11.4 Analysis and troubleshooting of alarms displayed on console (Supervisor)

- **INPUT MISMATCH**
  Cause:
  The Supervisor microcontroller has different values of the inputs from the Master microcontroller.

  Troubleshooting:
  Compare the values read by Master and Slave by tester menu of console.
  Ask the assistance of a Zapi technician
  If the problem is not solved it is necessary to replace the logic board

- **W.SET. TG-EB**
  Cause:
  Supervisor uC has detected a Master uC wrong setpoint for TG or EB output

  Troubleshooting:
  - Checks the correspondence of the parameters between Master and Supervisor
  - Ask the assistance of a Zapi technician.

  If the problem is not solved it is necessary to replace the logic board

- **LOGIC FAILURE #1**
  Cause:
  This fault is displayed when the controller detects an under voltage condition, less than 8.5V, on KEY input pin CNA-1 (M1) or CNA-5 (L1).

  Troubleshooting of fault displayed at startup or in standby:
  - Key input signal down-going pulses (below under voltage threshold) due to external loads, like DC/DC converters starting-up, relays or contactor switching, solenoids energizing / de-energizing.
  - If no voltage transient is detected on the supply line and the alarm is present every time the key is switched ON, the failure is probably in the controller hardware, so it is necessary to replace the logic board.

  Troubleshooting of fault displayed during motor driving:
  - If the alarm happens during traction acceleration or driving hydraulic functions, check battery charge condition, power cable connection.

- **LOGIC FAILURE #3**
  Cause:
  Hardware problem in the logic card circuit for high current (overload) protection. An overcurrent condition is triggered even if the power bridge is not driven.

  Troubleshooting:
  This type of fault is not related to external components, so, when it is present it is necessary to replace the logic board.

- **VDC LINK OVERV.**
  Cause:
  This fault is displayed when the controller detects an over voltage condition in hardware. The threshold is normally set at 720V.
  As soon as the condition occurs the IGBT bridge is switched off and the main contactor is opened.

  Troubleshooting:
  If the alarm happens during release braking, check line contactor contact, check battery power cable connection looking for high series resistance.

- **WATCH DOG**
  Cause:
  This is a safety related test. It is a self diagnosis test within the logic between Master and Supervisor microcontrollers.

  Troubleshooting:
  This alarm could be caused by a CAN Bus malfunctioning, which blinds master-supervisor communication

- **WRONG RAM MEM. 05**
  Cause:
  The algorithm implemented to check the main RAM registers finds a wrong contents: the register is “dirty”. This alarm inhibit the machine operations.

  Troubleshooting:
  Try to switch the key off and then on, if the alarm is still present replace the logic board.

- **SP MISMATCH XX**
  Cause:
  This is a safety related test. The Master µC has detected a Supervisor µC wrong set point.

  Troubleshooting:
  - Checks the correspondence of the parameters between Master and Supervisor
  - Ask the assistance of a Zapi technician.
  - If the problem is not solved it is necessary to replace the logic board.

- **OUT MISMATCH XX**
  Cause:
  This is a safety related test. Supervisor µC has detected that the Master µC is driving traction motor in a wrong way (not correspondent to the status of operator commands).

  Troubleshooting:
  - Checks the correspondence of the parameters between Master and Supervisor
  - Ask the assistance of a Zapi technician.
  - If the problem is not solved it is necessary to replace the logic board.

- **CONTROLLER MISM.**
  Cause:
  The software is not compatible with the Hardware. Each controller produced

is “signed” during the end of line test with a specific mark code saved in eeprom according to the customized Part Number.
According with this “sign”, only the customized firmware can be uploaded.

Troubleshooting:
- Upload the correct firmware
- Ask the assistance of a Zapi technician in order to verify that the firmware is correct

- **ANALOG INPUT**
  Cause:
  This alarm occurs when the A/D conversion of the analog inputs gives frozen value, on all of the converted signals, for more than 400msec. The goal of this diagnosis is to detect a failure of the A/D converter or a problem in the code flow that omits the refreshing of the analog signal conversion.

  Troubleshooting:
  If the problem occurs permanently it is necessary to substitute logic board.

- **WRONG ENC SET**
  Cause:
  Mismatch between ENCODER PULSES 1 parameter and ENCODER PULSES 2 parameter (see paragraph 9.2.6).

  Troubleshooting:
  Set the correct parameters.

- **NO CAN MSG.**
  Cause:
  This is a safety related test. It is a self diagnosis test within the logic between Master and Supervisor microcontrollers.

  Troubleshooting:
  This alarm could be caused by a CAN Bus malfunctioning, which blinds master-supervisor communication

11.5 Microcontroller Master warnings overview

Error Code	Effect	Machine status When the test is done	Restart procedure	Indication			
				CAN OPEN CODE	MDI CODE	ZAPI CODE	LED BLINKS
WAITING FOR NODE	MC is opened, EB is applied, Traction/Pump stopped	Start-up, stand-by, running	Key re-cycle	0x0000	0	224	0
BATTERY LOW	The maximum current is reduced to half and speed is reduced (if CHECK OPTION=1)	Start-up, standby, running	Battery recharge, key re-cycle	0XFF42	0	66	0
DATA ACQUISITION	Traction is stopped	Controller calibration	Traction request	0x0000	0	247	0
CHECK UP NEEDED		Start-up, stand-by, running	Check-up done, key re-cycle	0x0000	0	249	0
WARNING SLAVE	It depends by Supervisor uC			0xFF01	1	244	1
ACQUIRING A.S.		Sensor Acquiring	Key re-cycle	0xFFAB	2	171	2
ACQUIRE END		Sensor Acquiring	Key re-cycle	0xFFAD	2	173	2
ACQUIRE ABORT				0xFFAC	2	172	2
SIN/COS D.ERR	Traction is stopped	running	Traction request	0xFFA8	3	168	3
ENCODER D.ERR	Traction is stopped	running	Traction request	0xFFA9	3	169	3
HOME SENS.ERROR	Traction is stopped	running	Traction request	0xFFAA	3	173	3
EVP DRIVER OPEN	MC is opened (the command is released), EB is applied, Traction/Pump, valves stopped	Start-up, stand-by, running	Valves Request	0xFFF8	9	240	9
EVP COIL OPEN	MC is not closed, EB is applied, Traction/Pump, valves stopped	Start-up, stand-by, running	Valves or Traction/Pump Request	0x5002	9	214	9
HW FAULT EV XX	MC is not closed , EB is applied, Traction/Pump stopped	Start-up	Key re-cycle	0x00EE	16	238	16
STALL ROTOR	Traction/Pump stopped	Start-up, stand-by, running	Key-on recycle	0xFFD3	11	211	11
EEPROM KO	Controller works using Deafult parameters	Continuous		0x3610	13	208	13
PARAM RESTORE	No effect	Start-up	Traction/ Pump request	0X000	14	209	14
COIL SHOR. EVAUX	EV stopped	EV on	Valve EV request	0xFFF1	21	241	21

Error Code	Effect	Machine status When the test is done	Restart procedure	Indication			
				CAN OPEN CODE	MDI CODE	ZAPI CODE	LED BLINKS
CONT DRIV EV XX	MC is opened (the command is released), EB is applied, Traction/Pump, valves stopped	Start-up, stand-by, running	Valves Request	0xFFE8	21	232	21
DRIV SHOR EV XX	MC is opened (the command is released), EB is applied, Traction/Pump, valves stopped	Start-up, stand-by, running	Valves or Traction/Pump Request	0xFFF9	21	234	21
PEV NOT OK	MC is not closed, EB is applied, Traction/Pump, valves stopped	Start-up, stand-by, running	MC is not closed, EB is applied, Traction/Pump, valves stopped	0xFFDB	25	217	25
HW FAULT EB 01 HW FAULT EB 02 HW FAULT EB 03	MC is closed , EB is applied, Traction/Pump stopped	Start-up	Key re-cycle	0xFFE5	34	229	34
EB. DRIV.SHRT.	MC remain closed, EB is applied (the command is released), Traction/Pump, valves stopped	Stand-by, running	Valves or Traction/Pump Request	0x3222	40	254	40
EB.DRIV.OPEN	MC remain closed, EB is applied (the command is released), Traction/Pump, valves stopped	Running	Valves or Traction/Pump Request	0x3224	42	246	42
EB. COIL OPEN	MC remain closed, EB is applied (the command is released), Traction/Pump, valves stopped	Start-up, Stand-by, running	Valves or Traction/Pump Request	0xFFD8	43	216	43
HANDBRAKE	Traction/ Pump motor is stopped	Start-up, stand-by, running	Traction/ Pump request	0xFFDD	46	221	46
THROTTLE PROG.	MC remain closed, EB is applied (the command is released), Traction stopped	Start-up, Stand-by,	Valves or Traction/Pump Request	0xFFF3	48	243	48
LIFT + LOWER	Pump is stopped	Start-up, stand-by, Running	Pump request	0xFFBB	49	187	49
TILLER OPEN	MC opens	Start-up, stand-by, running	Valves or Traction/Pump Request	0x0000	51	228	51
WRONG ZERO	valve, pump, traction stopped, Lc opened, Eb applied	Start-up	Valves or Traction/ Pump request	0x3210	58	252	58
THERMIC SENS. KO	The maximum current is reduced to half and speed is reduced	Start-up, stand-by, running		0x4211	61	250	61
TH. PROTECTION	Traction controller reduces the max current linearly from Imax (85°C) down to 0A (105°C)	Start-up, stand-by, running		0x4210	62	62	62
BRAKE RUN OUT	No effect, the warning is only displayed through the console	Continuous	or Traction/Pump Request	0xFFCC	63	204	63

Error Code	Effect	Machine status When the test is done	Restart procedure	Indication			
				CAN OPEN CODE	MDI CODE	ZAPI CODE	LED BLINKS
MOTOR TEMPERAT.	The maximum current is reduced to half and speed is reduced	Start-up, stand-by, running		0x4110	65	65	65
MOTOR TEMP. STOP	MC stays closed, EB is applied, Traction/Pump, valves stopped	Continuous		0xFFB2	65	178	65
SENS MOT TEMP KO	The maximum current is reduced to half and speed is reduced	Start-up, stand-by, running		0x4311	68	218	68
EPS RELAY OPEN	Traction/ Pump motor is stopped	Start-up, stand-by, Running	Valves or Traction/ Pump request	0xFFCD	70	205	70
VACC NOT OK	Traction/ Pump motor is stopped	Start-up, stand-by, running	Traction/ request	0xFF4E	78	78	78
INCORRECT START	Traction/ Pump motor is stopped	Start-up, stand-by	Traction request	0xFF4F	79	79	79
FORW + BACK	Traction is stopped	Start-up, stand-by, running	Traction request	0xFF50	80	80	80
VACC OUT OF RANGE	Traction/ Pump motor is stopped	Start-up, Stand-by, Running	Traction/ Pump request	0xFFE2	85	226	85
PEDAL WIRE KO	Traction is stopped	Start-up, Stand-by, Running	Traction request	0xFF56	86	86	86
WRONG SLAVE VER	MC opened, EB is applied, Traction/Pump, valves stopped	Start-up	Key re-cycle	0Xffc5	91	197	91
CURRENT GAIN	Controller works, but with low maximum current	Start-up, stand-by		0x6302	92	236	92
PARAM TRANSFER	MC stays closed, EB is applied, Traction/Pump, valves stopped	Continuous	Key re-cycle	0xFFC7	93	199	93
Reload HM		Start-up	Key re-cycle	0x0000	94	0	94
STEER SENS KO	EB is applied, traction/pump stopped	Continuous	Key re-cycle	0xFFB3	95	179	95
M/S PAR CHK MISM	MC stays closed, EB is applied, Traction/Pump, valves stopped	Start-up	Save again the parameter and Key re-cycle	0xFFC6	97	198	97
TORQUE PROFILE	EB is applied, Traction/Pump motor is stopped	Start-up, stand-by	Valves or Traction/ Pump request	0xFFC9	98	201	98

11.6 Analysis and troubleshooting of warnings displayed on console (Master)

- **WAITING FOR NODE**
  Cause:
  The controller receives from the CAN the message that another controller in the net is in fault condition; as a consequence the controller itself cannot enter an operative status, but has to WAIT for the other controller

coming out from the fault status.

- **TORQUE PROFILE**
  Cause:
  There is an error on the choice of the parameters of the torque profile.

  Troubleshooting:
  Check in the hardware setting menu the value of those parameter.

- **DATA ACQUISITION**
  Cause:
  Controller in calibration state.

  Troubleshooting:
  The alarm ends when the acquisition is done.

- **ACQUIRING A.S.**
  Cause:
  Controller is acquiring the absolute feedback sensor.

  Troubleshooting:
  The alarm ends when the acquisition is done.

- **ACQUIRE END**
  Cause:
  Absolute feedback sensor acquired. A re-cycle of the KEY input is needed

- **ACQUIRE ABORT**
  Cause:
  Absolute feedback sensor acquiring aborted.

- **SIN/COS D.ERR**
  Cause:
  This alarm occurs only when the controller is configured as PMSM and the feedback sensor selected is Sin / Cos.
  The Sin/Cos signal direction is not correct.

  Troubleshooting:
  - Check the wiring
  - If the motor direction is correct swap the sin and cos signals. As a last resource, there's a special parameter in the HW SETTING menù. Ask a Zapi technician before modify this parameter.
  - If the motor direction is not correct swap the motor phases. As a last resource, there's a special parameter in the HW SETTING menù. Ask a Zapi technician before modify this parameter.

  If the problem is not solved contact Zapi Technician.

- **ENCODER D.ERR**
  Cause:
  This alarm occurs only when the controller is configured as PMSM and the feedback sensor selected is Encoder.
  The A and B signals direction is not correct.

  Troubleshooting:
  - Check the wiring

- If the motor direction is correct swap A and Bsignals
- If the motor direction is not correct swap the motor phases

If the problem is not solved contact Zapi Technician.

- **HOME SENS.ERR**

  Cause:

  This alarm occurs only when the controller is configured as PMSM and the feedback sensor selected is Encoder ABI.
  The home signal is not present (diagnosis made for speed higher than 20Hz).

  Troubleshooting:
  - Check the wiring
  - Verify if the sensor works properly

  If the problem is not solved the failure is in the logic board, replace it.

- **BRAKE RUN OUT**

  Cause:

  The CPOTBRAKE input read by the microcontroller is at the maximum value without the hand brake request.

  Troubleshooting:

  Check the mechanical calibration and the functionality of the brake potentiometer.
  If the alarm is not disappeared the failure is in the logic board, replace it.

- **RELOAD HM FROM MDI**

  Cause:

  The HourMeter of the Controller is transferred and recorded on the HourMeter of the Standard MDI.

- **CHECK UP NEEDED**

  Cause:

  This is just a warning to call for the time programmed maintenance.

  Troubleshooting:

  It is just enough to turn the CHECK UP DONE option to level ON after the maintenance is executed.

- **PARAM TRANSFER**

  Cause:

  Master uC is transferring parameter to Supervisor.

  Troubleshooting:

  Wait for the end of procedure.
  If the alarms doesn't disappear try to re-cycle the key.

- **THROTTLE PROG.**

  Cause:

  A wrong profile in the Throttle map has been set.

  Troubleshooting:
  - Set properly the Throttle parameters (see paragraph 10.6).

- **INCORRECT START**

  Cause:

This is a warning for an incorrect starting sequence. Possible reasons are (use the readings in the TESTER to facilitate the troubleshooting):

- A travel demand active at key on
- Presence man sensor active at key on

Troubleshooting:
The Check the wirings

- Check the microswitches for failures.
- It could be also an error sequence made by the operator. A failure in the logic is possible too; so when all of the above conditions were checked and nothing was found, replace the logic board.

- **VACC OUT RANGE**
  Cause:
  The CPOT input read by the microcontroller is not comprised in the range Vacc_min ÷ Vacc_max, programmed through the “PROGRAMM VACC” function (see paragraph 10.1).

  Troubleshooting:
  - Acquire the maximum and minimum potentiometer value through the PROGRAM VACC function. If the alarm is still present, check the mechanical calibration and the functionality of the potentiometer.
  - If the problem is not solved it is necessary to replace the logic board.

- **PEDAL WIRE KO**
  Cause:
  This alarm appears if a fault is detected in the accelerator unit wiring (only if A30 is used and properly configured as NPOT).

  Troubleshooting:
  - Check the pedal connection
  - If the problem is not solved it is necessary to replace the logic board.

- **FORW + BACK**
  Cause:
  This alarm occurs when both the travel demands (Fwd and Bwd) are active at the same time.

  Troubleshooting:
  - Check the wiring of the Fwd and Rev travel demand inputs (use the readings in the TESTER to facilitate the troubleshooting).
  - Check the microswitches for failures.
  - A failure in the logic is possible too. So, when you have verified the travel demand switches are fine working and the wiring is right, it is necessary to replace the logic board.

- **M/S PAR CHK MISM**
  Cause:
  At Start-up, there is a mismatch in the parameter checksum between the Master and the Slave.

  Troubleshooting:
  Try to restore and save again the parameters list.

- **TILLER OPEN**
  Cause:
  When the tiller is released, after a fixed period of time of standby (30 seconds) the main contactor open.

  Troubleshooting:
  At the next travel request the warning disappear.

- **CURRENT GAIN**
  Cause:
  The Maximum current gain parameters are at the default values, which means the maximum current adjustment procedure has not been carried out yet.

  Troubleshooting:
  Ask the assistance of a Zapi technician to do the correct adjustment procedure of the current gain parameters**.**

- **COIL SHOR. EVAUX**
  Cause:
  This alarm occurs when there is an overload of one or more EV driver. As soon as the overload condition has been removed, the alarm exits automatically by releasing and then enabling a travel demand.

  Troubleshooting:
  - The typical root cause is in the harness or in the load coil. So the very first check to carry out concerns connections between controller outputs and loads.
  - Collect information about characteristics of the coils connected to the two driver and ask the assistance of a Zapi technician in order to verify that the limit of the hardware is not exceeded .
  - In case no failures/problems have been found, the problem is in the controller, which has to be replaced.

- **LIFT+LOWER**
  Cause:
  This alarm occurs when both the travel demands (Lift and Lower) are active at the same time.

  Troubleshooting:
  - Check the wiring of the Lift and Lower travel demand inputs (use the readings in the TESTER to facilitate the troubleshooting).
  - Check the microswitches for failures.
  - A failure in the logic is possible too. So, when you have verified the travel demand switches are fine working and the wiring is right, it is necessary to replace the logic board.

- **WRONG SLAVE VER.**
  Cause:
  Wrong software version on Supervisor uC

  Troubleshooting:
  Upload the correct software version or ask the assistance of a Zapi technician

- **SENS MOT TEMP KO**
  Cause:
  The output of the motor thermal sensor is out of range.

  Troubleshooting:
  - Check the sensor ohmic value and the sensor wiring.
  - If the sensor is OK, then the problem is inside the logic board, replace it.

- **HW FAULT EV 01/02/04/08/10/20/40/80**
  Cause:
  The controller checks, at each start up that the hardware circuit for enabling and disabling of EV drivers by Supervisor uC works properly.
  The number indicates the source of the failure

  Troubleshooting:
  This type of fault is not related to external components, so, when it is present it is necessary to replace the logic board.

- **WRONG ZERO**
  Cause:
  The outputs of the amplifiers (used to measure the motor voltage) are cheked this alarm occurs when voltage signals >3V o <2V at the init

  Troubleshooting:
  This type of fault is not related to external components; replace the logic board

- **VACC NOT OK**
  Cause:
  The test is made at key-on and immediately after that both the travel demands have been turned off. This alarm occurs if the ACCELERATOR reading in the TESTER menù is 1V higher than PROGRAM VACC min acquisition when the accelerator is released.

  Troubleshooting:
  - Check the wirings
  - Check the mechanical calibration and the functionality of the potentiometer
  - Acquire the maximum and minimum potentiometer value through the PROGRAM VACC function.
  - If the alarm is not disappeared the failure is in the logic board, replace it.

- **STALL ROTOR**
  Cause:
  The traction rotor is stuck or the encoder signal is not correctly received by the controller.

  Troubleshooting:
  Please check if the sign of FREQUENCY and ENCODER on the tester menu are the same and different than zero during a traction request.

- **PARAM RESTORE**
  Cause:
  This warning appears when the controller restored the default values.

Troubleshooting:
If a CLEAR EEPROM has been made before the last keyon-recycle, this warning just means that the EEPROM was correctly cleared. A travel demand or a pump request cancel the alarm. If this alarm appears at keyon without any CLEAR EEPROM request by the operator, there could be a problem inside the controller

- **CONT DRIV EV 02/04/08/20**
  Cause:
  One or more on/off valve drivers is not able to drive the load (cannot close).
  - 02 → EV1
  - 04 → EV2
  - 08 → EV3
  - 20 → EV4

  If more than one output is affected by the this fault condition the codes shown will correspond to the sum of the single codes.

  Troubleshooting:
  The device or its driving circuit is damaged, replace the controller.

- **DRIV SHORT EV 02/04/08/20**
  Cause:
  One or more on/off valve driver is shorted.
  - 02 → EV1
  - 04 → EV2
  - 08 → EV3
  - 20 → EV4

  If more than one output is affected by the this fault condition the codes shown will correspond to the sum of the single codes.

  Troubleshooting:
  - Check if there is a short or a low impedance between the negative of the coil and -BATT.
  - If the problem is not solved it could be necessary to replace the controller.

- **EEPROM KO**
  Cause:
  It's due to a HW or SW defect of the non-volatile embedded memory supporting the controller parameters. This alarm does not inhibit the machine operations, but the truck will work with the default values.

  Troubleshooting:
  Try to execute a CLEAR EEPROM operation (refer to Console manual). Switch the key off and on to check the result. If the alarm occurs permanently, it is necessary to replace the controller. If the alarm disappears, the previously stored parameters will have been replaced by the default parameters

- **WARNING SLAVE**
  Cause:
  Warning on supervisor uC

  Troubleshooting:
  Connect to Supervisor uC and check what alarms is shown

- **TH. PROTECTION**
  Cause:
  This warning message indicate that maximum current is limited under the nominal value because of high temperature in the IGBT module.
  Current derating in HV Inverter depends on IGBT temperature and output frequency, when it's very low current limiting is bigger.
  For output frequency greater than 10Hz, current start decreasing at 70° and reach zero at 120°

  Troubleshooting:
  If the warning message appears after a significative time that HV Inverter is working, is not really a problem, but just an information. If condition is reached soon, perhaps it's necessary to improve controller cooling. To realize an adequately cooling in case of finned heat sink are important factor the flux [m3/h] and temperature [°C] of cooling air. For liquid cooling, it's necessary that flux is at the nominal speed and temperature. In case of thermal dissipation realized with the controller base plate installed on truck frame it is important the thickness of frame and the planarity and roughness of its surface, normally is not allowed for HV Inverter. If the alarm is signaled when the controller is cold, the possible reasons are a thermal sensor failure or a failure in the logic card. In this case, it is necessary to replace the controller.

- **MOTOR TEMPERAT.**
  Cause:
  This warning occurs when the temperature sensor is opened (if digital) or has overtaken the threshold of "MAX MOTOR TEMP" (if analogue) (see paragraph 9.2.2).
  There are up to two motor sensors in the HV Inverter

  Troubleshooting:
  - Check the thermal sensor inside the motor (use the MOTOR TEMPERATURE reading in the TESTER menu).
  - Check the sensor ohmic value and the sensor wiring.
  - If the sensor is OK, improve the cooling of the motor.
  - If the warning is present when the motor is cool, then the problem is inside the controller.

- **THERMIC SENS. KO**
  Cause:
  The output of the controller thermal sensor is out of range.

  Troubleshooting:
  This type of fault is not related to external components; replace the controller.

- **MOTOR TEMP. STOP**
  Cause:
  This warning occurs when the temperature sensor is has overtaken the threshold of "TEMP.MOT.STOP". (if analogue, see paragraph 9.2.2).

  Troubleshooting:
  - Check the thermal sensor inside the motor (use the MOTOR TEMPERATURE reading in the TESTER menu).
  - Check the sensor ohmic value and the sensor wiring.
  - If the sensor is OK, improve the cooling of the motor.

- If the warning is present when the motor is cool, then the problem is inside the controller.

- **BATTERY LOW**

  Cause:

  It occurs when the battery charge is calculated being less than or equal to 10% of the full charge and the BATTERY CHECK setting is other than 0 (refer to SET OPTION menu).

  Troubleshooting:

  - Get the battery charged
  - If it doesn't work, measure with a voltmeter the battery voltage and compare it with the value in the BATTERY VOLTAGE parameter. If they are different adjust the value of the ADJUST BATTERY function connected to the right connector pin and if it is not interrupted.
  - If the alarm is not disappeared the failure is in the ACE logic board, replace it.

11.7 Microcontroller Supervisor warnings overview

Error Code	Effect	Machine status When the test is done	Restart procedure	Indication			
				CAN OPEN CODE	MDI CODE	ZAPI CODE	LED BLINKS
EEPROM KO	Controller works using Default parameters	Continuous		0x3610	13	208	13
PARAM RESTORE	No effect	Start-up	Traction/ Pump request	0X000	14	209	14
STEER SENS KO	EB is applied, traction/pump stopped	Continuous	Key re-cycle	0xFFB3	95	200	95

11.8 Analysis and troubleshooting of warnings displayed on console (Supervisor)

- **PARAM RESTORE**
  Cause:
  This warning appears when the controller restored the default values.

  Troubleshooting:
  If a CLEAR EEPROM has been made before the last keyon-recycle, this warning just means that the EEPROM was correctly cleared. A travel demand or a pump request cancel the alarm. If this alarm appears at keyon without any CLEAR EEPROM request by the operator, there could be a problem inside the controller

- **EEPROM KO**
  Cause:
  It's due to a HW or SW defect of the non-volatile embedded memory supporting the controller parameters. This alarm does not inhibit the machine operations, but the truck will work with the default values.

  Troubleshooting:
  Try to execute a CLEAR EEPROM operation (refer to Console manual). Switch the key off and on to check the result. If the alarm occurs permanently, it is necessary to replace the controller. If the alarm disappears, the previously stored parameters will have been replaced by the default parameters

12 RECOMMENDED SPARE PARTS FOR INVERTER

Part number	Description
C16500	10A 20 mm Control Circuit Fuse

13 PERIODIC MAINTENANCE

Check the wear and condition of the Contactors' moving and fixed contacts. Electrical Contacts should be checked every **3 months.**

Check the Foot pedal or Tiller microswitch. Using a suitable test meter, confirm that there is no electrical resistance between the contacts by measuring the volt drop between the terminals. Switches should operate with a firm click sound. Microswitches should be checked every **3 months.**

Check the Battery cables, cables to the inverter, and cables to the motor. Ensure the insulation is sound and the connections are tight.
Cables should be checked every **3 months.**

Check the mechanical operation of the pedal or tiller. Are the return springs ok. Do the potentiometers wind up to their full or programmed level.
Check every **3 months.**

Check the mechanical operation of the Contactor(s). Moving contacts should be free to move without restriction.
Check every **3 months.**

Checks should be carried out by qualified personnel and any replacement parts used should be original. Beware of NON ORIGINAL PARTS.
The installation of this electronic controller should be made according to the diagrams included in this Manual. Any variations or special requirements should be made after consulting a Zapi Agent. The supplier is not responsible for any problem that arises from wiring methods that differ from information included in this Manual.

During periodic checks, if a technician finds any situation that could cause damage or compromise safety, the matter should be bought to the attention of a Zapi Agent immediately. The Agent will then take the decision regarding operational safety of the machine.

Remember that Battery Powered Machines feel no pain.

NEVER USE A VEHICLE WITH A FAULTY ELECTRONIC CONTROLLER.



IMPORTANT NOTE ABOUT WASTE MANAGEMENT:
This controller has both mechanical parts and high-density electronic parts (printed circuit boards and integrated circuits). If not properly handled during waste processing, this material may become a relevant source of pollution. The disposal and recycling of this controller has to follow the local laws for these types of waste materials.
Zapi commits itself to update its technology in order to reduce the presence of polluting substances in its product.

14 APPENDICES

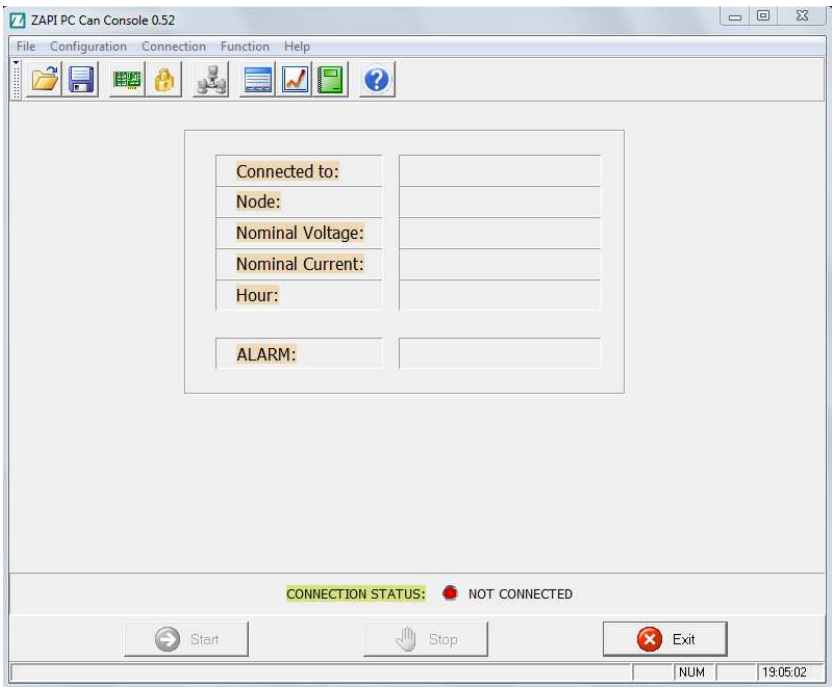
The goal of this chapter is to give to the operator a general overview relating the use of Zapi PC CAN Console and Zapi Smart Console.
The description contained in the next paragraph focuses on the basic information about the connection and change of parameters.
For additional functionalities available for both tools it is suggested to contact Zapi technicians in order to received more detailed information or dedicated documentation.

14.1 Appendix A: PC CAN Console user guide

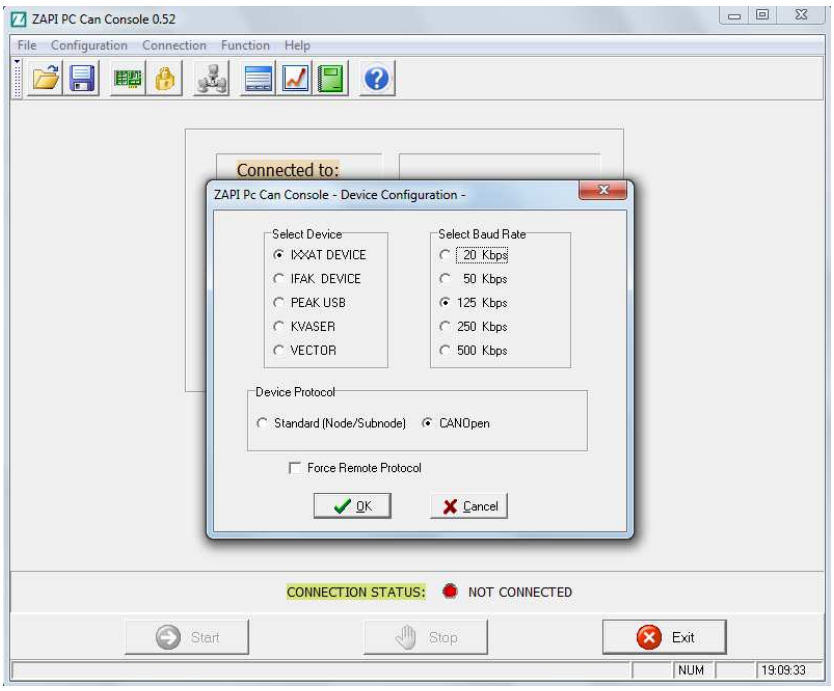
Windows PcConsole uses standard Zapi communication protocol to display chopper's information. It provides all standard Zapi Console functions with the easier handling of Windows devices. Besides, PcConsole offers another function: the possibility to save parameter configurations to a file and to restore them to the chopper.
Before running PcConsole, user must install it launching "setup.exe".

14.1.1 PC CAN Console configuration

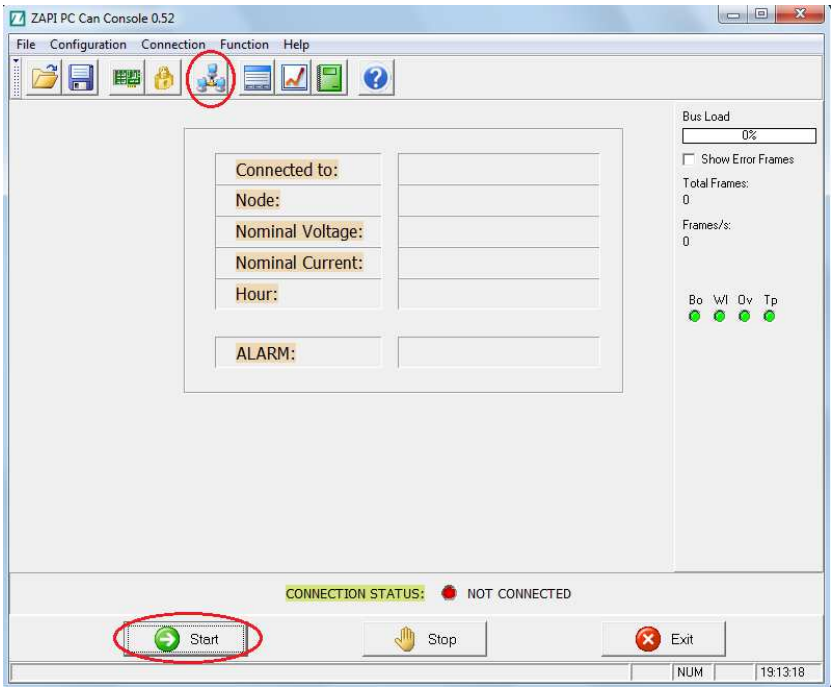
Running the PC Can Console software, the following form will appear:



the first step to accomplish is to define the CAN device attached to the PC, so select the "Configuration" (Alt-C) -> Can Device (Ctrl-C) menu or click on Can Device icon.



From this form you can define the CNA-device used (IXXAT or IFAK or Peak) and the CAN communication speed then press the OK button.
Once you have define the CAN interface, you have to choose which CNA-device you need to connect to, then choose “Connection” -> “Set Node” menu (or push the “Set Node” icon).



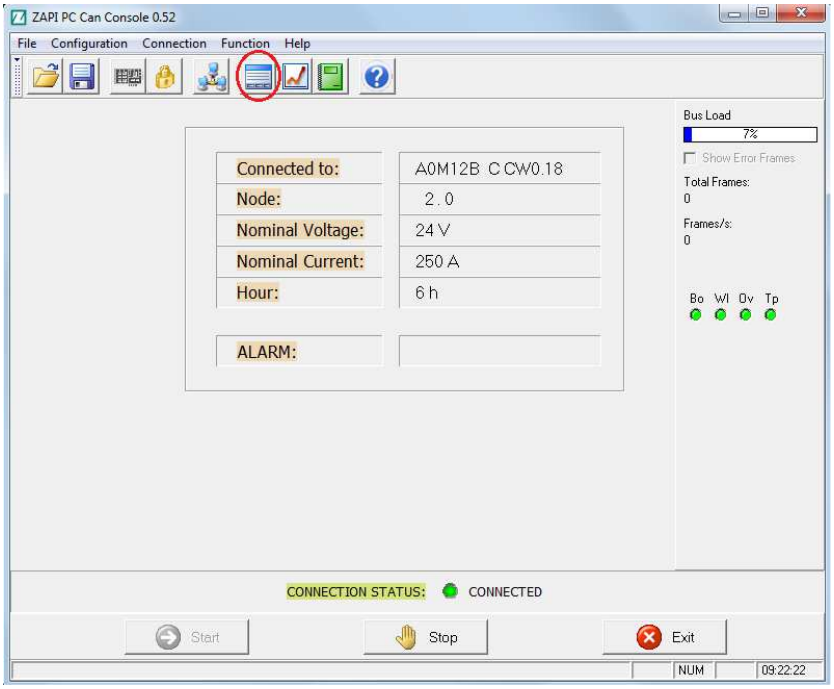
Once you have chosen the node, start the connection and insert the password in order to have the possibility to change the parameters.
Choose “Configuration” -> “Enter Password”.



Write the Password -> “ZAPI”

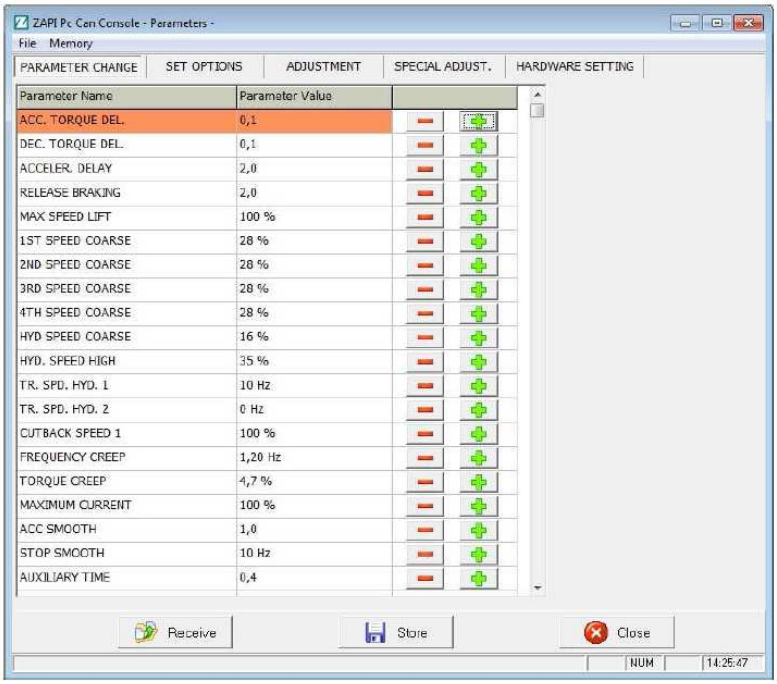
14.1.2 Parameter download

Once you are connected you need to download the parameter; choose “Function” > “Parameter” menu (or push the “Parameter” icon).



and then click on Receive button. The parameters will be downloaded automatically.
When the device as finish to send the device parameter you can change they.

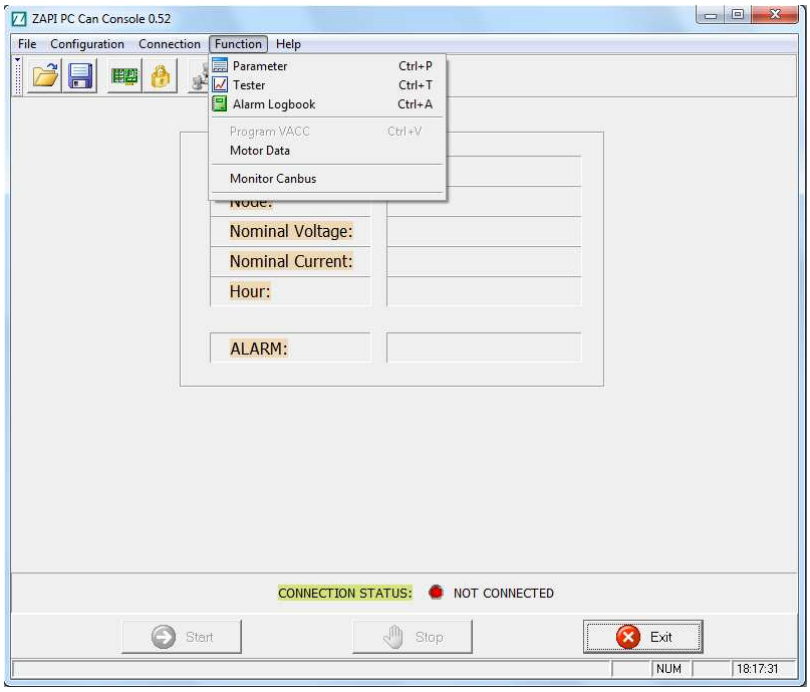
14.1.3 How to modify the parameters

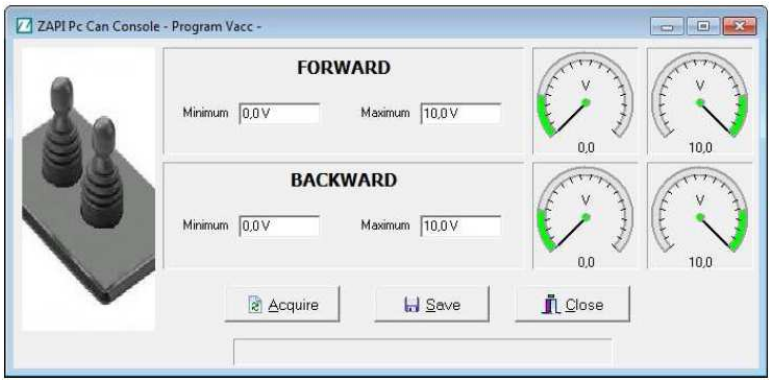


Once you are selected the menu and the parameter to change, using the “+” and “–“ button it is possible to modify the parameter value.
Click on the button “Store” to save the change on EEPROM.

14.1.4 Program Vacc

Choose “Function” > “Program VACC” menu.





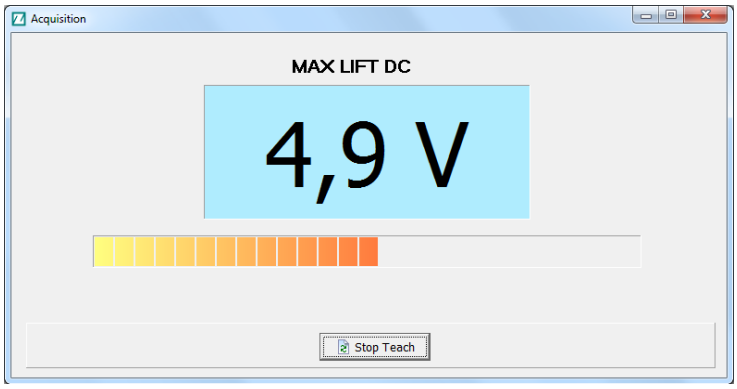
When “Acquire” is pressed PROGRAM VACC procedure will start:

- To select the Enable switch, if any
- To select the direction switch (either forward or backward)
- To press the pedal up to its maximum excursion.

Displayed values will vary accordingly to operator inputs.

14.1.5 Lift & Lower command acquiring

Once you are connected you need to download the parameter; choose “Function” > “Parameter” menu (or push the “Parameter” icon).
Choose “Adjustment” menu.
Select the value to acquire pressing “acquiring” button and the acquisition will start:



- Select the Enable switch, if any
- Select the command switch (either lift or lower)
- Move the command sensor (lift/lower potentiometer) to the correct position according to what you are acquiring
- Press “Stop Teach” command

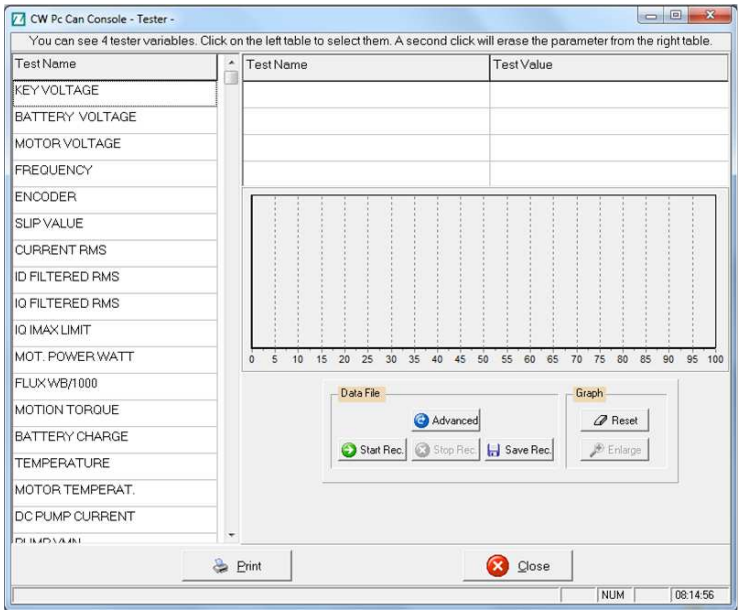
The procedure is the same for all the lift and lower command values.

14.1.6 Steer acquiring

Once you are connected you need to download the parameter; choose “Function” > “Parameter” menu (or push the “Parameter” icon).
Choose “Adjustment” menu.
Select the value to acquire pressing “acquiring” button and the acquisition will start: the procedure is the same described for Lift & Lower commands at paragraph 14.1.5

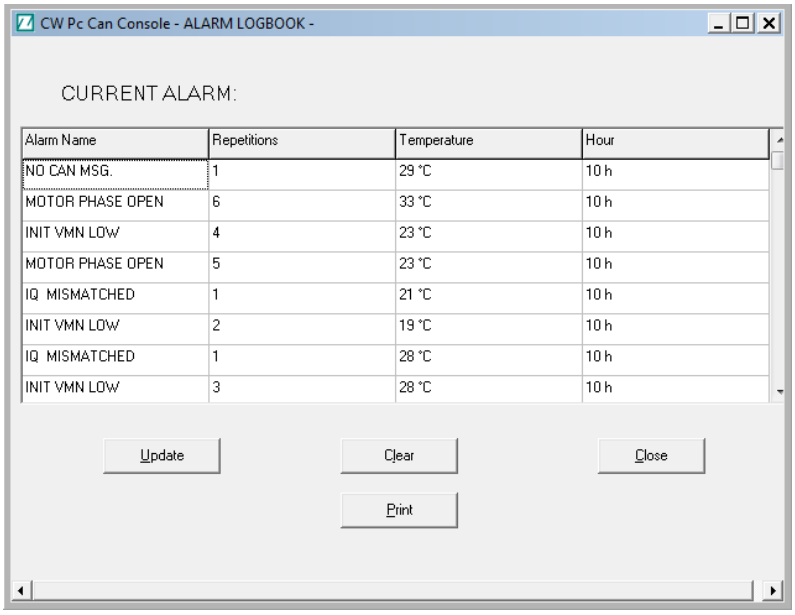
14.1.7 Tester Functionality

From the main page you can also access to the chopper TESTER menu from the Function (Alt-u)->Tester Ctrl-T menu where you can check some inverter information



14.1.8 Alarm Logbook

This window will display the alarm stored in the controller.
For every alarm are related the working hour, motor temperature and the number of repetitions.



Four buttons are present:
Update →user can update alarm logbook,
Clear →user clear alarm logbook on chopper EEPROM,
Close →closes the window,

Print →prints alarm logbook data on the selected printer.

14.2 Appendix B: Zapi Smart Console user guide



14.2.1 Operational Modes

The Smart Console has been designed to have multiple ways of operation. Three modes can be identified:

- Serial connection powered by four standard AA size batteries placed in the battery holder of the console.
- CAN BUS connection powered by four standard AA size batteries placed in the battery holder of the console.
- CAN BUS connection with Smart Console supplied by an external dc source. This source may be a standard battery (lead-acid or other type) or a dc/dc converter

Current-loop serial connection

The Smart Console offers the same serial connection as the well-known Console Ultra.
Main characteristics of this operational mode:

- Current-loop serial communication
- Console is connected to a *single* controller only (even if Remote Console option is available)
- Baud-rate selectable
- Zapi can provide the serial cable compatible with Molex SPOX connector used in Console Ultra .

CAN BUS connection

The Smart Console can connect to an existing CAN line and connect with any Zapi controller inside this line.
Main characteristics of this operational mode:

- It can be connected to a CAN line composed of any combination of modules, both Zapi ones and not-Zapi ones
- Supported speeds: 125, 250, 500kbps
- It sees the entire CAN line and all CAN modules.

14.2.2 The keyboard

The keyboard is used to navigate in the different menu. It features some key with special functions and a green led.
Different button functions are shown below.

UP and DOWN keys
In most cases a menu is a list of items: these items are ordered in rows. Selected item is highlighted in light blue .
Up and down keys are used to move the selection up and down: in other words they are used to “scroll”.

LEFT and RIGHT keys
Normally used to increase and decrease the value associated with the selected item inside a menu.

OK and ESC keys
OK key is used either to confirm actions or to enter a submenu.
ESC is used either to cancel an action or to exit a menu.

F1, F2, F3 keys
These buttons have a contextual use. The display will say which F button can be used and its function.

ON key
Used while operating with internal batteries.

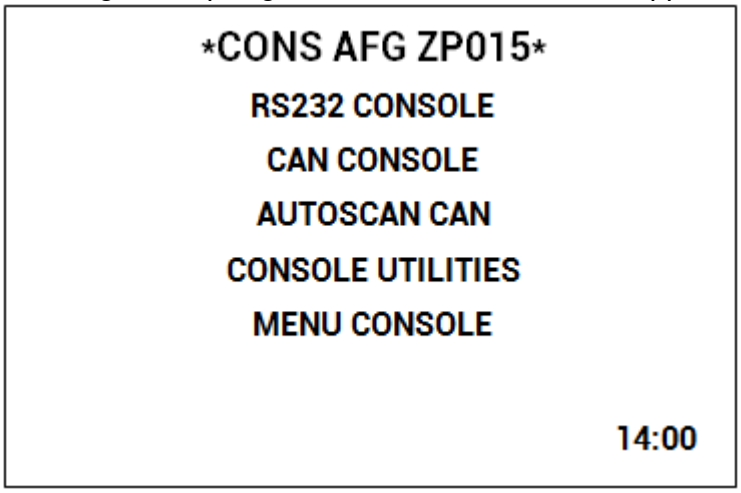


While the Smart Console is powered from external sources on pin CNX8 the button ON is deactivated regardless of the presence of the batteries.

Green LED
When the console is powered on and running the green led is on.
Green led can blink in certain cases which will be described better in following sections.

14.2.3 Home Screen

After showing the Zapi logo, the HOME SCREEN will appear on the display:



From top:

- First line tells which firmware version is running inside the console, in this case ZP 0.15
- RS232 Console: enter this menu to start a serial connection as in the Console Ultra
- CAN Console: enter this menu to start a CAN connection
- AUTOSCAN CAN: another way to start a CAN connection
- Console Utilities and Menu Console: ignore them at the moment
- The current hour is shown in the bottom right.

Moreover the green led must be turned on and still.

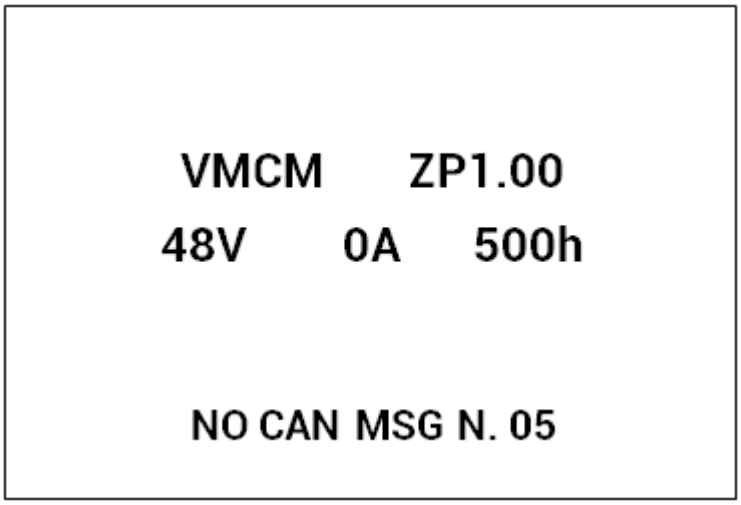
The “RS232” line is already highlighted so press OK key.
Display prompts a message to inform you that a connection attempt is ongoing.

If serial connection fails a “NO COMMUNICATION” warning will be shown after some second: press ESC key and look for what is preventing the connection.

Please notice the red dot appearing in the top right of the display every time you press a button. It indicates that the console has received the command and it is elaborating the request. If the red dot does not appear when a button is pressed, there is probably a failure inside the keyboard or the console has stalled.

14.2.4 Connected

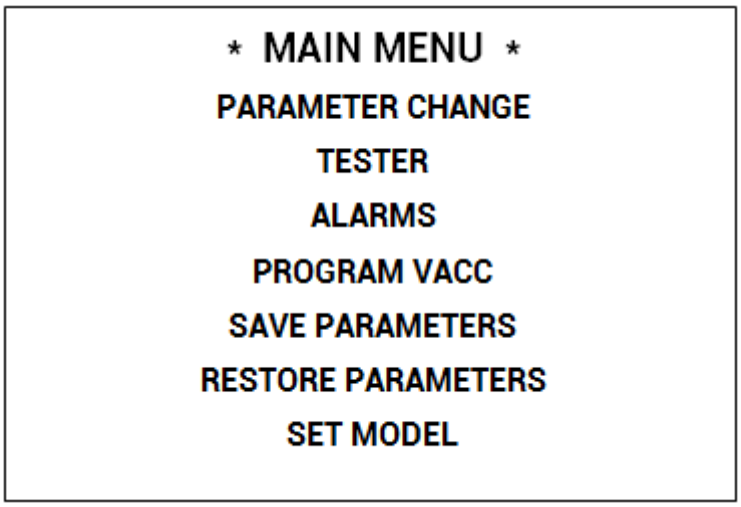
If connection is successful the display will show an image similar to the next one.



This menu shows basic information about the controller, in a way similar to console Ultra.

- First line tells the controller firmware
- Second line shows controller voltage, controller current and hourmeter
- Last line shows the current alarm code, if present

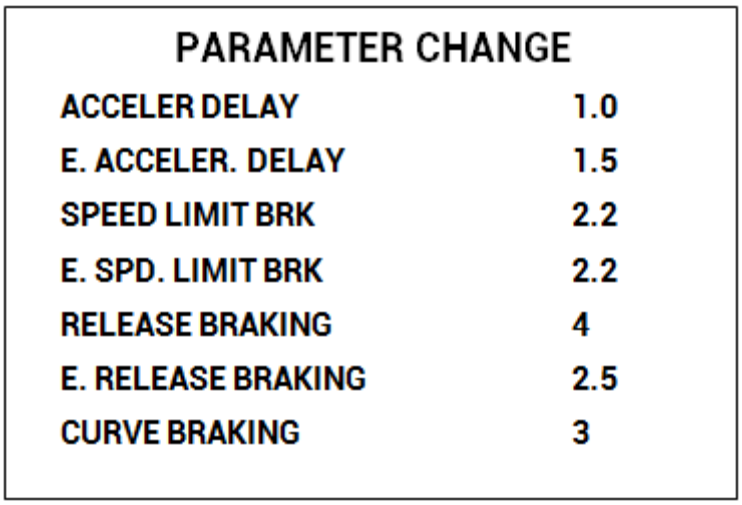
Press OK to access the MAIN MENU



MAIN MENU contains the complete list of menus available in the controller. Contrary to Console Ultra there are no “hidden” menus which must be reached by pressing many buttons at once: now all menus are visible.
Use UP and DOWN keys to navigate the list: once you find the desired menu press OK to enter it.

14.2.5 How to modify a parameter

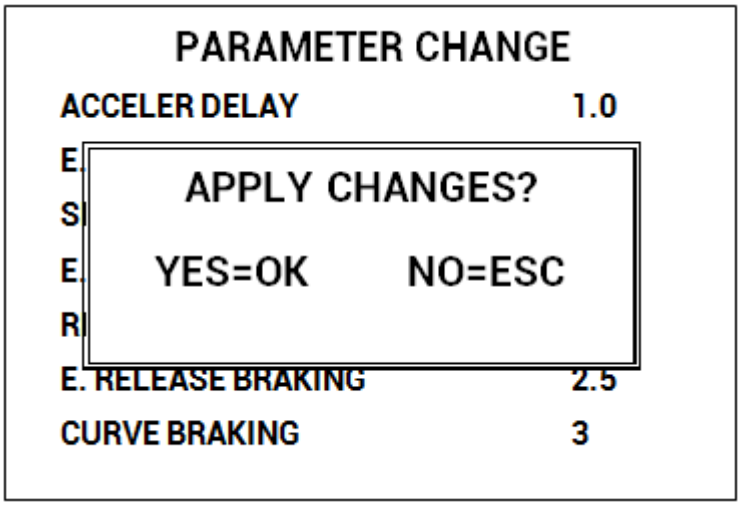
From MAIN MENU enter the PARAMETER CHANGE menu.



With UP and DOWN keys you can scroll the whole list: once you have highlighted a parameter that you want to modify, use LEFT or RIGHT keys to decrease or increase the parameter value.

Keep LEFT/RIGHT button pressed to continuously repeat the value modification (“autorepeat” function): this function will speed up the procedure in case many parameter values must be changed.

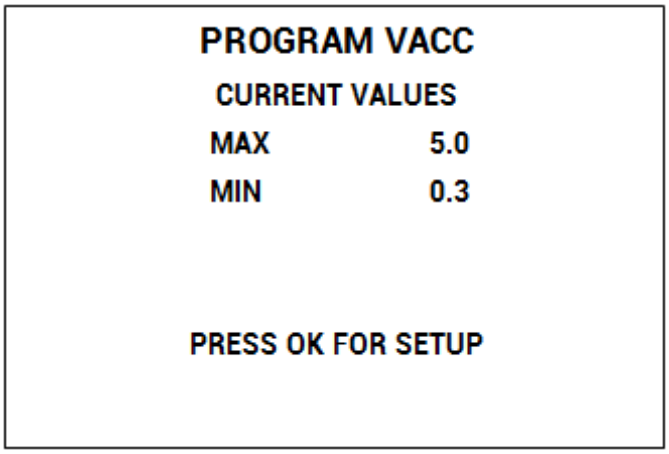
You can press ESC to exit the menu at any time. In case some parameter has been modified, the console will prompt a request to confirm/discard changes.



Description above is valid for every menu which contains parameters and options like SET OPTIONS, ADJUSTMENTS, HARDWARE SETTINGS, etc.

14.2.6 Program VACC

Program VACC menu has been slightly modified compared with old console. Upon entering this menu the console will show the current programmed values.

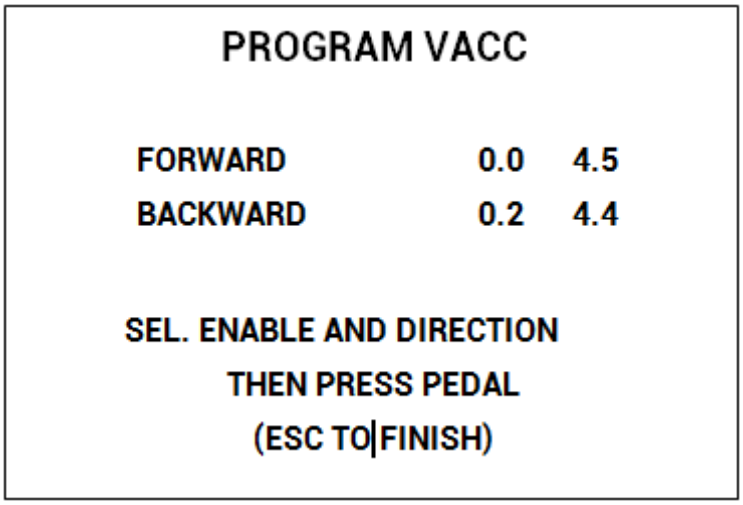


When OK is pressed PROGRAM VACC procedure will start: console will invite you:

- Select the Enable switch, if any
- Select the direction switch (either forward or backward)
- Press the pedal up to its maximum excursion.

Displayed values will vary accordingly to operator inputs.

Sequence above can slightly vary depending on controller firmware. Anyway the logic will remain the same: before programming the min/max values, execute any tarting sequence which is necessary, then press the pedal/push the joystick.



When ESC is pressed, console will ask if programmed values must be saved or discarded.

14.2.7 Lift and Lower commands acquiring

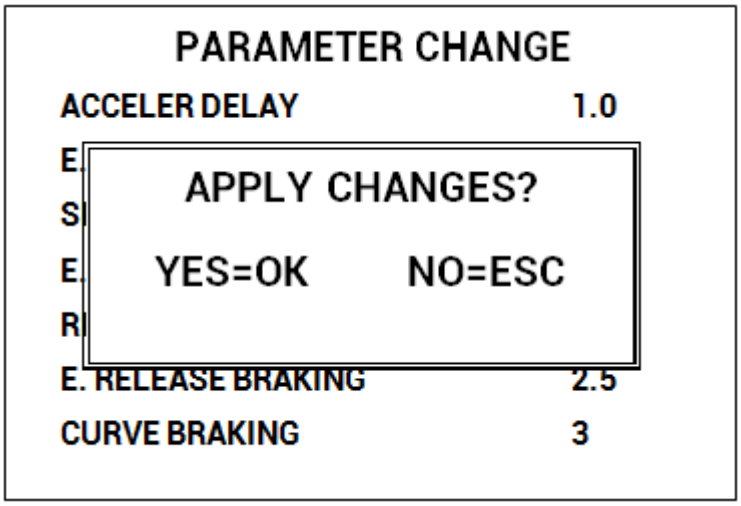
From MAIN MENU enter the Adjustment menu.
With UP and DOWN keys you can scroll the whole list: once you have highlighted a value that you want acquire press OK.
When OK is pressed the procedure will start:

- Select the Enable switch, if any
- Select the command switch if any (either lift or lower)
- Move the command sensor (lift/lower potentiometer) to the correct position according to what you are acquiring.

Displayed values will vary accordingly to operator inputs.

Sequence above can slightly vary depending on controller firmware. Anyway the logic will remain the same: before programming the min/max values, execute any tarting sequence which is necessary, then press the pedal/push the joystick.

It is possible step by step acquire all the value in only one session.
At the end you can press ESC and the console will prompt a request to confirm/discard changes.

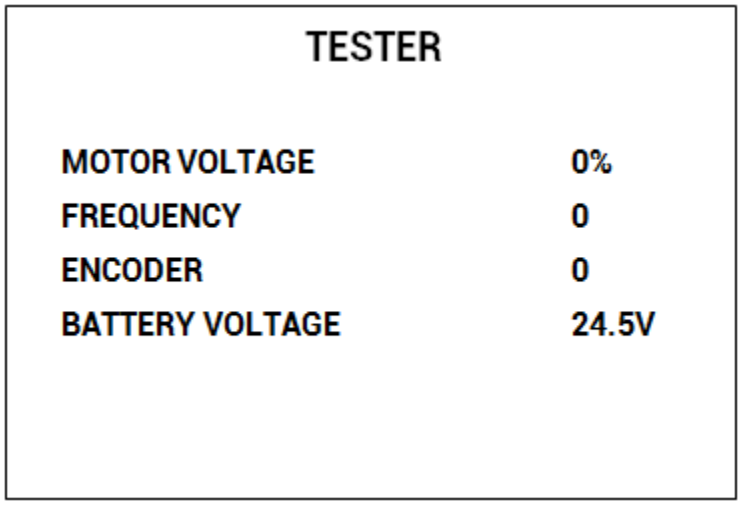


14.2.8 Steer command acquiring

From MAIN MENU enter the Adjustment menu.
The procedure to follow is the same described in paragraph 14.2.7.

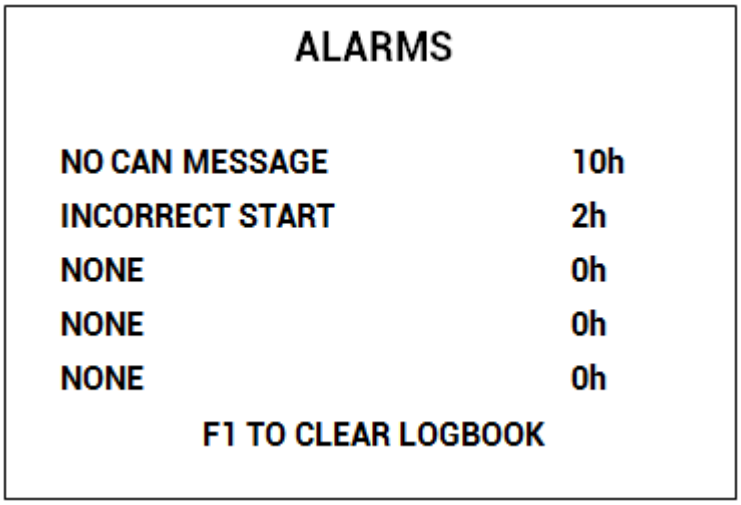
14.2.9 Tester

Compared to standard console Ultra, the TESTER menu has been modified deeply. Now it shows four variables at once: use UP/DOWN keys to scroll the list as usual.



14.2.10 Alarms

ALARMS menu has changed from Console Ultra. Display shows all controller alarms at once.



Five is the maximum number of alarm codes which is stored inside the controller.

Colours are used to separate recurrent alarm codes from rare events. In order of increasing frequency alarm names can be:

- White: up to 5 occurrences
- Yellow: up to 20,
- Orange: up to 40,
- Red: more than 40.

Use UP/DOWN to select a certain alarm in the list: if OK is pressed, additional information about that alarm will be displayed.

Press F1 to cancel the alarm logbook of the controller: once pressed, the console will ask for confirmation.